

UNIVERSITY OF AUCKLAND

MASTER'S THESIS

**Comparative Analysis of forecasting
techniques for the Implied Volatility
Surface**

Author:
PETER BRIESCH

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UNIVERSITY OF AUCKLAND

Abstract

Engineering
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Masters of Engineering

Comparative Analysis of forecasting techniques for the Implied Volatility Surface

by Peter BRIESCH

The predictability of asset returns is a foundational topic in finance, crucial for informed investment and risk management. For options, this predictability is intrinsically linked to the forecasting of implied volatility (IV). The collection of IVs across all strike prices and maturities forms the Implied Volatility Surface (IVS), a complex, dynamic structure whose behavior is critical to understand for institutional and retail investors alike. While there is extensive literature exploring forecasting different facets of the IVS, whether forecasting the surface point by point, cross-sectionally or as a whole, a comprehensive comparison of feature representations and predictive models has been lacking, leaving open questions about whether cross-sectional information can be used to improve forecasting accuracy. This thesis addresses this gap by conducting a large-scale comparative analysis of models for forecasting the S&P 500 index options IVS. The methodology compares seven distinct feature encoding techniques and seven predictive models, introducing several methods not previously applied in this context, including Canonical Covariance Analysis (CCA), Surface Stochastic Volatility Inspired (SSVI) parameterization, and the Heterogeneous Autoregressive (HAR) model. The out-of-sample forecasting performance is rigorously evaluated using multiple statistical metrics, including R^2 , Root Mean Square Error (RMSE), and directional accuracy. Results demonstrate that non-linear models, particularly Support Vector Regression (SVR), often outperform simpler linear models, challenging some previous findings in the literature. Furthermore, this research extends prior work by showing that the predictive power of historical data is not limited to point-wise forecasts but extends to entire cross-sections of the IVS, indicating that information from the entire surface—both term structure and moneyness skew—is crucial for accurate predictions. Finally, the economic significance of the best-performing models is validated through a simple volatility trading strategy, which demonstrates that the developed forecasting models can generate significant excess returns over a benchmark.

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Contents

Abstract	i
Acknowledgements	ii
1 Introduction	1
1.1 State-of-the-art work on forecasting the implied volatility surface	7
1.1.1 Problem Statement	8
1.2 Contributions	8
1.3 Thesis structure	9
2 Related Work	10
2.1 Options implied volatility	10
2.2 Model frameworks for forecasting implied volatility	13
2.2.1 Equilibrium Framework	13
2.2.2 Data Driven	14
2.2.3 Parametric	14
2.2.4 Machine Learning	15
2.3 Background: Machine learning techniques	16
2.3.1 Feature encoding techniques	16
Principal Component Analysis (PCA)	16
Canonical Correlation Analysis (CCA)	17
Nelson-Siegel (NS) model	18
Augmented Dumas Nelson Siegel (ADNS) model	18
Surface Stochastic Volatility Inspired (SSVI) model	19
Heterogeneous Autoregressive (HAR) model	19
2.3.2 Prediction models	20
Linear regression models	20
Ridge regression	21
Lasso regression	21
ElasticNet regression	22
Long Short-Term Memory (LSTM) model	22
Random Forest (RF) model	24
Support Vector Machine (SVM) model	25
Attention model	26

3	Methodology for Forecasting Implied Volatility Surfaces	30
3.1	Descriptive Statistics and IVS Construction	30
3.1.1	Descriptive statistics	30
3.1.2	Implied Volatility Surface Interpolation	32
4	Feature construction and encoding	36
4.1	Feature matrix construction	36
4.1.1	Point features	36
4.1.2	Moneyness Skew features	37
4.1.3	Term structure features	37
4.1.4	Surface features	37
4.2	Feature encoding models	38
4.2.1	Sampling input feature matrix (SAM) encoder	39
4.2.2	Principal Component Analysis (PCA) latent space encoder	39
4.2.3	Canonical Covariance Analysis (CCA) latent space encoder	40
4.2.4	Nelson Siegel parametric encoder	40
4.2.5	Augmented Dumas Nelson Siegel (ADNS) parametric encoder	41
4.2.6	Surface Stochastic Volatility Inspired (SSVI) parametric encoder	42
4.2.7	Heterogenous Autoregressive (HAR) latent space encoder	43
4.3	Prediction models	44
4.3.1	Linear models	45
4.3.2	Non-linear models	46
5	Empirical study for forecasting the IVS	51
5.1	Experimental setup	51
5.2	Quantitative measures	51
5.2.1	The R^2 Statistic	52
5.2.2	Root Mean Square Error (RMSE)	53
5.2.3	Directional accuracy	54
5.3	Results and Discussion	55
5.3.1	Out of sample forecasting	55
	Directionality Analysis	67
	Forecast comparison across varying lags	73
	Comparing best feature input models	75
	Forecast performance across years	76
	Moneyness and Time to maturity forecast comparisons	77
5.3.2	Function of features	79
5.3.3	Robustness Check	82
6	Economic Significance	90
7	Conclusion	93
7.1	Future work	94

A	Comparative out of sample forecasting results	95
A.1	Call Options Contracts	95
A.2	Put Options Contracts	104
B	Peseran-Timmermann statistic	112
B.1	Call Options Contracts	112
B.2	Put options	116
C	Trading results for put options	119
D	Interpolation example surfaces	120
D.1	Call Options	120
D.2	Put Options	121

Chapter 1

Introduction

The following thesis will perform a comparative analysis of different techniques for forecasting the implied volatility surface using the S&P 500 index options. First, in order to understand the importance of forecasting the implied volatility surface, a discussion on the importance of forecasting asset returns is required. What is an asset? An asset is any resource owned by a business, institution, or individual that has trade value and/or is expected to provide future economic benefits. There are many different ways to classify assets, such as by their liquidity, physical form, risk/return profile, etc. A most useful classification in the context of this thesis would be by their usage, in this case investment assets are of interest. However, investment assets are still a broad category that include many different types of assets such as real estate, stocks, bonds, commodities, currencies, and more recently cryptocurrencies. Now, the importance of forecasting asset returns (the interval change in the value of an asset over time) for the average investor is that it allows them to make informed decisions about which assets to buy, hold, or sell in order to maximize their returns and minimize their risks.

Options are a type of investment asset that are derived or built on top of another asset, such as stocks, bonds, or currencies, hence they are called *derivative* assets. An options contract gives the buyer the right, but not the obligation, to buy or sell some other underlying asset at a predetermined price (known as the strike price) and date (known as the maturity date or expiration date). The price paid for the option contract is known as the option premium. At some point in the future, the option can be exercised, meaning the owner makes the decision to exercise the terms of the option contract otherwise the option expires worthless. In this case the owner loses the premium paid for the option. Intuitively, options can be used to speculate on the future price of the more expensive underlying asset, without having to actually own it. There are many different types of options, that slightly differ in the way they are structured and executed, such as American options, European options, and exotic options. This thesis will focus on European options, which can only be exercised at the maturity date. There are two types of European option contracts that can be bought or sold:

- **Buying call options** give the buyer the right to buy the underlying asset at a predetermined date for a given strike price.
- **Selling call options** give the seller the obligation to sell the underlying asset at a predetermined date for a given strike price.

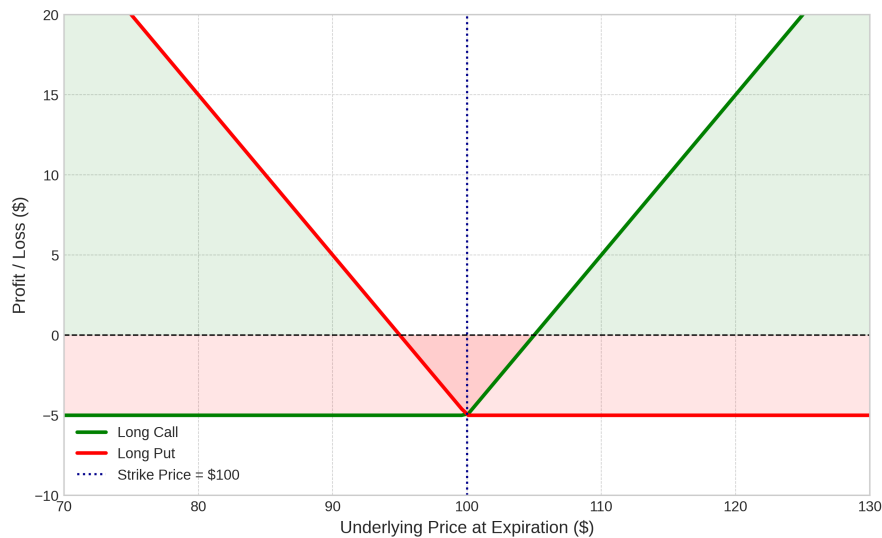


FIGURE 1.1: An example of the profit and loss (P&L) profile at expiration for **buying** a call and put option with a strike price at \$100 and a premium of \$5. The loss is bounded by the premium paid for the option as there is no obligation to exercise the option. The profit is unbounded as the underlying asset can increase in the case of a call option and decrease in the case of a put option indefinitely.

- **Buying put options** give the buyer the right to sell the underlying asset at a pre-determined date for a given strike price.
- **Selling put options** give the seller the obligation to buy the underlying asset at a predetermined date for a given strike price.

To more effectively illustrate the execution of an option contract, consider the following example: suppose an investor buys a call option on a stock with a strike price of \$100 and a premium of \$5, with an expiration date in 30 days. If the stock price rises to \$120 by the expiration time, the investor can exercise the option and buy the stock for \$100, then immediately sell it for \$120, making a profit of $\$15 = \$120 - \$100 - \5 (the difference between the profit on the sale and the premium paid for the options contract). If the stock price falls to \$80 at expiration, the investor can choose not to exercise the option, and only lose the premium paid of \$5. In this case, the option contract provides a way for the investor to profit from the stock price increase while limiting their potential losses to the premium paid. Figure 1.1 illustrates this concept. The same is true for the put option as illustrated in figure 1.1. However, the put option profits when the underlying asset price decreases. Now in order to facilitate this trade there must be someone taking the opposite position, thereby selling the put option. This investor takes an equal and opposite position to the buyer of the option and will only profit by the premium paid for the option if the underlying stock price is above the strike price at expiration. The same is true for the call option as shown in figure 1.2.

Figure 1.3 shows an example of how options are typically presented at a brokerage or on a trading platform. The middle strike column contains the strike prices of the option contracts, and divides the table into two sections, the left side shows the call

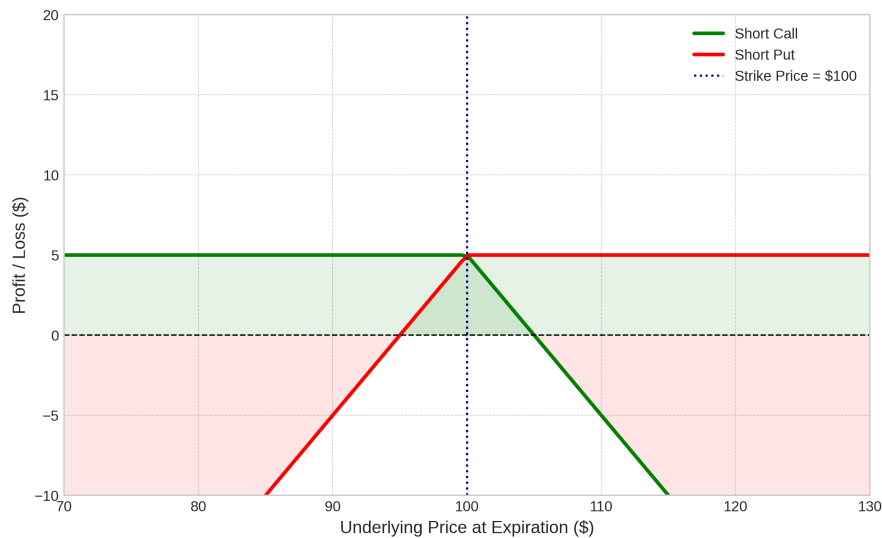


FIGURE 1.2: An example of the profit and loss (P&L) profile at expiration for **selling** a call or put option with a strike price at \$100 and a premium of \$5. The profit is bounded by the premium paid for the option. The loss is unbounded as the underlying asset can increase in the case of a call option and decrease in the case of a put option indefinitely and the seller is obligated to buy/sell the stock at the strike price if the option is exercised.

options and the right side shows the put options. Each row represents an option contract for a different strike price at the same expiration date. This is known as an options chain. Additionally, every expiration date has its own options chain. This means that for N different strike prices and M different expiration dates, there are $N \times M$ different option contracts for call and put options. There are several different factors when it comes to pricing an option contract. Most obviously is the price of the underlying asset (S), secondly is the strike price (K) of the option contract. Together these two values determine the intrinsic value, which is the profit that an investor would make if they were to buy the contract and immediately execute the contract. Given the previous example of a call option with a strike of \$100, if the underlying stock price is \$110 the intrinsic value would be $\$110 - \$100 = \$10$. Considering this information the seller of the contract would not sell the contract for less than \$10. Next is the time to expiration (T), which is the time remaining until the option contract expires. The longer the maturity the more time there is for the underlying asset to move in the direction that is profitable for the option holder. Therefore, the longer the time to expiration, the more expensive the option contract. Finally, one of the most important factors in the options price (premium) is the volatility of the underlying asset. The general definition of volatility is that it is the standard deviation of the returns of the underlying asset. Volatility is a measure of the uncertainty or risk associated with the underlying asset's price movements. During periods of high volatility, the underlying asset's price is expected to fluctuate more drastically. During periods of low volatility, the underlying asset's price stays relatively stable. This is important for options pricing because the more volatile the underlying asset, the more uncertainty there is that the given option will expire worthless.

EXPIRATION

Friday, June 27, 2025 (4) (W)

Show All Columns

Strike Range: 176 - 220

In The Money

Calls

Puts

BID	ASK	MID	VOL	OI	IV	DELTA	STRIKE	BID	ASK	MID	VOL	OI	IV	DELTA
23.85	24.50	24.18	48	24	24.98%	1.00	177.5	0.03	0.04	0.04	271	879	23.70%	0.00
21.60	21.80	21.70	93	1,653	24.98%	0.9999	180	0.04	0.05	0.04	1,595	6,097	23.70%	0.00
19.05	19.35	19.20	21	241	24.98%	0.9999	182.5	0.05	0.06	0.06	3,677	2,900	23.70%	0.00
16.35	17.00	16.68	55	430	24.98%	0.9995	185	0.07	0.08	0.07	3,953	9,421	23.70%	-0.0003
13.95	14.40	14.17	134	2,673	24.98%	0.9973	187.5	0.10	0.11	0.11	2,162	2,187	23.70%	-0.0017
11.60	11.95	11.78	480	1,362	24.98%	0.9885	190	0.14	0.16	0.15	7,870	10,070	23.70%	-0.0082
9.30	9.50	9.40	557	1,711	24.98%	0.9621	192.5	0.23	0.24	0.23	5,734	4,656	23.70%	-0.0307
7.00	7.15	7.08	2,172	10,484	24.98%	0.9002	195	0.41	0.43	0.42	18,578	10,028	23.70%	-0.0884
4.85	5.10	4.97	2,954	4,815	24.98%	0.7868	197.5	0.75	0.78	0.77	20,730	5,779	23.70%	-0.2012
3.00	3.15	3.08	25,261	15,435	24.98%	0.6234	200	1.40	1.43	1.42	31,320	5,701	23.70%	-0.3706
1.66	1.70	1.68	35,749	12,793	24.98%	0.4362	202.5	2.45	2.53	2.49	7,997	1,563	23.70%	-0.5677
0.75	0.77	0.76	45,378	28,659	24.98%	0.2644	205	4.00	4.20	4.10	1,883	3,366	23.70%	-0.7470
0.27	0.30	0.29	17,674	11,961	24.98%	0.1371	207.5	6.05	6.45	6.25	305	129	23.70%	-0.8757
0.10	0.11	0.11	21,869	23,768	24.98%	0.0604	210	8.40	8.75	8.57	540	1,045	23.70%	-0.9491
0.03	0.05	0.04	13,646	4,270	24.98%	0.0225	212.5	10.80	11.65	11.22	29	37	23.70%	-0.9827
0.02	0.03	0.02	4,218	10,097	24.98%	0.0071	215	13.15	13.85	13.50	14	30	23.70%	-0.9951
0.01	0.02	0.01	962	2,619	24.98%	0.0019	217.5	15.70	16.50	16.10	2	10	23.70%	-0.9988
	0.01	0.01	2,701	12,055	24.98%	0.0004	220	18.20	18.75	18.48	30	7	23.70%	-0.9997

FIGURE 1.3: An example of an options chain for Apple Inc. (AAPL) stock. The table shows the available call and put options for different strike prices and maturities for 4 days to expiration [1].

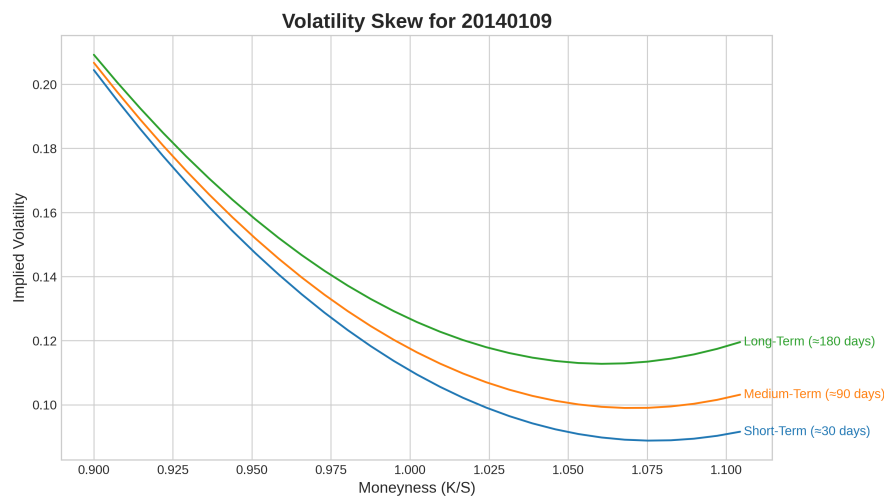


FIGURE 1.4: An example of the implied volatility skew for the S&P 500 index options on January 9th, 2014 for 3 different maturities. The x-axis shows the moneyness (strike/underlying price) of the option contract, the y-axis shows the implied volatility of the option contract

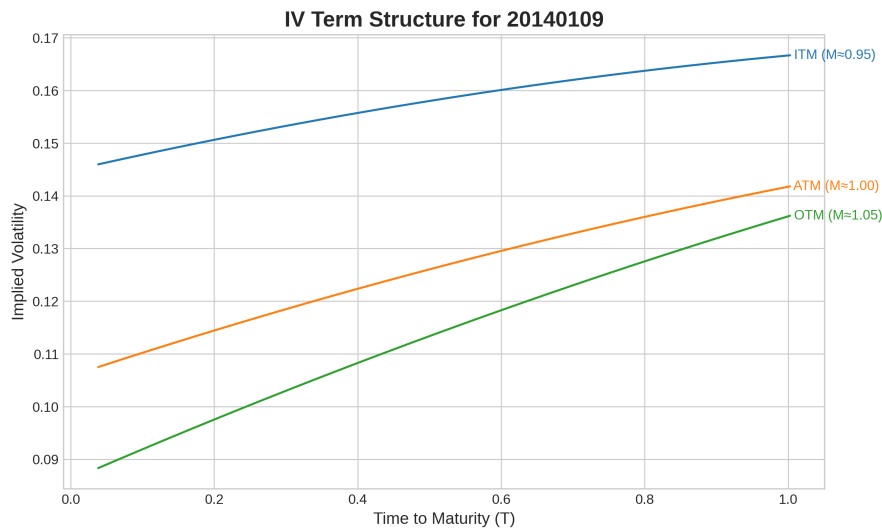


FIGURE 1.5: An example of the implied volatility term structure for the S&P 500 index options on January 9th, 2014 for 3 different strike prices. The x-axis shows the annualised time to maturity ($T = \text{days to maturity} / 365$) of the option contract, the y-axis shows the implied volatility of the option contract

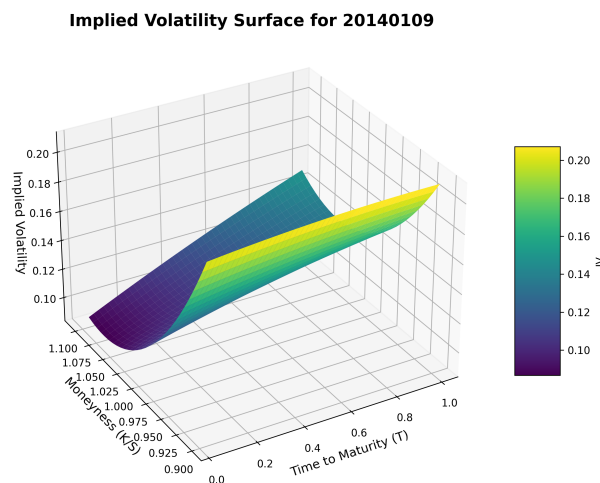


FIGURE 1.6: An example of the Implied Volatility Surface (IVS) for the S&P 500 index options on January 9th, 2014. The x-axis shows the strike price of the option contract, the y-axis shows the implied volatility of the option contract, and the z-axis shows the maturity of the option contract.

Now, given that options contracts for all strikes and maturities are written on the same underlying asset, one might expect that the volatility of the underlying asset would be the same for all options contracts. However, this is not the case in practice. As shown in the options chain in figure 1.3, each option contract has an implied volatility (IV) associated with it. This implied volatility is derived from the market price/premium of the option contract, and reflects the market's expectations of the future volatility of the underlying asset. In most cases, this implied volatility value is calculated using the Black-Scholes model, which will be further discussed in section 2.1. When plotting the implied volatility against the strike price—the "Strike" column against the "IV" column—the resulting plot is typically not flat, but instead displays a smile or skew shape known as the implied volatility skew (see figure 1.4). Similarly, plotting implied volatility against time to maturity for a fixed strike price produces the implied volatility term structure (see figure 1.5). Stitching together these plots across all strikes and maturities forms a three-dimensional surface, known as the implied volatility surface (IVS), as illustrated in figure 1.6.

With this understanding, whether options assets can be predictably forecasted arises. Whether asset returns can be predictably forecasted has been a longstanding one in finance, with extensive research focusing on primary investment assets such as stocks, corporate bonds, and treasury bonds [2]–[11]. Compared to these asset classes, forecasting the returns of options is considerably more challenging. The excess future returns from holding options, as opposed to holding the underlying asset, are heavily influenced by the volatility of the underlying asset [12], [13]. Thus, the predictability of option returns is closely tied to the predictability of option volatility.

This relationship was formalized in the Black-Scholes option pricing model [14], where the volatility implied by observed option prices—implied volatility (IV)—is the main determinant of the option price as this is the only factor that is not directly observable. Correctly forecasting future IV is therefore crucial for accurate option pricing and for anticipating future option returns. The implied volatility surface (IVS) can be visualized by plotting IV against strike price and maturity, resulting in a 3D surface (Figure 1.6). While the Black-Scholes model assumes a flat IVS, empirical evidence shows that the surface is far from flat, exhibiting a curve shape in both strike price and maturity. The IV for varying strike prices at a given maturity is referred to as the moneyness skew, and the IV for varying maturities at a given strike is known as the term structure. Each point on the IVS corresponds to a specific combination of strike and maturity. Understanding the evolving dynamics of the IVS is valuable for traders and risk managers, as it can inform investment decisions and improve risk management strategies [15].

Forecasting the IVS can be approached in several ways: by predicting each point on the surface individually, forecasting slices of the term structure or moneyness skew, or by predicting the entire surface at once. Prior research has explored point-based forecasting using past IV data for single moneyness and maturity [16]–[18], as well as approaches that consider broader cross-sectional information [19]–[22]. This thesis

builds on these foundations, aiming to compare and evaluate different techniques for forecasting the implied volatility surface of S&P 500 index options.

1.1 State-of-the-art work on forecasting the implied volatility surface

There have been several works in the literature investigating the predictability of the IVS [16], [19]–[21], [23]–[25]. Fewer works investigate the time varying dynamics of the IVS structure. Wen, Zhai, Wang, *et al.* [16] use lagged points of the S&P 500 index options to compare out of sample forecasting of linear and non-linear machine learning methods. Through their study they find that the options IVS is past-dependent. By investigating the out-of-sample forecast performance they show that simple linear models achieve R^2 scores up to 81.91% with the best performing model making use of 20-day historical sequences. Wen, Zhai, Wang, *et al.* show the importance of historical Implied Volatility (IV) in predicting future IV. At the same time they show that more parsimonious linear models can more effectively capture total variation of the IVS out of sample. While Wen, Zhai, Wang, *et al.* explore the past-dependence of the IVS, they only reconstruct the IVS by doing point by point forecasting. So, no cross-sectional information of the IVS is included.

A recent study by Kumar, Agarwalla, Varma, *et al.* [23] investigates whether modeling the relationship across the strike prices of the IV surface can improve forecasting of future IVS's. They make use of the very popular Nelson and Siegel model [26], which first found prominence in successfully modeling the term structure of the yield curve [27]. Their approach to modeling the volatility smile is in the same vain as Chalamanan and Tsekrekos [19] and many others [20], [24], [28]. However, in contrast to these other studies, they make modifications to apply the Dumas Nelson Siegel (DNS) model to the volatility skew instead of the term structure. Kumar, Agarwalla, Varma, *et al.* make an argument that more traditional measures of model performance through statistical forecast measures like R^2 , are not sufficient in evaluating model performance in the context of real-world trading performance. Instead of using traditional statistical measures, they focus on the models utility for trading desks and use portfolio performance to measure forecasting accuracy of models. Using the top 30 single stock options from the NSE of India they employ a long short trading strategy. They perform model forecasting then buy (sell) options the model deems undervalued (overvalued). If there is information to be gained from the term structure of options IV, then the portfolio should show excess returns over the benchmark. They show that the DNS model outperforms the random walk benchmark model with a trade win rate of 85% compared to 50%. All dynamic models show improved portfolio results over their static counterparts.

The three dimensions of the Implied Volatility Surface (IVS) have been thoroughly evaluated in the literature: beginning with the investigations into the dynamics of the implied volatility surface, many attempt to account for the biases presented in the Black

and Scholes model [14] with stochastic volatility models [29]–[32], and behavioural models [33], [34]. There is work on using the whole past IVS to forecast the future IVS [19], [20], [35]. There is work on predicting the term structure of the IVS alone [24], and even predicting just the IVS moneyness skew [23].

1.1.1 Problem Statement

All these works show that there is some information contained in all these different features of the past IVS surfaces. However, ① there has not been a comprehensive comparison between the predictive ability of each of these feature representations, ② it is still not clear which forecast technique provides the greatest predictive ability. Is the past IV good enough for future IV forecasting or does information across the entire IVS need to be included? Do dimensionality reduction techniques lead to better performance than parametric models or should direct sampling of features be preferred? Are simple linear models better predictors of future IVS than non-linear models or neural networks? These are all questions that this thesis explores.

1.2 Contributions

The research questions and contributions follow from the motivations above and to investigate them the market data from the S&P 500 (SPX) options is used.

1. Firstly, several feature encoding techniques are compared including Principal Component Analysis (PCA), Nelson Siegel (NS) [22], [26] to simply using the lagged IVS features. New feature encoding techniques are also introduced: Surface Stochastic Volatility Inspired (SSVI) [36], HAR [37], and Canonical Covariance Analysis (CCA) [38]. The results show that the HAR method of feature encoding shows the best out of sample performance out of all the encoding methods. However, simply using the sampled lagged IVS features achieves better if not equal out of sample performance than the other feature encoding methods.
2. Secondly, there is some evidence to suggest that long and short-dated term structures contain different information and both are needed for good prediction of future IVS [24]. Is this also true for information across the moneyness skew and entire surface of the IVS? An analysis of the regression feature weighting of the best performing linear models is performed. Our observations further support the work of Guo, Han, and Lin [24] for the term structure. The results also show that this extends to information across the moneyness skew. In other words the entire implied volatility surface (IVS) contains information for forecasting future IVS. Understanding the importance of lagged IVS features for predicting future IVS is one of the major contributions of this work.
3. A comparison of the out of sample forecast performance is performed for long, medium and short lagged input features. The results contradict the findings of

Wen, Zhai, Wang, *et al.* [16]. They show that longer lag lengths lead to better out of sample performance for point by point prediction of the IVS. In contrast, this thesis shows that in most cases shorter lags achieve better R^2 and Root Mean Square Error (RMSE) scores and increasing the lagged input past 10 lags may even lead to deterioration in model performance.

4. A comprehensive comparison is also performed between the best performing models for each feature of the IVS. This includes comparing the predictive power of each model for each region of the IVS. The results show that overall the support vector regression model with the sampling method of feature encoding achieves a better Diebold and Mariano (DM) statistic and R^2 statistic than the other models. However, when comparing each model's performance on every region of the IVS it is clear that there is no single model best suited to predicting every region of the IVS. The result reinforces the findings in [24] that different regions of the IVS contain different information and all regions are needed for good prediction of future IVS.
5. Finally, an analysis of the economic significance is performed to compare the developed feature encoding and prediction models in order to provide guidance and recommendations to finance professionals.

1.3 Thesis structure

The rest of the thesis will be organised as follows. Chapter 2 goes through the related literature beginning with a brief history of options pricing and options pricing theory. It then moves onto discussing previous and recent work on options price and implied volatility forecasting. The literature is categorized into 4 model frameworks; general equilibrium, data-drive, parametric, and machine learning. The state of the art is discussed in each of these and compared to the work of the thesis.

The methodology is then presented in chapter 3. It begins with descriptive statistics about the dataset and overall IVS construction 3.1, which includes data preprocessing 3.1.1, and interpolation techniques 3.1.2 used in the work.

Chapter 4 forms the core of the study, describing the various feature construction, encoding techniques and prediction models. Firstly, in section 4.1, the four feature construction methods are described for point, term structure, volatility smile, and surface prediction. Each of the feature encoding/dimensionality reduction techniques are described in section 4.2. The prediction models are described in section 4.3.

The empirical study along with the statistical measures are described in chapter 5. This is then followed by a discussion about the out of sample forecasting results in section 5.3 and the effect of IVS features on forecasting performance in section 5.3.2. A discussion about the economic significance results follows in chapter 6. The thesis concludes in chapter 7 where the findings are summarised, followed by a discussion of potential future work 7.1.

Chapter 2

Related Work

This chapter will provide an overview of the current literature for the prediction of implied volatility. However, for one to have an appreciation of the importance of implied volatility and the forecasting thereof an explanation of options pricing and implied volatility is needed. The early works on options pricing forms the foundation for the current understanding of underlying market dynamics that drive options prices this includes the seminal work of Black and Scholes [14] and Merton [39]. In which some assumptions are made about the underlying market dynamics for example the volatility of the underlying is assumed to be constant across different options maturities and strike prices. In reality these assumptions do not hold true. It has been shown empirically that the skew and term structure of the implied volatility surface (IVS) is not constant but rather shows smirk and smile effects [40]. There have been several evolutions and attempts to improve upon the Black-Scholes options pricing model [29], [32], [41], [42]. However, the Black-Scholes model still remains the benchmark for options pricing.

The rest of this chapter will proceed as follows: ① A brief introduction to the literature surrounding options pricing and implied volatility dynamics. ② Then moving on to popular methods of forecasting implied volatility broken up into 4 categories; equilibrium, data driven, machine learning and parametric models.

2.1 Options implied volatility

The foundation of options pricing comes from the seminal work of Black and Scholes who proposed the well known Black-Scholes model for options pricing [14], here after referred to as Black-Scholes model (BS). The BS model makes assumptions about the underlying market dynamics including that the underlying asset of the options contracts follows a geometric Brownian motion in its returns, and that the volatility of the underlying asset is constant across strikes and maturities. The price of a European call option under the BS model is given by:

$$\begin{aligned}
C(S, t) &= SN(d_1) - Ke^{-r(T-t)}N(d_2) \\
d_1 &= \frac{\ln \frac{S}{K} + \left(r + \frac{\sigma^2}{2}\right) \tau}{\sigma \sqrt{\tau}} \\
d_2 &= d_1 - \sigma \sqrt{\tau}
\end{aligned}$$

Where:

- $C(S, t)$ is the price of the call option at time t with underlying asset price S
- S_t is the spot price at time t
- K is the strike price
- T is the maturity time, with $\tau = T - t$ being the time to maturity
- σ is the volatility of the asset
- r is the risk-free interest rate
- $N(\cdot)$ is a standard normal cumulative distribution function.

As one can see, the solved BS formula for the price of a call option is a function of the underlying price and time. It assumes that for every strike and maturity the volatility is constant which is not observed in actual market data [40]. These idiosyncrasies observed in market data often manifest as smiles or smirks when plotting the implied volatility against the strike price of options (see figure 3.2). This smile (smirk) phenomenon can also be observed when plotting IV against the maturity of options. These initial biases discovered in the Black and Scholes formulation have drawn much attention, with many attempting to improve upon it [29], [30], [40]–[42].

Merton [42] first proposed a jump diffusion model in order to explain the pricing inaccuracies in the BS model. Merton attempts to improve upon the BS model by relaxing the assumption that the underlying assets returns are continuous in time, and instead allows for large "jumps" in the process that drives the underlying asset. He finds that this model more aptly explains the price discrepancies found between the BS model prices and market prices. However, in a study by Rubinstein [40], competing alternate pricing models are compared against the BS model. These include models that relax the assumption of constant volatility, and jump based models including the model presented in [42]. He does not find that such models sufficiently captures all the biases observed from the BS model. Furthermore, Das and Sundaram [43] find that jump diffusion and stochastic volatility models do not sufficiently capture or explain all the structural anomalies observed in the IVS data. However, of the two classes of models, the stochastic volatility models are found to be more successful at capturing the structural dynamics of the IVS.

Other popular models include the binomial options pricing model introduced in [44]. The binomial model is a discrete time model that allows for the underlying asset to move up or down. It divides the time to maturity of the option into discrete time steps, where at each time step the underlying asset price can either increase or decrease with some probability. The entire price path of the underlying is modeled as a binomial tree, where the current price of the option is the expected value of the option price at maturity, discounted back to the current time. The binomial model is more flexible than the BS model, as it allows for the underlying asset to have a non-constant volatility, and can also be used to price American options.

Beside parametric approaches, there have also been several non-parametric approaches to options pricing, including the use of neural networks [45]–[49]. Much like regression models which learn a linear combination of features that map some input to an output, neural networks learn a set of weights and biases that pass through non-linear activation functions to map some input to an output. In theory these neural networks can learn any non-linear approximation of a function. Hutchinson, Lo, and Poggio [46] are one of the first to apply neural networks to options pricing. They show that learning networks can be used to approximate the BS model and compare several networks including, radial basis functions, ordinary least squares (OLS), and multi-layer perceptrons (MLP). However, they only show that the neural networks can approximate the BS model, and do not show that they can outperform the BS model in terms of pricing accuracy. Anders, Korn, and Schmitt [47] extends upon the work of [46] and focuses solely on the use of MLPs for options pricing. An elimination strategy is used to specify the parameters and architectures of the network. Starting with a small network, successive hidden units are added and tested for significance. Once the number of hidden units is determined, each input to the hidden units is tested and if not significant to the 5% level the input with the highest p-value is removed. This process leads to a highly parsimonious model and show higher pricing accuracy than the BS model. Understandably, such a model is highly specific to the data used for training and is computationally expensive.

Despite several proposed models that more accurately capture the real options market prices [29], [32], [41], [42], the Black-Scholes model (BS) still reigns as *the* benchmark model for options pricing. This is primarily due to the computational complexity of the more complex models. Given the breadth of work attempting to account for the structural dynamics of the implied volatility surface. It is clear that being able to accurately forecast the IVS is of great importance options pricing, risk management, and trading.

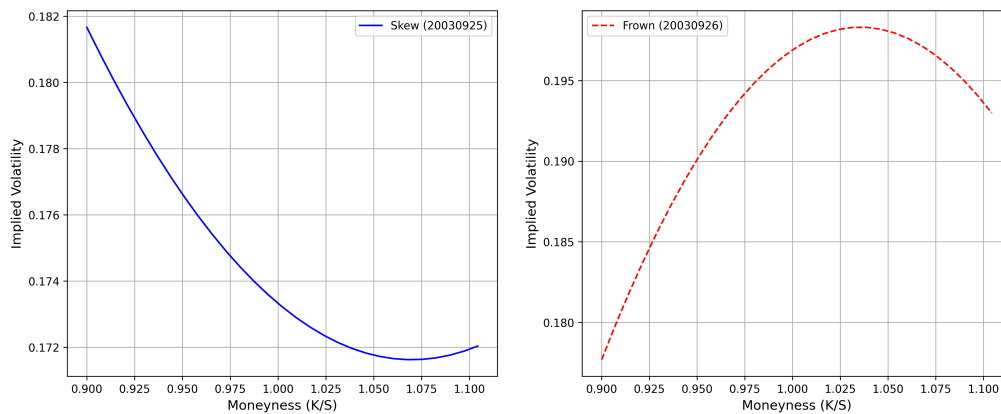
2.2 Model frameworks for forecasting implied volatility

2.2.1 Equilibrium Framework

Equilibrium models [15], [33], [34], [50] attempt to explain the observed IVS money-ness skew and its evolution overtime via investor's uncertainty, economic fundamentals (e.g., dividends), etc. For example David and Veronesi [50] attempt to explain some of the puzzling features of call options. They point out that the IV smile observed is often "highly unstable over time with sudden changes in slope and shape". This can be observed in the data as shown in figures 2.1a and 2.1b.

FIGURE 2.1: Example of the extreme changes in the structure of implied volatility skew over time present in the S&P 500 index options data

(A) Skew for 2003-09-25 at time to maturity of 14 days. (B) Skew for 2003-09-26 at time to maturity of 14 days.



They also point out that options prices are sometimes negatively correlated to changes in the underlying asset. This is strange because one would assume that a call option, which gives the purchaser the right but not the obligation to buy an asset at some date in the future for a stated price, would increase in value as the difference between the stated price and the actual price of the asset becomes more and more positive. David and Veronesi [50] attempt to explain this by constructing an options pricing model that models investors estimation of dividend drift rate and their confidence in an equilibrium model. Their findings show that investor uncertainty in dividend drift rate can somewhat account for the volatility smile phenomenon. They also develop an options pricing model based on their findings, that shows during periods of high uncertainty, volatility smiles across strike price flip from smiles to frowns due to increases in investor risk aversion. There types of flips in the volatility smile are very difficult to forecast even when using behavioral models such as the one proposed by David and Veronesi [50]. Simply using the past values of the implied volatility surface may not be sufficient to capture these dynamics and other economic features may be needed.

2.2.2 Data Driven

The pure data-driven approach [19], [20], [35], [51], [52] captures the IVS dynamics (moneyness skew, term structure, etc) in a statistical latent space. Usually the latent space are the Principal Components (PCs). These techniques first create a regular-sized IVS grid and then use the daily time series of PCs to predict the future PC and reconstruct the IVS.

Particularly, Chalamandaris and Tsekrekos [20] investigate the predictability of the IVS using the historic time series of implied volatility from the OTC options in forex markets. They do this through the use of three different models. Two of the models use parametric specifications to capture the behaviours of the implied volatility surface. The last model used is the principal component method of decomposing the features of the IVS into an often lower dimensional feature space. This way they hope to capture the dynamics of the IVS into m number of latent factors. First they show that each of the models adequately describe the IVS time series. They then compare forecast accuracy of each model against univariate benchmark models that only use previous values of implied volatility as features for forecasting future IV. They found that for short horizon forecasting of 5 days, neither the parametric nor data driven model consistently outperformed any of the benchmark models. However, for forecasts 10–200 days ahead some models showed more promising results but still struggled to outperform the random walk benchmark up to the 50-day forecast horizon. For much longer horizons (100–200 days) they saw much greater improvements over the benchmark models. Notably the principal component method of forecasting consistently underperforms against the random walk benchmark across almost all currency markets.

Techniques inspired by these approaches will be exploited and compared against other techniques in the literature to see if such a statistical latent space encoding of IVS features is *more* effective for IVS forecasting. Furthermore, as far as the literature shows, this thesis also extends these works to include so called Canonical Covariance Analysis (CCA) [38]. Unlike Principal Component Analysis (PCA), which only projects the inputs to a latent space, that explains most of the variability of the input *only*; CCA projects the input and output together into latent space such that the resulting (usually smaller) latent factors explain most of the variation in outputs and inputs together.

2.2.3 Parametric

The parametric approach to forecasting IVS (or its components) is one of the most popular. Parametric models capture the IVS dynamics (moneyness skew, term structure) as coefficients of different functions. Dumas, Fleming, and Whaley [53], is one of the most used example of a parametric model of the option IVS, where the IVS dynamics are captured as functions of moneyness, curvature, and term structure. A time series of the coefficient of Dumas et al.'s model has been shown to be a good predictor of future IVS [15], [28]. The work of Dumas, Fleming, and Whaley [53] is extended in Chalamandaris and Tsekrekos [19] and Guo, Han, and Zhao [22] to capture the term structure of

IVS explicitly using the Nelson-Siegel yield curve model [26]. The work in Guo, Han, and Lin [24] and Kearney, Shang, and Sheenan [28] shows that incorporating explicit term structure via Nelson-Siegel leads to improved forecasting accuracy of the IVS. Recent work in Kumar, Agarwalla, Varma, *et al.* [23] models the IVS moneyness skew using a variation of the Nelson-Siegel model, and uses Kalman filter [54] to predict the one-day ahead IVS moneyness skew. All these works show that parametric specifications of the IVS can be used to effectively and parsimoniously capture the dynamics of the IVS and can be used to forecast future IVS.

2.2.4 Machine Learning

Recently more attention has been given to exploring whether machine learning techniques can provide better forecasting performance than traditional econometric models. However, the machine learning literature focusing on forecasting the entire Implied Volatility Surface (IVS) is far sparser than that focusing on IV indices, such as the VIX, or volatility in general. The following will look at the general literature for forecasting volatility and then focus will be given to the works on forecasting the IVS.

The machine learning and non-parametric approach to predicting the IVS has become popular recently. Wen, Zhai, Wang, *et al.* [16] compare linear and non-linear models on the lagged point IV, for a single moneyness and time to maturity, to predict the future IV point. They clearly show that regularized linear models [55] outperform the non-linear models of Long Short Term Memory (LSTM) neural networks, and regression trees (XGBoost [56]). Wen, Zhai, Wang, *et al.* [16] use data from both the S&P 500 and SSE 50 ETF options for the period from January 2012 - December 2022 and February 2015 - December 2022 respectively. Due to the sparse nature of the options chain data used, interpolation techniques are required to construct smooth IVS. They compare two different interpolation techniques, a parametric technique called deterministic volatility function (DVF) model [53] and a non-parametric technique based on the Nadaraya-Watson (NW) estimator [57] or Nadaraya-Watson kernel regression. They state their aim is not to provide a comparative analysis of these interpolation techniques. However they do, show both the DVF and NW techniques achieve comparable results. Notably, for the S&P 500 data some models have negative R^2 performance with the NW interpolation techniques. Overall, they found that the DVF method only outperforms the NW estimator for call options in the S&P 500. For the predictor models they compared three linear models; lasso regression, elastic net regression, and ridge regression and three non-linear models; extreme gradient boosting (XGBoost), deep neural network and a hybrid long-short term memory deep neural network. The primary contribution of their paper was showing that future IV for a given moneyness and time to maturity is informed by past values of IV.

In contrast, the work in Zhang, Li, and Zhang [21] uses a two step-approach of first extracting daily IVS features, then using the extracted features as input for training a LSTM model [58], they predict next days features. They then reconstruct the complete arbitrage-free IVS from the predicted feature set. The work in Zhang, Li, and Zhang

[21] does not compare the prediction accuracy of their approach with any linear models. Similarly, to this thesis, they compare the PCA method of feature extraction with simply sampling the IVS. They find that simply sampling the IVS leads to better performance than PCA. This leads to the conclusion along with the findings in Chalamandaris and Tsekrekos [20] that the IVS features show a non-linear relationship that are not easily captured by feature reduction techniques such as PCA.

2.3 Background: Machine learning techniques

The following section provides some preliminary information on the machine learning techniques used in this thesis. The techniques are grouped into two categories, feature encoding techniques and prediction models.

2.3.1 Feature encoding techniques

Principal Component Analysis (PCA)

Principal component analysis (PCA) is a statistical technique commonly used to reduce the number of features in a dataset. This method of dimensionality reduction is achieved by projecting the original features of the dataset onto a new set of orthogonal axes, called principal components. The principal components are calculated to maximize the variance of the projected data, while keeping each principal component uncorrelated with each other. In other words, the technique seeks to project the features into a lower-dimensional space where most of the information in the data is retained. The first principal component often captures the most variance in the data, while the subsequent principal components capture the remaining variance in decreasing order.

In order to calculate the principal components, first the data needs to be centered by subtracting the mean of each feature from the data. Then the covariance matrix Σ of the centered data is calculated.

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \cdots & \text{Var}(X_n) \end{bmatrix} \quad (2.1)$$

Where X_1 to X_n are the features of the dataset. The diagonal elements of the covariance matrix represent the variance of each feature, since the covariance of a feature with itself is its variance. Since the covariance matrix is symmetrical and non-invertible, it can be decomposed into its eigenvalues and eigenvectors.

$$\Sigma v = \lambda v$$

Where λ is the eigenvalue and v is the eigenvector. The eigenvectors of the covariance matrix represent the directions of the principal components, that capture the most

variance in the data and the eigenvalues represent the amount of variance captured by each principal component. By sorting the eigenvalues in descending order, one can select the top k eigenvectors that correspond to the largest eigenvalues, which represent the most important principal components. The matrix of eigenvectors can be denoted as V_k , where k is the number of principal components selected. Once the appropriate number of principal components are selected, the original data can be projected onto the new set of axes by multiplying the centered data by the selected eigenvectors.

$$X_{proj} = X_{centered} V_k$$

This new lower dimensional representation of the data can be used for further analysis or modeling.

Canonical Correlation Analysis (CCA)

Canonical Correlation Analysis (CCA) much like PCA is a statistical technique used to find the relationships between two multivariate datasets. However, unlike PCA which only finds the relationships within a single dataset, CCA finds the relationships between two datasets by projecting them onto a common latent space. The goal of CCA is to find a linear combination of the features in each dataset that maximizes the correlation between the two datasets. Again much like PCA the canonical variables are calculated by first finding the covariance matrix however, in this case the covariance matrix is calculated between the two datasets, X and Y .

$$\Sigma_{XY} = \begin{bmatrix} Cov(X_1, Y_1) & Cov(X_1, Y_2) & \cdots & Cov(X_1, Y_m) \\ Cov(X_2, Y_1) & Cov(X_2, Y_2) & \cdots & Cov(X_2, Y_m) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(X_n, Y_1) & Cov(X_n, Y_2) & \cdots & Cov(X_n, Y_m) \end{bmatrix} \quad (2.2)$$

Where n is the number of samples of matrix X and m is the number of samples of matrix Y . The objective of CCA is to find a sequence of vectors a and b such that the correlation between the linear combinations of the two datasets, $a_k^T X$ and $b_k^T Y$, is maximized. Where k is the number of canonical variables. Once the first pair of canonical variables a_1 and b_1 are found, the next pair of canonical variables a_2 and b_2 are found by maximizing the correlation between the linear combinations of the two datasets, subject to the constraint that they must be uncorrelated to the first pair of canonical variables. This can be repeated until a minimum of the number of features $\min(n, m)$ in either dataset is reached.

The original datasets can then be projected into the new latent space by multiplying the original datasets by the canonical variables.

$$X_{proj} = X a_k$$

$$Y_{proj} = Y b_k$$

In the context of this thesis CCA is used to find the most predictive relationship between the IVS features and the future IVS features by projecting both the input and output features into a common latent space.

Nelson-Siegel (NS) model

The Nelson-Siegel (NS) model, originally proposed by Nelson and Siegel [26] for modeling the term structure of interest rates, has been successfully adapted to capture the term structure of the IVS [19], [22]. The model is parsimonious yet flexible, capable of fitting a variety of curve shapes observed in financial markets. It achieves this by decomposing the term structure into three intuitive components: a long-term level, a short-term slope, and a medium-term curvature.

The NS model is defined as:

$$f(\tau) = \beta_0 + \beta_1 \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) + \beta_2 \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) \quad (2.3)$$

where:

- τ is the time to maturity.
- β_0 represents the long-term level factor, which the curve converges to as maturity increases.
- β_1 represents the short-term slope factor, capturing the difference between short-term and long-term rates.
- β_2 models the medium-term curvature or "hump" in the curve.
- λ is a decay factor that controls the rate of convergence and the position of the maximum curvature.

By estimating the time-varying coefficients $(\beta_0, \beta_1, \beta_2)$, usually by ordinary least squares (OLS) regression, for the IVS on a daily basis, these coefficients can be used as a low-dimensional feature set to represent the dynamics of the term structure for forecasting. The coefficients for the next day can be forecasted and used to reconstruct the IVS for that day.

Augmented Dumas Nelson Siegel (ADNS) model

Building on the work of Dumas, Fleming, and Whaley [53], who used polynomial functions of moneyness and maturity to model the IVS, the augmented Dumas Nelson Siegel (ADNS) model extends the NS framework to model the entire surface by incorporating moneyness dimensions [19], [22]. This allows the model to capture not only the term structure across maturities but also the volatility smile or skew across different strike prices for a given maturity.

The ADNS model adds linear and quadratic moneyness terms to the NS formulation:

$$\begin{aligned} \sigma_t^{(m_i, \tau_j)} = & \beta_t^0 + \beta_t^1 \mathbb{1}_{\{m_i \geq 1\}} m_i^2 + \beta_t^2 \mathbb{1}_{\{m_i < 1\}} m_i^2 + \beta_t^3 \left(\frac{1 - e^{-\lambda \tau_j}}{\lambda \tau_j} \right) \\ & + \beta_t^4 \left(\frac{1 - e^{-\lambda \tau_j}}{\lambda \tau_j} - e^{-\lambda \tau_j} \right) + \beta_t^5 \mathbb{1}_{\{m_i \geq 1\}} m_i \tau_j + \beta_t^6 \mathbb{1}_{\{m_i < 1\}} m_i \tau_j + \epsilon \end{aligned} \quad (2.4)$$

where m represents moneyness (i.e., K/S , the ratio of strike price to spot price) and τ represents time to maturity. The additional learned coefficients β^1 and β^2 capture the curvature (smile) of the IVS across different strike prices and the terms β^5 and β^6 right and left smile attenuation respectively. $\mathbb{1}$ is an indicator function that is 1 if the condition is true and 0 otherwise and ϵ is a random error term. The full set of coefficients $(\beta_0, \dots, \beta_6)$ provides a concise, low-dimensional representation of the entire surface, suitable for time-series forecasting.

Surface Stochastic Volatility Inspired (SSVI) model

The surface stochastic volatility inspired (SSVI) model, proposed by Gatheral and Jacquier [36], is a parametric model designed specifically to produce IVSs that are free of static (or calendar-spread) arbitrage by construction. This is a crucial property for practical applications, as arbitrage opportunities in a model can lead to inconsistent pricing and flawed risk management.

The SSVI model parameterizes the total implied variance $w(k, \theta) = \sigma^2 \tau$ as a function of log-moneyness $k = \log(K/S)$ and the at-the-money total variance $\theta_t = w(0, t)$:

$$w(k, \theta) = \frac{\theta}{2} \left(1 + \rho \phi(\theta) k + \sqrt{(\rho \phi(\theta) k)^2 + (1 - \rho^2) \phi(\theta)^2} \right) \quad (2.5)$$

where ρ is a parameter controlling the at-the-money skew and $\phi(\theta)$ is a function of at-the-money variance, chosen to ensure no-arbitrage conditions are met. The model's parameters can be fitted to market data to provide an arbitrage-free, smooth representation of the IVS, which can then be used for forecasting.

Heterogeneous Autoregressive (HAR) model

The heterogeneous autoregressive (HAR) model, introduced by Corsi [37], is a simple yet remarkably effective time-series model that captures the long-memory property often observed in financial volatility. It is inspired by the heterogeneous market hypothesis, which posits that the market is composed of agents with different trading horizons (e.g., daily, weekly, monthly). The model mimics this structure by using an additive cascade of volatility components calculated over different time scales.

The standard HAR model predicts future volatility using a linear combination of past daily, weekly, and monthly realized volatility components:

$$RV_{t+h}^{(d)} = \beta_0 + \beta_d RV_t^{(d)} + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \epsilon_{t+h} \quad (2.6)$$

where $RV_t^{(d)}$ is the daily realized volatility, and $RV_t^{(w)}$ and $RV_t^{(m)}$ are the average volatilities over the past week (5 days) and month (22 days), respectively. In the context of this thesis, the HAR framework can be adapted as a feature extraction technique that encodes the IVS features into short-term, medium-term, and long-term components, which can then be used as inputs to prediction models.

2.3.2 Prediction models

The features extracted by the encoding techniques described above are used as inputs to various prediction models to forecast the future IVS. The following models are considered in this thesis, ranging from simple linear benchmarks to more complex neural networks.

Linear regression models

Linear regression is a fundamental supervised learning model that serves as a crucial baseline for comparison. It models the target variable as a linear combination of the input features. A simple case of linear regression on a single feature is given by:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon \quad (2.7)$$

in matrix notation:

$$\mathbf{y} = \mathbf{x}\beta + \epsilon \quad (2.8)$$

Where $\mathbf{y}$ is the column vector of target variables, $\mathbf{x}$ is the column vector of input features, and β is the vector of coefficients β_0, β_1 . The coefficients are most often estimated using the ordinary least squares (OLS) method, which minimizes the sum of squared errors between the predicted and actual values of the target variable, given by:

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - x_i \beta)^2 \quad (2.9)$$

The model assumes a linear relationship between the input features and the target variable. Despite its simplicity, recent work by Wen, Zhai, Wang, *et al.* [16] has shown that regularized linear models can outperform more complex non-linear models for IVS forecasting, highlighting their use as a good benchmark. Multivariate linear regression is a more general form of linear regression that allows for multiple independent variables to predict multiple dependent variables. It is given by the following equation:

$$y_{ij} = \beta_{0j} + \beta_{1j}x_{i1} + \beta_{2j}x_{i2} + \dots + \beta_{pj}x_{ip} + \epsilon_{ij} \quad (2.10)$$

Where y_{ij} is the j -th dependent variable for the i -th observation, for $i = 1, \dots, n$, and $j = 1, \dots, m$. Using linear regression the first order autoregressive (AR(1)) model can be estimated for time series data, such as the IVS features. This model will be used as the baseline for comparison with the more complex models in this thesis. The AR(1) model is given by the following equation:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \epsilon_t \quad (2.11)$$

One can see that this is very similar to the simple case of linear regression described above, however, the input feature is the previous value of the target variable.

Ridge regression

Ridge regression is a regularized version of linear regression that adds an L2-norm penalty to the loss function. This technique is particularly useful when features are highly correlated (multicollinearity), a common characteristic of IVS data. The penalty term shrinks the coefficients of correlated predictors towards each other, improving the stability of the model. The strength of the penalty is controlled by a hyperparameter, α , which balances the trade-off between fitting the data and keeping the model weights small. This regression problem can be formulated as:

$$\min_{\beta} \left(\sum_{i=1}^n (y_i - X_i \beta)^2 + \alpha \sum_{j=1}^p \beta_j^2 \right) \quad (2.12)$$

This penalty term added to the loss function helps to prevent the coefficients from growing too large, and prevents overfitting to the training data.

Lasso regression

Lasso (least absolute shrinkage and selection operator) regression is another regularized linear model that uses an L1-norm penalty on the coefficients [59]. A key property of the L1 penalty is that it can shrink some coefficients to exactly zero. This effectively performs feature selection, producing a more sparse model allowing it to focus on the most relevant predictors. The lasso regression problem can be formulated as:

$$\min_{\beta} \left(\sum_{i=1}^n (y_i - X_i \beta)^2 + \alpha \sum_{j=1}^p |\beta_j| \right) \quad (2.13)$$

So, instead of penalizing the sum of the squares of the coefficients, as in ridge regression, lasso penalizes the sum of the absolute values of the coefficients. This encourages the estimation procedure to shrink some coefficients, and with a large enough penalty term α , some coefficients will be shrunk to exactly zero.

ElasticNet regression

ElasticNet regression, proposed by Zou and Hastie [60], is a regularized regression method that linearly combines the L1 and L2 penalties of the lasso and ridge methods. It benefits from the stability of ridge regression when dealing with correlated features while also performing the automatic feature selection characteristic of lasso. This makes it a robust and flexible choice for high-dimensional financial data where groups of correlated predictors may exist. The ElasticNet regression problem can be formulated as:

$$\min_{\beta} \left(\sum_{i=1}^n (y_i - X_i \beta)^2 + \alpha_1 \sum_{j=1}^p |\beta_j| + \alpha_2 \sum_{j=1}^p \beta_j^2 \right) \quad (2.14)$$

where α_1 and α_2 are hyperparameters that control the strength of the L1 and L2 penalties, respectively.

Long Short-Term Memory (LSTM) model

The LSTM is a type of recurrent neural network that was first introduced by Hochreiter and Schmidhuber [61]. The LSTM is an innovation over the traditional recurrent neural network that addresses the vanishing gradient problem and introduces LSTM memory cells allowing the network to more effectively learn long-term dependencies. The problem of vanishing gradients has plagued traditional recurrent neural networks, effectively limiting their ability to learn long-term dependencies in sequential data. The LSTM architecture addresses this by introducing the memory cell (c.a. 2.2).

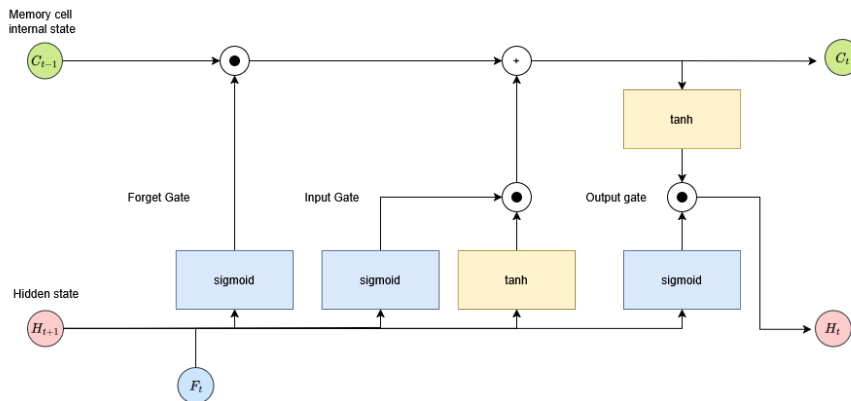


FIGURE 2.2: LSTM architecture

The LSTM memory cell architecture with input, forget, and output gates.

The LSTM memory cells consist of three gates: the input gate, the forget gate, and the output gate. The forget gate determines which information to consider from the previous time step. The output of the forget gate can be expressed as:

$$f_t = \sigma(\mathbf{W}_{Ff} \cdot \mathbf{F}_t + \mathbf{W}_{hf} \cdot \mathbf{h}_{t-1} + \mathbf{b}_f) \quad (2.15)$$

where $\mathbf{F}_t$ is the input feature matrix at time t , $\mathbf{h}_{t-1}$ is the hidden state from the previous time step, $\mathbf{W}_{Ff}$ and $\mathbf{W}_{hf}$ are the feature weights and hidden weight matrices respectively, and $\mathbf{b}_f$ is the bias vector. The sigmoid function σ ensures that the output of the forget gate is between 0 and 1, where 0 means to forget all information and 1 means to keep all information. Weights and biases are learned to minimise the loss function and effectively determine which information to retain or discard for long-term dependencies.

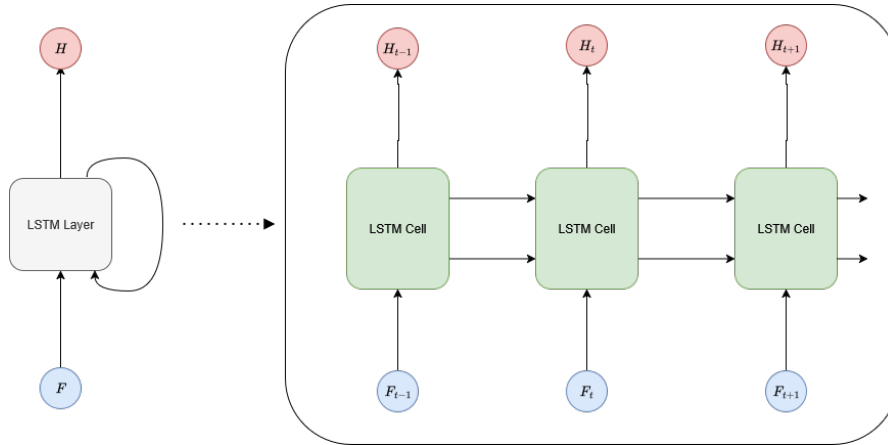


FIGURE 2.3: LSTM architecture

An illustration of the LSTM architecture unfolded over time. This highlights the recurrent nature of the LSTM, where the fixed length hidden state is updated every time step thereby compressing the temporal relationship of the input features into a fixed size representation. In essence the "memory" of the LSTM is limited by the size of the hidden state.

The input gate determines which information to add to the memory cell. The output of the input gate can be expressed as:

$$i_t = \sigma(\mathbf{W}_{Fi} \cdot \mathbf{F}_t + \mathbf{W}_{hi} \cdot \mathbf{h}_{t-1} + \mathbf{b}_i) \quad (2.16)$$

The input gate uses the same structure as the forget gate, but it learns to add relevant information to the memory cell. The input gate's output is also between 0 and 1, where 0 means to not add any information and 1 means to add all information.

The output gate determines which information is relevant for the next hidden state. The output of the output gate can be expressed as:

$$o_t = \sigma(\mathbf{W}_{Fo} \cdot \mathbf{F}_t + \mathbf{W}_{ho} \cdot \mathbf{h}_{t-1} + \mathbf{b}_o) \quad (2.17)$$

The output gate uses the same structure as the forget and input gates, but it learns to output relevant information for the next hidden state. The output gate's output is also between 0 and 1. The next memory cell state is then computed by applying the forget gate to the previous memory cell state and adding the element-wise product $\odot$ of the output of the input gate and the new memory cell input vector.

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (2.18)$$

where $\tilde{c}_t$ is the new memory cell input vector:

$$\tilde{c}_t = \tanh(\mathbf{W}_{Fc} \cdot \mathbf{F}_t + \mathbf{W}_{hc} \cdot \mathbf{h}_{t-1} + \mathbf{b}_c) \quad (2.19)$$

and finally the next hidden state is computed by applying the output gate to the new memory cell state:

$$h_t = o_t \odot \tanh(c_t) \quad (2.20)$$

Unlike the traditional recurrent neural network, the LSTM should be able to more effectively capture long-term dependencies in the data.

Random Forest (RF) model

Random forest, introduced by Breiman [62], is a versatile ensemble learning method used for both classification and regression tasks. It operates by constructing a multitude of decision trees at training time. For regression, the final prediction is the average of the predictions from all the individual trees. The model combines two key ideas to reduce variance and improve prediction accuracy: bootstrap aggregating (bagging) and feature randomness.

In the bagging process, each tree in the forest is trained on a different random subset of the training data. Furthermore, when splitting a node in a tree, the split is chosen from a random subset of the features rather than all of them. This process helps to reduce correlation between the trees, making the ensemble more robust. A visual representation of the random forest model is shown in figure 2.4.

At each node of a decision tree, of which random forest computes many, the algorithm selects a random subset of features and computes the best split that separates the subset of data into two groups. For every possible split, the algorithm calculates some evaluation metric, in the case of regression this is the sum of squared errors (SSE) for the two groups.

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y})^2$$

Where $\hat{y}$ is the mean of the target variable in the group and the total SSE is the sum of the SSE for each group. The split that minimizes the SSE is chosen as the best split for that node and the process is repeated recursively until the tree is fully grown, or some stopping criteria is met. In most cases, this is determined by the maximum depth variable set by the user.

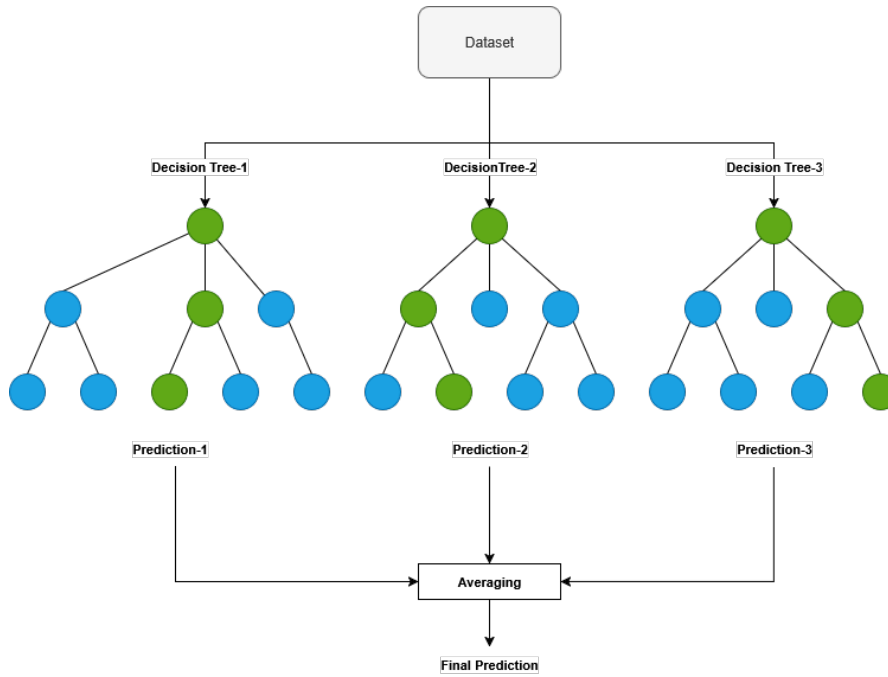


FIGURE 2.4: Illustration of a random forest model. Each tree in the forest is trained on a random subset of the training data, and the final prediction is the average of the predictions from all the individual trees.

For a regression task, if the forest consists of B trees $\{T_1, T_2, \dots, T_B\}$, the final prediction for a new input point $\mathbf{x}$ is the average of the predictions from all the individual trees:

$$\hat{y}_{RF} = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}) \quad (2.21)$$

where $T_b(\mathbf{x})$ is the prediction of the b -th tree. This averaging process helps to smooth out the predictions and reduce the risk of overfitting.

Support Vector Machine (SVM) model

Support vector machines (SVMs) are a class of supervised learning models originally developed for classification [63], which can be adapted for regression tasks, a method known as support vector regression (SVR). The core idea of SVR is to find a function $f(\mathbf{x})$ that deviates from the actual targets y_i by a value no greater than a specified margin ϵ for all training data, while at the same time being as flat as possible. This concept is illustrated in figure 2.5.

For a linear function $f(\mathbf{x}) = \mathbf{w} * \mathbf{x} + b$, the goal is to find the weight vector $\mathbf{w}$ and bias b by solving the following optimization problem:

$$\min_{\mathbf{w}, b, \xi, \xi^*} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^n (\xi_i + \xi_i^*) \quad (2.22)$$

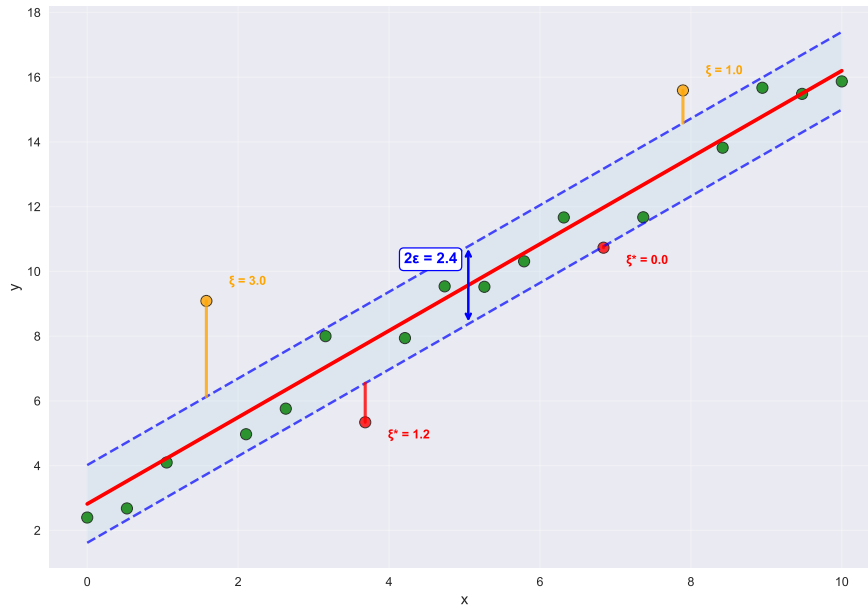


FIGURE 2.5: illustration of support vector regression (svr) with an ϵ margin for one input feature x . The blue line represents the predicted function, while the shaded area indicates the ϵ -tube around the function. Points outside this tube are penalized by the slack variables ξ_i and ξ_i^* , which allow for some deviation from the predicted function. The goal is to find a function that minimizes the error while keeping the function as flat as possible.

subject to:

$$\begin{aligned} y_i - \mathbf{w} * \mathbf{x}_i - b &\leq \epsilon + \xi_i \\ \mathbf{w} * \mathbf{x}_i + b - y_i &\leq \epsilon + \xi_i^* \\ \xi_i, \xi_i^* &\geq 0 \end{aligned}$$

Here, ξ_i and ξ_i^* are slack variables that allow for some errors (points outside the ϵ -tube), and $C > 0$ is a regularization parameter that controls the trade-off between the flatness of $f(\mathbf{x})$ and the tolerance for errors. To handle non-linear relationships, SVMs employ the "kernel trick," which maps the input data into a higher-dimensional feature space, without explicitly computing the transformation, where a linear separation might be possible. This improves the model's ability to capture complex patterns in the data by searching for a hyperplane that best captures the relationship between the input features and the target variable.

Attention model

Attention mechanisms, first introduced in the context of neural machine translation by Bahdanau, Cho, and Bengio [64], have become a fundamental component in modern deep learning, particularly for sequence to sequence modeling tasks. The core idea is to allow a model to dynamically focus on the most relevant parts of an input sequence when producing an output. Unlike the LSTM predecessor, which processes the

input sequence by sequence, transformers process the entire input sequence simultaneously. In other words they do not rely on a recurrent memory which is limited by the fixed size hidden state which is updated for each processed sequence. Although the LSTM mitigates this vanishing gradient problem by using memory cells as described in section 2.3.2. The attention mechanism solves the memory problem by maintaining a dynamic context vector (embedding) for each token of the input sequence, allowing the model to focus on different parts of the input sequence at each output step. This also allows attention models such as GPT, BERT and the transformer to process long sequences in parallel, significantly improving training efficiency and performance.

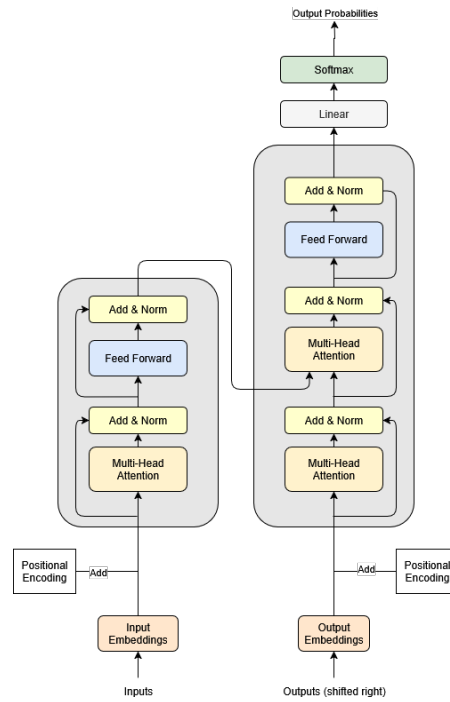


FIGURE 2.6: Transformer architecture

An illustration of the transformer architecture. The left side shows the encoder that only has a self-attention sublayer, and a feed forward layer, while the right side shows the decoder with an additional cross-attention sub-layer before the feed forward layer and finally an output linear layer with softmax activation.

The most common form of attention is the scaled dot-product attention, popularized by the transformer architecture [65]. It operates on three inputs: a set of queries (Q), keys (K), and values (V). The attention score is computed by taking the dot product of a query with all keys. These scores are then scaled and passed through a softmax function to obtain weights, and finally used to compute a weighted sum of the values. A diagram of this mechanism is shown in figure 2.7.

Mathematically, the output of the attention mechanism is computed as:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (2.23)$$

where d_k is the dimension of the keys. The scaling factor $\sqrt{d_k}$ prevents the dot products

from growing too large, which could push the softmax function into regions with extremely small gradients. In time-series forecasting, attention can be used to weigh the importance of different positions in the sequence when predicting a future value.

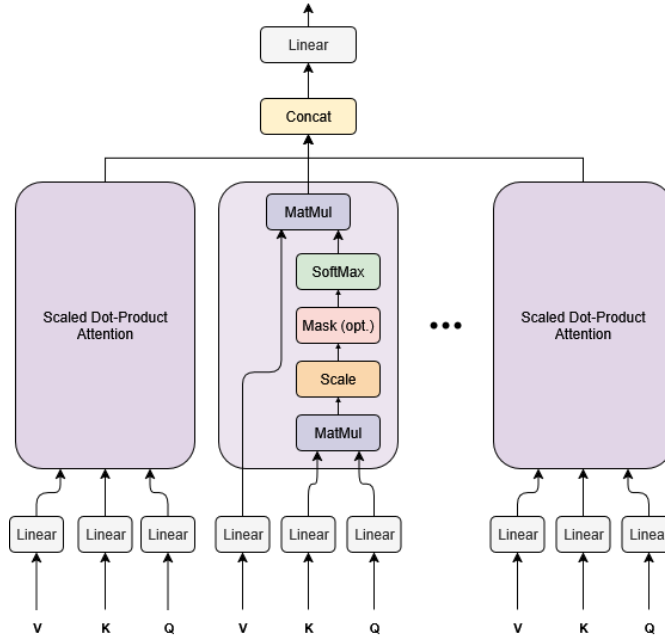


FIGURE 2.7: Multi-head attention with scaled dot product attention

An illustration of the attention mechanism, where queries, keys, and values are used to compute attention scores. Computation can be parallelized across multiple heads, allowing the model to jointly attend to information from different representation subspaces.

Due to the lack of recurrence in the transformer architecture, there needs to be some way for the model to know the order of the input sequence. In the original transformer architecture introduced by Vaswani, Shazeer, Parmar, *et al.* [65], this is achieved by adding positional encodings to the input embeddings. These encodings are calculated using sine and cosine functions of different frequencies, allowing the model to learn the relative positions of tokens in the sequence. One can think of the positional encodings as a pseudo binary encoding of the position of each token in the sequence, which is added to the input embeddings. Figure 2.8 illustrates the positional encoding used in the transformer architecture. One can see that the first position has been encoded as a vector of alternating 1s and 0s, while the 20th position has been encoded as a vector of values between -1 and 1, with a more complex pattern. The positional encoding is calculated as follows:

$$PE_{(pos, 2i)} = \sin \left(\frac{pos}{10000^{\frac{2i}{d_{model}}}} \right) \quad (2.24)$$

$$PE_{(pos, 2i+1)} = \cos \left(\frac{pos}{10000^{\frac{2i}{d_{model}}}} \right) \quad (2.25)$$

The positional encoding is added to the input embedding before being passed to the transformer model. This allows the model to learn the relative positions of tokens in the input sequence and capture the temporal relationships in the data. As shown in figure 2.6, after passing the input sequence through the encoder, which "learns" the contextual relationships between the tokens, the value and key outputs are passed to the decoder's cross-attention sub-layer. The decoder passes the embeddings of the output sequence through a self-attention sub-layer, which allows the model to "learn" the contextual relationship between the output tokens. The queries from this self-attention sub-layer are then passed to the cross-attention sub-layer, which computes the attention scores between the queries of the decoder and the keys of the encoder. This allows the transformer to learn which parts of the input sequence are relevant for predicting the next token in the output sequence. The output of the cross-attention sub-layer is then passed to a feed-forward layer, which applies a non-linear transformation to the output. Finally, the output of the feed-forward layer is passed through a linear layer with a softmax activation function, which produces the final output probabilities for the next token in the sequence.

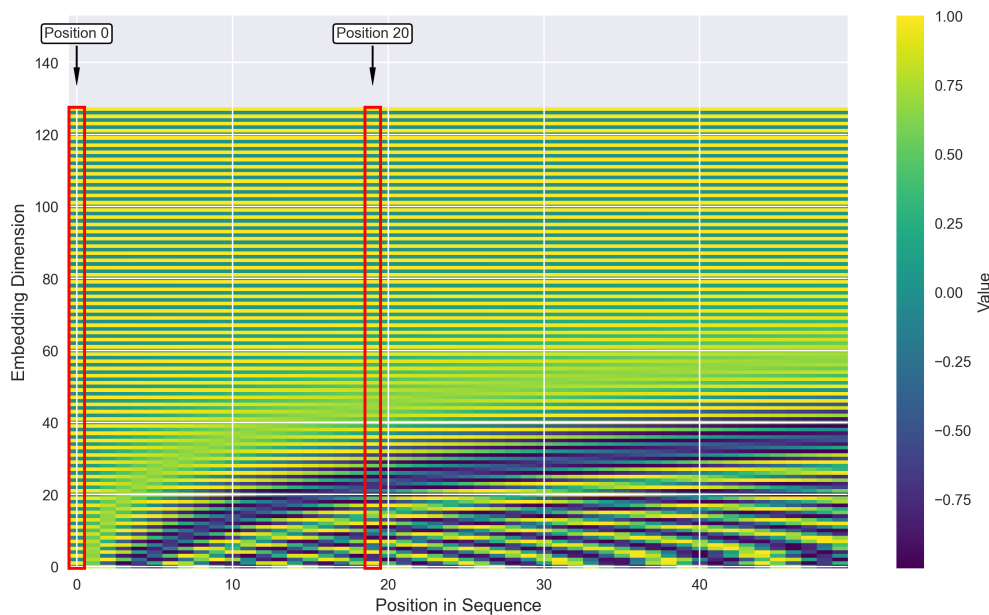


FIGURE 2.8: Positional encoding

An illustration of the positional encoding used in the transformer architecture. The positional encodings are added to the input embeddings to provide information about the position of each token in the sequence.

Chapter 3

Methodology for Forecasting Implied Volatility Surfaces

The overall methodology is shown in figure 3.1. Daily market call and put option data which includes IV is obtained from [deltaneutral](http://deltaneutral.com)¹. After preprocessing, the discrete points are interpolated to get an IVS. Next, four feature matrices are constructed from the IVS: ① IVS point feature matrix, ② IVS moneyness skew feature matrix, ③ IVS term structure feature matrix, and ④ complete the IVS matrix. These feature matrices are then encoded using eight different feature encoders, which include latent space models, parametric models, and sampling the IVS features directly. Prediction models are developed to forecast the one-day ahead IVS from lagged feature encodings. Overall, 225 different forecasting techniques are compared, using three different statistical measures. Robustness tests using put options are also carried out.

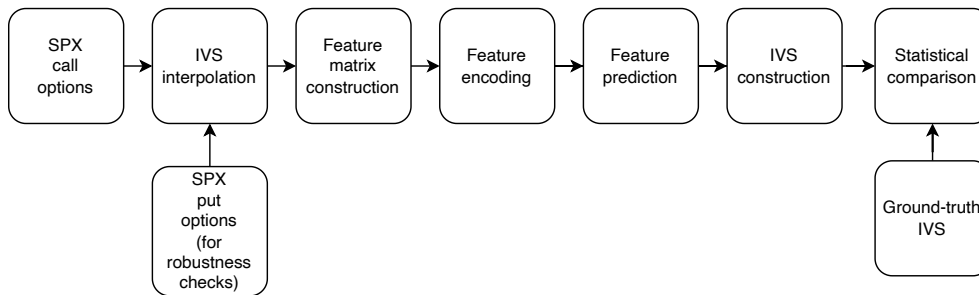


FIGURE 3.1: Overall methodology

3.1 Descriptive Statistics and IVS Construction

In this section a description of the dataset is given along with the preprocessing that is carried out on the dataset. The rationale for this preprocessing is also described.

3.1.1 Descriptive statistics

Daily S&P500 options data is obtained from [deltaneutral](http://deltaneutral.com). The original datasets include option prices, option expiration dates, strike prices, implied volatility, the option greeks, etc., calculated using the Black–Scholes formula [14]. The S&P500 sample covers

¹www.deltaneutral.com

the period from January 2002 to December 2022. The raw dataset is filtered according to the work in [16], [66]. Initially, option data that is bereft of critical option greeks and zero or lower IV is removed. Following this precise selection criterion, options from the dataset that demonstrate a lack of market activity, specifically those with a trading volume of zero are eliminated. This step ensures that the analysis is grounded in options that exhibit tangible market engagement. Next, options whose time to maturity is within 14 to 366 days, respectively, are selected. Finally, from amongst this dataset only those options whose moneyness (strike price/current underlying's price) is between 0.9 and 1.1 are selected. These are the most traded options in the market.

Two important metrics are calculated for each option contract: m and τ . The moneyness m is calculated by dividing the option's strike price (K) by the current price of the underlying (S_0), offering insight into the option's relative costliness. On the other hand, τ , representing the time to maturity normalized over a year, is calculated by dividing the option's remaining days to expiration by 365. This normalization allows us to compare the time value across options with differing expiration dates on a standardized basis. These calculated metrics, m and τ , are pivotal in furthering our understanding of the option's pricing dynamics and its term structure dynamics.

Call Options	Count	Mean	Std	Min	25%	50%	75%	Max
Moneyness	907795.0	1.017588	0.045833	0.900000	0.987654	1.019409	1.053129	1.10000
Time-to-maturity	907795.0	0.241488	0.217573	0.038356	0.090411	0.161644	0.295890	1.00274
Implied Volatility	907795.0	0.155112	0.062058	0.005300	0.111300	0.141400	0.186400	1.00070
Put Options	Count	Mean	Std	Min	25%	50%	75%	Max
Moneyness	955433.0	0.973549	0.044854	0.900000	0.936934	0.971530	1.003639	1.099998
Time-to-maturity	955433.0	0.243830	0.217131	0.038356	0.090411	0.164384	0.306849	1.002740
Implied Volatility	955433.0	0.187551	0.065672	0.016100	0.144200	0.178800	0.220200	0.862300

TABLE 3.1: Dataset statistics

The table reports the descriptive statistics for the S&P 500 daily option dataset used for training and testing the forecasting models. The data is obtained from *deltaneutral*. The period of the dataset is from January 2002 — December 2022. The option dataset is filtered for moneyness ($m \in [0.9, 1.1]$), time-to-maturity ($\tau \in [14/365, 366/365]$), implied volatility being greater than 0 and trading volume of greater than 1.

The dataset characteristics obtained after applying the filters is shown in table 3.1 the number of put options is significantly higher than the number of call options. The various forecasting models will be trained and tested using the filtered call option dataset. The put option dataset will only be used for robustness tests. Furthermore, the options from January 2002—December 2013 will be used for training the prediction models. The rest of the option dataset from January 2014—December 2022 will be used for out-of-sample testing and analysis. Notice that the training period includes the Global Financial Crisis (GFC), while the testing dataset includes the covid pandemic.

A further classification of dataset according to the moneyness (m) and time-to-maturity (τ) is shown in table 3.2. In terms of moneyness; the options are classified as In-The-Money (ITM), At-The-Money (ATM), and Out-The-Money (OTM). ITM options

Category	Call Options		Put Options	
	Definition	Count	Definition	Count
Moneyness	ITM ($m \leq 0.99$)	240,963	ITM ($m \leq 0.99$)	609,947
	ATM ($0.99 < m \leq 1.01$)	146,558	ATM ($0.99 < m \leq 1.01$)	149,323
	OTM ($m > 1.01$)	520,274	OTM ($m > 1.01$)	196,163
Time-to-maturity	Short-term ($\tau < 90/365$)	619,207	Short-term ($\tau < 90/365$)	643,623
	Medium-term ($90/365 \leq \tau < 180/365$)	171,276	Medium-term ($90/365 \leq \tau < 180/365$)	186,471
	Long-term ($\tau \geq 180/365$)	117,312	Long-term ($\tau \geq 180/365$)	125,339

TABLE 3.2: The table reports the dataset statistics in terms of moneyness and term structure of the implied volatility. The data is obtained from deltaneutral. The period of the dataset is from January 2002 — December 2022. The option dataset is filtered for moneyness ($m \in [0.9, 1.1]$), time-to-maturity ($\tau \in [14/365, 366/365]$), implied volatility being greater than 0 and trading volume of greater than 1.

	Parametric Interpolation	Nadaraya-Watson Interpolation
MSE	1.04e-4 (7.82e-5)	3.79e-6 (3.37e-6)
MAE	6.03e-3 (6.62e-3)	6.21e-4 (6.57e-4)
R ²	0.8421 (0.9050)	0.9926 (0.9954)

TABLE 3.3: Interpolation results

The table shows the interpolation results for the Dumas, Fleming, and Whaley [53] parametric interpolation and the Nadaraya-Watson interpolation on the call options dataset. The results are obtained by taking the average fit score for the entire dataset.

The results for the put options are shown in brackets.

are those, whose option strike price (K) is less than the current underlying's price (S_0). ATM options are those whose strike price is within a very short range of the current underlying's price. Finally, OTM options are those, whose strike price is higher than the current underlying's price. As expected, the number of OTM call options outnumber the ATM and ITM call options and reverse for the put options. This is because, market participants expect the price of the underlying to rise in general and hedgers buy put options to hedge against uncertainty. Similar to moneyness; the options are classified according to their term-structure (τ). The number of short-term options eclipses the medium-term and long-term options. This is expected, because most market traders trade in the short term. The moneyness and term-structure based classification will be used to understand and analyse the forecasting performance of different models in the upcoming sections.

3.1.2 Implied Volatility Surface Interpolation

On any given trading day, there is a discrete number of options being traded. Hence, an interpolation scheme is needed to generate a complete Implied Volatility Surface (IVS) from these discrete option prices and their respective implied volatilities. There are two well known interpolation techniques for interpolating the IVS given the discrete set of implied volatilities $\{\sigma_t^{(m_i, \tau_j)} : i = 1, \dots, I, j = 1, \dots, J\}$ for a given day $t \in \mathbb{Z}^{\geq 1}$. The first is the parametric approach of Dumas, Fleming, and Whaley [53] and the second is the non-parametric approach of using the Nadaraya-Watson interpolation [57].

The parametric interpolation technique of Dumas, Fleming, and Whaley [53] hypothesizes:

$$\sigma_t^{(m_i, \tau_j)} = \max(0.01, a_0 + a_1 m_i + a_2 \tau_j + a_3 m_i^2 + a_4 \tau_j^2 + a_5 m_i \tau_j + \epsilon) \quad (3.1)$$

Where ϵ is random error term for some day $t \in \mathbb{Z}^{\geq 1}$. Here, the coefficients a_1 through to a_5 are estimated using Ordinary Least Square (OLS) for any given day. Coefficient a_0 captures the flat Black-Scholes implied volatility. Coefficients a_1 and a_2 relate the implied volatility with moneyness and term-structure, respectively. The coefficients a_3 and a_4 relate the implied volatility with the curvature in moneyness and term-structure, respectively. Finally, coefficient a_5 captures any relations between moneyness and term-structure. The clip at 0.01, guarantees that the estimated implied volatility at any given point, on the IVS, is not unexpectedly negative or zero.

The Nadaraya-Watson estimator (NW) is a non-parametric interpolation technique, that performs a local weighted average of the dependent variable with respect to the independent variables. This can be formally defined as:

$$\hat{y}(x) = \frac{1}{N} \sum_{i=1}^N W_i y_i, \quad W_i = \frac{K\left(\frac{x-x_i}{b}\right)}{N^{-1} \sum_{i=1}^N K\left(\frac{x-x_i}{b}\right)} \quad (3.2)$$

Where W_i is what determines the weight for each dependent variable. The numerator of the weight function W_i adjusts the weights according to the distance of the independent variable (x_i) from the point being evaluated (x). The denominator normalizes the weights so that they sum to 1. The hyperparameter b is also added to control the degree of smoothing. The kernel function in the case of estimating the IVS is a Gaussian kernel function:

$$K(x - x_i) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-x_i)^2}{2b^2}}$$

For multivariate data as in the case of the IVS where both the moneyness (m) and time to maturity (τ) are used as independent variables, a multidimensional product kernel function can be used:

$$K(x_1, \dots, x_d) = \prod_{j=1}^d K(x_j)$$

The Nadaraya-Watson interpolation technique can then be used to interpolate the IVS for each day t for a given moneyness m and time to maturity τ as follows:

$$\sigma_t^{(m, \tau)} = \frac{\sum_{i=1}^N K(m - m_i, \tau - \tau_i) \sigma_t^{(m_i, \tau_i)}}{\sum_{i=1}^N K(m - m_i, \tau - \tau_i)} \quad (3.3)$$

where

$$K(m - m_i, \tau - \tau_i) = \frac{1}{2\pi} e^{-\frac{(m-m_i)^2}{2b_1^2}} e^{-\frac{(\tau-\tau_i)^2}{2b_2^2}} \quad (3.4)$$

In order to ensure the best fit, least squares cross validation is used to find the optimal bandwidth parameters b_1 and b_2 for the moneyness and time to maturity, respectively.

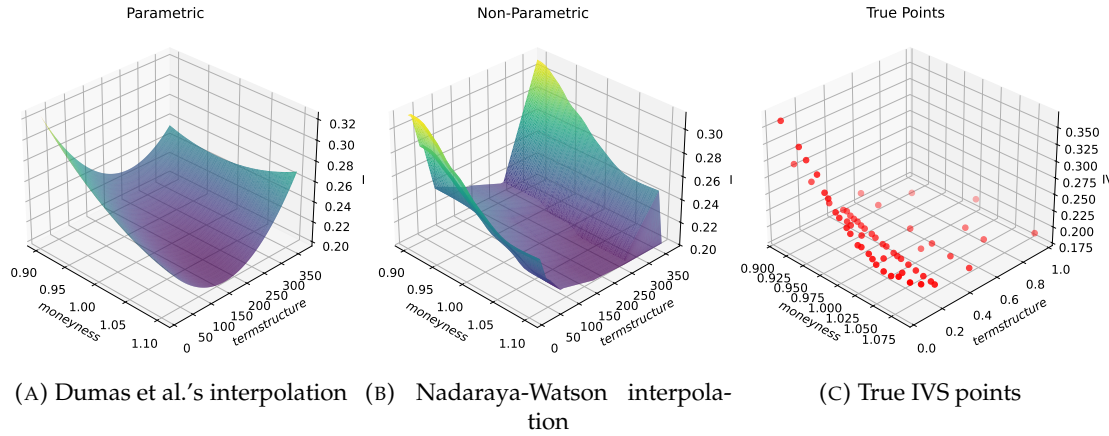


FIGURE 3.2: Interpolated IVS for 2002-07-05

Interpolated IVS surface, for call options. The X-axis and Y-axis (in (b) and (a)) indicates the moneyness ($m_i \in [0.9 : 0.0073 : 1.1]$) dynamics (moneyness skew) and the term-structure dynamics ($\tau_j \in [(14 : 1 : 366)/365]$) respectively. The Z-axis (in (a)) gives the IV — $\sigma_t^{(m_i, \tau_j)}$ for day t .

The results of the two interpolation techniques are shown in table 3.3. It is clear that the Nadaraya-Watson interpolation technique performs better than the Dumas et al.'s parametric interpolation technique in terms of mean squared error (MSE), mean absolute error (MAE), and R^2 . However, note the example in figure 3.2 shows that the Dumas et al.'s parametric interpolation technique captures the moneyness skew and term-structure dynamics better than the Nadaraya-Watson interpolation technique. This may be due to the Nadaraya-Watson interpolation technique overfitting to the data. The Dumas et al.'s interpolation technique seems to capture the shape of the IVS better and provides a much smoother surface than the Nadaraya-Watson interpolation technique. Further examples of the two interpolation techniques are shown in the appendix D. Figure D.1 shows an especially egregious cases where there are discontinuous jumps in IV for the Nadaraya-Watson estimator. These cases are due to the Nadaraya-Watson interpolation technique generating NaN values that require further processing to remove. In contrast, Zhang, Li, and Zhang [21] find for their dataset that the Dumas interpolation technique shows better performance than the non-parametric approach. However, they do acknowledge that their findings may be due to the use of the same b_1 and b_2 bandwidth parameters for all points of the IVS. Due to the Dumas technique producing much smoother surfaces, this thesis focuses on the parametric Dumas interpolation technique with supporting evidence from the non-parametric techniques as all the analysis and forecasting is also performed with the Nadaraya-Watson interpolation technique, and it gives the same overall conclusions.

The moneyness skew is easily visible along the X-axis for the Dumas et al.'s interpolation technique. Similarly, one can also see the dynamics of the term-structure

along the Y-axis. Every point on the graph is the IV ($\sigma_t^{(m_i, \tau_j)}$) for a single $m_i \in [0.9...1.1]$ and $\tau_j \in [14/365...366/365]$, respectively, for some day $t \in \mathbb{Z}^{\geq 1}$. The moneyness and term-structure variables m_i , and τ_j are incremented with a step of 0.0073 and 1 days, respectively, to form a very fine uniform 2D grid of (m_i, τ_j) . Formally, the uniform grid is defined as $\mathcal{I} = \{(m_i, \tau_j), m_i \in [0.9 : 0.0073 : 1.1], \tau_j \in [(14 : 1 : 366)/365]\}$. For sake of clarity in the rest of the paper the moneyness and term-structure vectors are referred to as I and J . Furthermore, the length of these individual vectors is denoted as $|I|$ and $|J|$. Hence, the size of the uniform grid $\mathcal{I}$ is $|\mathcal{I}| = |I| \times |J|$. This grid forms the input to Dumas' parametric formula for interpolating the IVS.

Chapter 4

Feature construction and encoding

This thesis investigates what features of the past IVSs are best suited to predict future IVS. The very first step in carrying out this work is to construct input feature matrices that capture different aspects of the IVS. Then, encoding these feature matrices for building forecasting models. These two tasks are performed in the upcoming sections.

4.1 Feature matrix construction

There are four different ways to capture and predict the IVS: ① predicting every IV point $(\sigma_t^{(m_i, \tau_j)}, (m_i, \tau_j) \in \mathcal{I})$ on the IVS_t for a single moneyness (m_i) and time-to-maturity (τ_j) for some day $t \in \mathbb{Z}^{\geq 1}$. ② Predicting the term-structure (the complete Y-axis in figure 3.2a) for a given moneyness m_i . ③ Predicting the moneyness skew (the complete X-axis) for a given time-to-maturity (τ_j) . ④ Predicting the whole IVS at once. These four different features are captured as input matrices.

4.1.1 Point features

Let a point on the IVS_t (c.f. Figure 3.2a) for moneyness m_i and time-to-maturity τ_j , for some day t , where $t \in \mathbb{Z}^{\geq 1}$ be represented as $\sigma_t^{(m_i, \tau_j)}$. Then the *single sample* input feature vector predicting $\sigma_t^{(m_i, \tau_j)}$ is given by $X_t^{(m_i, \tau_j)} = [\sigma_{t-1}^{(m_i, \tau_j)}, \dots, \sigma_{t-l}^{(m_i, \tau_j)}]$, for some lag $l \in \mathbb{Z}^{>1}$. Hence, the one sample input feature matrix (Z_t^p) predicting every point on the IVS_t is given in equation (4.1), where m_I and τ_J are the end points of the uniform grid ($\mathcal{I}$) as defined in section 3.1.2.

$$Z_t^p = \left(m_1, \tau_1, X_t^{(m_1, \tau_1)}; m_2, \tau_1, X_t^{(m_2, \tau_1)}; \dots; m_1, \tau_2, X_t^{(m_1, \tau_2)}; \dots; m_I, \tau_2, X_t^{(m_I, \tau_2)}; \dots; m_I, \tau_J, X_t^{(m_I, \tau_J)} \right) \quad (4.1)$$

Each row in the column vector Z_p (Equation (4.1)), gives the moneyness (m_i) and the time-to-maturity (τ_j) along with the (l) lagged values of volatility that will be used for prediction. Equation (4.1) forecasts each point on the IVS using lagged points for the given moneyness and time-to-maturity. This is akin to autoregression along with two external predictor variables moneyness and time-to-maturity. No information other than lagged points are used in this input feature matrix. For illustrative purposes, a single point in figure 3.2c can be thought of as $\sigma_t^{(m_i, \tau_j)}$, and $X_t^{(m_i, \tau_j)}$ is the vector that contains l previous values of $\sigma_t^{(m_i, \tau_j)}$.

4.1.2 Moneyness Skew features

Let the vector Implied Volatility Moneyness Skew (IVM) $_{t_i}^{\tau_i}$ denote the moneyness skew slice (Figure 1.4) for a given time-to-maturity (τ_i), on day $t \in \mathbb{Z}^{\geq 1}$. Then the *single sample* vector is defined as $IVM_t^{\tau_j} = [\sigma_t^{(m_1, \tau_j)}, \dots, \sigma_t^{(m_l, \tau_j)}]$. One can predict the complete IVS on day t , moneyness skew slice by moneyness skew slice using the lagged ($l \in \mathbb{Z}^{>1}$) input feature matrix (Z_t^m) in equation (4.2).

$$Z_t^m = \begin{pmatrix} \tau_1, IVM_{t-1}^{\tau_1}, \dots, IVM_{t-l}^{\tau_1} \\ \vdots \\ \tau_2, IVM_{t-1}^{\tau_2}, \dots, IVM_{t-l}^{\tau_2} \\ \vdots \\ \tau_J, IVM_{t-1}^{\tau_J}, \dots, IVM_{t-l}^{\tau_J} \end{pmatrix} \quad (4.2)$$

Equation (4.2) predicts the complete moneyness skew using the lagged skews. Each point in the lagged moneyness skew vectors ($IVM_{t-1}^{\tau_j}, \dots, IVM_{t-l}^{\tau_j}$) will affect the forecast of every point in the predicted moneyness skew vector $IVM_t^{\tau_j}$. This is akin to vector autoregression, in the moneyness, along with an external predictor time-to-maturity.

4.1.3 Term structure features

Similar to the moneyness skew based prediction, one can also predict the complete IVS term-structure slice by term-structure slice (Figure 1.5). Let the term-structure slice of the IVS be defined by the vector Implied Volatility Term Structure (IVT) $_{t_i}^{m_i}$ for some moneyness (m_i) on day $t \in \mathbb{Z}^{\geq 1}$. A single sample of this vector is given by $IVT_t^{m_i} = [\sigma_t^{(m_i, \tau_1)}, \dots, \sigma_t^{(m_i, \tau_J)}]$. The complete IVS $_t$ can be predicted using the input feature matrix (Z_t^T) in equation (4.3) using lagged term-structure slices, with lag $l \in \mathbb{Z}^{>1}$.

$$Z_t^T = \begin{pmatrix} m_1, IVT_{t-1}^{m_1}, \dots, IVT_{t-l}^{m_1} \\ \vdots \\ m_2, IVT_{t-1}^{m_2}, \dots, IVT_{t-l}^{m_2} \\ \vdots \\ m_l, IVT_{t-1}^{m_l}, \dots, IVT_{t-l}^{m_l} \end{pmatrix} \quad (4.3)$$

In equation (4.3), m_j gives some moneyness. The l^{th} lagged term-structure for this moneyness is given by the vector $IVT_{t-l}^{m_i}$. Equation (4.3) is also a vector auto-regressive process along with external predictor moneyness.

4.1.4 Surface features

Let IVS_t denote the matrix of the complete Implied Volatility Surface (IVS) for some day $t \in \mathbb{Z}^{\geq 1}$. The IVS_t matrix is defined in equation (4.4). Then, IVS_t can be predicted using

the lagged IVS feature vector (Z_t^s) in equation (4.5) with lag $l \in \mathbb{Z}^{>1}$.

$$\text{IVS}_t = \begin{pmatrix} \sigma_t^{(m_1, \tau_1)}, \dots, \sigma_t^{(m_1, \tau_J)} \\ \sigma_t^{(m_2, \tau_1)}, \dots, \sigma_t^{(m_2, \tau_J)} \\ \vdots \\ \sigma_t^{(m_I, \tau_1)}, \dots, \sigma_t^{(m_I, \tau_J)} \end{pmatrix} \quad (4.4)$$

$$Z_t^s = [\text{IVS}_{t-1}, \dots, \text{IVS}_{t-l}] \quad (4.5)$$

From equation (4.5) one can observe that all points on the lagged IVS effect the forecast of the future IVS.

Depending upon the IVS grid size $|\mathcal{I}|$, the IVS can be a very large matrix. Moreover, complete lagged IVS (or its components) are used to make forecasts. Learning relevant information from large input feature matrices requires careful feature encoding techniques. Different feature encoding techniques and regularised prediction models will be explored in the next sections, which help learn relevant information and prevent over-fitting.

4.2 Feature encoding models

	SAM	PCA	CCA	NS	ADNS	SSVI	HAR
Point (Z_t^p)	✓	✓	✓				✓
Moneyness Skew (MS) (Z_t^m)	✓	✓	✓	✓			✓
Term Structure (TS) (Z_t^T)	✓	✓	✓	✓			✓
Surface (Z_t^s)	✓	✓	✓		✓	✓	✓

TABLE 4.1: Feature encoding models

Table listing the input feature matrices (first column) and the encoding applied to these feature matrices. The different encoding techniques are: SAMpling feature matrix (SAM), Principal Component Analysis (PCA), Canonical Covariance Analysis (CCA), Nelson Siegel (NS), Augmented Dumas Nelson Siegel (ADNS), Surface Stochastic Volatility Inspired (SSVI), Heterogenous Autoregressive (HAR).

The various feature matrices (Z_t^p through to Z_t^s , section 4.1) extracted from the lagged IVS is encoded into a (possibly lower dimensional) feature matrix F_t^p through to F_t^s , such that $F_t^k = f(Z_t^k), \forall k \in \{p, m, T, s\}$. The following defines seven different feature encoders (f). Not all features encoders are applicable to all input feature matrices. The different encoders applicable to different input feature matrices point (Z_t^p), moneyness skew (Z_t^s), term-structure (Z_t^T), and surface (Z_t^s) are shown in table 4.1. Each of these encoders are described in detail in the next sections.

4.2.1 Sampling input feature matrix (SAM) encoder

The simplest method to encode the input feature matrices is sampling the complete feature matrix (SAM). This entails defining the feature encoding function f as $f \stackrel{\text{def}}{=} \text{id}$, where id is the identity function. Hence, after applying the identity function the four feature matrices are created $F_t^p = Z_t^p$, $F_t^m = Z_t^m$, $F_t^T = Z_t^T$, and $F_t^s = Z_t^s$ corresponding to the extracted input feature matrices from section 4.1. The work in [16], [21] has shown that using the lagged values as is, results in one of the best forecast outcomes. Hence, our very first feature encoder is the identity function. Of course, using the identity function would result in a very large set of input features, especially in case of feature matrices Z_t^m , Z_t^T and Z_t^s . The use of regularisation techniques are employed for learning in order to overcome potential over-fitting.

4.2.2 Principal Component Analysis (PCA) latent space encoder

There is a wealth of literature showing the effectiveness of applying Principal Component Analysis (PCA) for dimensionality reduction of the Implied Volatility Surface (IVS) [17], [51], [67]. The work in [17] shows that applying PCA for dimensionality reduction of the IVS for the purpose of forecasting can prove effective. In order to calculate the principal components as described in section 2.3.1, singular value decomposition (SVD) is used. When performing SVD on the input feature matrix X , this results in the decomposition:

$$X = U\Sigma V^T$$

Where the columns of the matrix V are called the right singular vectors and correspond to the principal components of the data and the columns of $U\Sigma$ correspond to the principal components. The example of encoding Z_t^p feature vector via PCA will be used, the technique is similarly applicable to other feature vectors.

Let $(m_i, \tau_j) \in \mathcal{I}$ be some point on the uniform grid. Then the input lagged value vector $X_t^{(m_i, \tau_j)}$ is as defined in section 4.1.1. For supervised learning, the following input feature matrix can be constructed $\mathcal{X} = [X_t^{(m_i, \tau_j)}, X_{t+1}^{(m_i, \tau_j)}, \dots, X_{t+M}^{(m_i, \tau_j)}]^T$, by rolling a window, of size M , on the daily interpolated IVS, for each point (m_i, τ_j) . Let $X = \mathcal{X}^T \mathcal{X}$ be the variance matrix. A Singular Value Decomposition (SVD) of $X = U\Sigma V^T$ is performed. Next, the diagonal elements of Σ are put in non-ascending order. The K largest components are chosen from Σ , and the input features are projected onto the K chosen left eigen vectors of V ($F_t^p = \text{PCA}(Z_t^p)$). This projection defines f and gives the feature vectors F_t^p .

K is chosen to explain 95% of the variance. Usually, a small value of $K = 2$ to 5 suffices. This leads to substantial reduction in the input feature vector space from thousands. Reconstructing the IVS during forecasting is a simple matter of projecting the predictions from the latent PCA feature space back to the original Z_t^p feature space.

4.2.3 Canonical Covariance Analysis (CCA) latent space encoder

Canonical Covariance Analysis (CCA) [38] is a relatively recent feature reduction technique. CCA projects the input features and output samples, in supervised learning, such that the covariance between the two is maximised as stated in section 2.3.1. This results in only those input feature dimensions being preserved that best explain the output. So far as the literature indicates, no one has applied CCA based encoding of feature matrices to the IVS prediction problem, and this thesis is the first to do so. The following description will use the example of encoding Z_t^p feature vector via CCA, the technique is similarly applicable to other feature vectors.

Let $(m_i, \tau_j) \in \mathcal{I}$ be some point on the uniform grid. Then the input lagged value vector $X_t^{(m_i, \tau_j)}$ is as defined in section 4.1.1. For supervised learning, the following input feature matrix is constructed $\mathcal{X} = [X_t^{(m_i, \tau_j)}, X_{t+1}^{(m_i, \tau_j)}, \dots; X_{t+M}^{(m_i, \tau_j)}]^T$, and the corresponding observed IV vector $\mathcal{Y} = [\sigma_{t+1}^{(m_i, \tau_j)}, \sigma_{t+2}^{(m_i, \tau_j)}, \dots, \sigma_{t+M+1}^{(m_i, \tau_j)}]$, by rolling a window, of size M , on the daily interpolated IVS, for each point (m_i, τ_j) . Let $Y = \mathcal{X}'\mathcal{Y}$ be the cross-covariance matrix. Then, a SVD of $Y = U\Sigma V$ is performed. Next, the diagonal elements of Σ are put in non-ascending order. The K largest components are chosen from Σ , and the input features are projected onto the K chosen left singular vectors of U and the outputs are projected onto K chosen right singular vectors of V ($F_t^p = \text{CCA}(Z_t^p)$). This projection defines f and gives the feature vectors F_t^p along with the external predictors m_i and τ_j .

K is chosen using Wilks Lambda test [68]. Usually, a small value of $K = 2$ to 5 suffices. This leads to substantial reduction in the input feature vector space from thousands. Reconstructing the IVS during forecasting is a simple matter of projecting the predictions from the latent CCA feature space back to the original Z_t^p feature space.

4.2.4 Nelson Siegel parametric encoder

Guo, Han, and Zhao [22] and Guo, Han, and Lin [24] have shown that the yield curve forecasting model of Nelson and Siegel [26] can be effectively applied to the prediction of the IVS term-structure. In a similar vein Kumar, Agarwalla, Varma, *et al.* [23] show that the Nelson and Siegel model is also effective at forecasting moneyness skew slices. Hence, this thesis uses the Nelson-Siegel model for encoding both the term-structure slices and the moneyness skew slices for future forecasting.

The following description will use the term-structure feature matrix Z_t^T for explaining the encoding principles. However, the principles are equally applicable to the moneyness skew slices. Consider the term-structure slice ($IVT_t^{m_i}$) for a given moneyness m_i and a given day t as defined in section 4.1.3. Each $IVT_t^{m_i}$ for each day t , is modelled using the Nelson and Siegel model in equation (4.6), where τ_j are the time-to-maturity points for the given term-structure, λ is a constant to be determined, and ϵ is the random error term.

$$\sigma_t^{(m_i, \tau_j)} = \beta_1^{m_i} + \beta_2^{m_i} \left(\frac{1 - e^{-\lambda \tau_j}}{\lambda \tau_j} \right) + \beta_3^{m_i} \left(\frac{1 - e^{-\lambda \tau_j}}{\lambda \tau_j} - e^{\lambda \tau_j} \right) + \epsilon \quad (4.6)$$

Each term-structure slice is captured using the parameter vector $\mathcal{B}_t^{m_i} = [\beta 1_t^{m_i}, \beta 2_t^{m_i}, \beta 3_t^{m_i}]$. Equation (4.6) is the feature encoder f that translates the term-structure slice feature input matrix Z_t^T to F_t^T , where F_t^T is a matrix (Equation 4.7) of the $\mathcal{B}_t^{m_i}$ parameters for each term-structure slice for some day t . This feature encoding substantially reduces the size of the term-structure feature matrix.

$$F_t^T = \begin{pmatrix} m_1, \mathcal{B}_{t-1}^{m_1}, \dots, \mathcal{B}_{t-l}^{m_1} \\ \vdots \\ m_2, \mathcal{B}_{t-1}^{m_2}, \dots, \mathcal{B}_{t-l}^{m_2} \\ \vdots \\ m_I, \mathcal{B}_{t-1}^{m_I}, \dots, \mathcal{B}_{t-l}^{m_I} \end{pmatrix} \quad (4.7)$$

In order to fit the term-structure slice non-linear optimisation is used to obtain λ_t and $\mathcal{B}_t^{m_i}$ parameter set for each slice on each day by reducing the sum of squared error. λ is then fixed to be the median from amongst all the daily values of λ_t obtained after non-linear optimisation as done in [28]. Once the λ value is fixed, then the daily parameter set ($\mathcal{B}_t^{m_i}$) is computed for each term-structure slice, using OLS. Once the feature set F_t^T is obtained, a prediction model is trained, on a rolling window, of size M samples, to predict each term-structure parameter set separately. The whole term-structure slice can be recreated from the predicted parameter set using equation (4.6). The complete IVS is then predicted by adjoining the individually predicted term-structure slices into a single matrix.

4.2.5 Augmented Dumas Nelson Siegel (ADNS) parametric encoder

The Nelson Siegel model can be applied to encode the term-structure and moneyness skew slices individually (Section 4.2.4). The predicted IVS is reconstructed using forecasts for the encoded features individually. Another approach to incorporating the Nelson Siegel model is augmenting the model proposed in Dumas, Fleming, and Whaley [53] with the Nelson and Siegel model [26] for capturing the whole IVS input feature matrix (Equation (4.5)) at once. This approach is presented in Chalamandaris and Tsekrekos [19] — Augmented Dumas Nelson Siegel (ADNS) feature encoder (f), which translates the input feature set Z_t^s to F_t^s . Consider IVS_t , for some day t , as defined in section 4.1.4, equation (4.4). Then, this whole surface can be encoded using the feature encoder f as defined in equation (4.8), where $(m_i, \tau_j) \in \mathcal{I}$, $\mathbb{1}_x$ is the indicator function, and ϵ is random error term.

$$\begin{aligned} \sigma_t^{(m_i, \tau_j)} &= \alpha_t^0 + \alpha_t^1 \mathbb{1}_{\{m_i \geq 1\}} m_i^2 + \alpha_t^2 \mathbb{1}_{\{m_i < 1\}} m_i^2 + \alpha_t^3 \left(\frac{1 - e^{-\lambda \tau_j}}{\lambda \tau_j} \right) \\ &+ \alpha_t^4 \left(\frac{1 - e^{-\lambda \tau_j}}{\lambda \tau_j} - e^{-\lambda \tau_j} \right) + \alpha_t^5 \mathbb{1}_{\{m_i \geq 1\}} m_i \tau_j + \alpha_t^6 \mathbb{1}_{\{m_i < 1\}} m_i \tau_j + \epsilon \end{aligned} \quad (4.8)$$

Equation (4.8) captures the non-linear dynamics of the term-structure (Figure 1.5) using the Nelson and Siegel 1987 model. In a similar vein, the asymmetry of moneyness skew, around ATM (Figure 1.4), is captured explicitly in the model by distinguishing the right ($\mathbb{1}_{\{m_i \geq 1\}}$) and left ($\mathbb{1}_{\{m_i < 1\}}$) sides of the moneyness skew around the ATM. A non-linear optimisation is first carried out on daily IVS_t data to compute the constant λ_t ; reducing the sum of squared errors. The median value from amongst the computed λ_t is chosen as the constant (λ). Next, given a constant λ ; the daily IVS_t is encoded into a seven feature vector: $\mathcal{A} = [\alpha_t^0, \alpha_t^1, \alpha_t^2, \alpha_t^3, \alpha_t^4, \alpha_t^5, \alpha_t^6]$, parameters of equation (4.8), using OLS.

$$F_t^s = [\mathcal{A}_{t-1}, \dots, \mathcal{A}_{t-l}]^T \quad (4.9)$$

The resultant latent feature encoding (F_t^s) is given by the column vector in equation (4.9). The very large surface feature set, with thousands of IV points on the surface, is reduced into *only* seven parameter vectors. Once the feature set F_t^s is obtained, supervised learning is used to train models on a rolling window of size M samples to predict one day ahead parameter set. The IVS can then be reconstructed from the predicted parameter set using equation (4.8).

4.2.6 Surface Stochastic Volatility Inspired (SSVI) parametric encoder

Gatheral and Jacquier [36] present a technique to capture the IVS, called Surface Stochastic Volatility Inspired (SSVI) implied volatility model. Unlike the ADNS model, the SSVI model is guaranteed to produce a static arbitrage free surface, which makes it more attractive for pricing exotic options. Although designed for pricing exotics; this thesis will use this model for feature reduction and prediction of future IVS. This thesis uses the SSVI model for feature reduction and IVS forecasting, which, given current literature, appears to be a novel application of this model.

The feature encoding technique for SSVI model is similar to the ADNS feature encoder. The SSVI feature encoder (f) converting the components of input feature matrix (IVS_t) in equation (4.4), on some day t , is shown in equation (4.10), where $(m_i, \tau_j) \in \mathcal{I}$.

$$w_t^{(\Lambda_i, \tau_j)} = \frac{\theta_{\tau_j}}{2} \left(1 + \frac{\rho \mu \Lambda_i}{\theta_{\tau_j}^{1/2}} + \sqrt{\left(\frac{\mu \Lambda_i}{\theta_{\tau_j}^{1/2}} + \rho \right)^2 + (1 + \rho^2)} \right) \quad (4.10)$$

$$\Lambda_i = \ln(m_i)$$

$$\theta_{\tau_j} = (\sigma_t^{(0, \tau_j)})^2 \tau_j \quad (4.11)$$

$$w_t^{(\Lambda_i, \tau_j)} = (\sigma_t^{(\Lambda_i, \tau_j)})^2 \tau_j \quad (4.12)$$

Each IV point on the IVS_t is first translated into so-called total variance (Equation (4.12)). The total variance matrix is then captured using the SSVI encoder in equation (4.10).

Note that moneyness (m_i) is translated into its logarithm. The resultant encoding results in a three parameter feature vector $\mathcal{S} = [\rho, \mu, \theta_{\tau_j}]$, where parameters ρ and μ are constant, and θ_{τ_j} is the ATM term-structure. These parameters are first estimated using non-linear optimisation for daily observed (historical) IVS. The objective of the optimisation is to reduce the sum of squared errors between the estimated $\widehat{IVS}_t$, estimated by equation (4.10), and the observed IVS_t . Once the daily parameters have been estimated. A supervised learner is trained using a rolling window, of sample size M , to learn the one-day ahead parameters using the feature matrix $F_t^s = [\mathcal{S}_{t-1}, \dots, \mathcal{S}_{t-l}]^T$. The whole IVS can then be reconstructed using the forecasted parameters and equations (4.10)—(4.12).

4.2.7 Heterogenous Autoregressive (HAR) latent space encoder

[69] have shown effective forecasting of historical volatility using the Heterogenous Autoregressive (HAR) feature encoding with OLS. To the best of our knowledge, HAR encoders have not been applied to the feature reduction of the IVS (or its components). This thesis proposes the use of the HAR model for encoding the point, Moneyness Skew, Term Structure, and surface input feature matrices using the HAR encoder. This is due to the work in [16], which has shown that future IV (point forecast) is past dependent, and the HAR encoder captures the short and long temporal dependencies with a reduced feature vector.

The following description will use the example of encoding Z_t^m feature vector via the HAR encoding model, the technique is similarly applicable to other feature vectors. Consider a Implied Volatility Moneyness Skew (IVM) slice ($IVM_t^{\tau_j}$) at day $t \in \mathbb{Z}^{\geq 1}$ as defined in section 4.1.2. For each MS slice for maturity τ_j in the uniform grid $\mathcal{I}$, we calculate the mean of the IVM slice over the past 1, 5, and 20 days. This can be defined as shown in equation (4.13):

$$\hat{\sigma}_N^{(m_i, \tau_j)} = \frac{1}{N} \sum_{l=1}^N \sigma_{t-l}^{(m_i, \tau_j)}, \forall m_i \in [1, \dots, |I|], \forall \tau_j \in [1, \dots, |J|] \quad (4.13)$$

Where $\hat{\sigma}_N^{(m_i, \tau_j)}$ is the mean value of point m_i in the slice $IVM_t^{\tau_j}$ for lag N . Each lagged moneyness skew feature, for some τ_j , is then given by equation (4.14).

$$IVM_N^{\tau_j} = [\hat{\sigma}_N^{(m_1, \tau_j)}, \hat{\sigma}_N^{(m_2, \tau_j)}, \dots, \hat{\sigma}_N^{(m_l, \tau_j)}] \quad (4.14)$$

For each lag $N \in \{1, 5, 20\}$. We can then define the transformed input feature vector as shown in equation (4.15).

$$F_t^M = \begin{pmatrix} \tau_1, IVM_1^{\tau_1}, IVM_5^{\tau_1}, IVM_{20}^{\tau_1} \\ \vdots \\ \tau_J, IVM_1^{\tau_J}, IVM_5^{\tau_J}, IVM_{20}^{\tau_J} \end{pmatrix} \quad (4.15)$$

For supervised learning, a rolling window of size M is used, and the models are trained with each row of F_t^M and then predicts the corresponding one-day-ahead $IVM_{t+1}^{\tau_j}$ moneyness skew slice. The whole IVS can be recreated by putting all the predicted moneyness skew slices into a single matrix.

4.3 Prediction models

Prediction Model	SAM	PCA	CCA	NS	ADNS	SSVI	HAR
Ridge	✓	✓	✓	✓	✓	✓	✓
Lasso	✓	✓	✓	✓	✓	✓	✓
Elastic-Net	✓	✓	✓	✓	✓	✓	✓
Random Forest	✓	✓	✓	✓	✓	✓	✓
LSTM	✓	×	×	✓	✓	†	✓
Transformer	✓	×	×	✓	✓	✓	✓
SVR	✓	✓	✓	✓	✓	✓	✓

TABLE 4.2: Prediction model compatibility with feature encoding techniques
Legend: ✓ = compatible, × = not compatible, † = limited compatibility due to computational constraints.

Feature Type	Encoding	Lag 5	Lag 10	Lag 20
Point	SAM	✓	✓	✓
	PCA	✓	✓	✓
	CCA	✓	✓	✓
	NS	×	×	×
	ADNS	×	×	×
	SSVI	×	×	×
	HAR	×	×	✓
Moneyness Skew	SAM	✓	✓	✓
	PCA	✓	✓	✓
	CCA	✓	✓	✓
	NS	✓	✓	✓
	ADNS	×	×	×
	SSVI	×	×	×
	HAR	×	×	✓
Term Structure	SAM	✓	✓	✓
	PCA	✓	✓	✓
	CCA	✓	✓	✓
	NS	✓	✓	✓
	ADNS	×	×	×
	SSVI	×	×	×
	HAR	×	×	✓
Surface	SAM	✓	✓	†
	PCA	✓	✓	†
	CCA	✓	✓	†
	NS	×	×	×
	ADNS	✓	✓	†
	SSVI	✓	✓	†
	HAR	×	×	†

TABLE 4.3: Feature encoding and lag length compatibility matrix

Table 4.2 shows the compatibility between different prediction models and feature encoding techniques. Linear models (ridge, lasso, elastic-net) are compatible with all

encoding methods, while non-linear models (LSTM and transformer) are restricted to SAM encoding due to computational constraints as mentioned in section 4.3.2.

Table 4.3 presents the compatibility matrix between feature representations, encoding techniques, and lag lengths. Parametric encoders (NS, ADNS, SSVI) have specific applicability constraints based on the feature type, while latent space encoders show varying performance with longer lag lengths due to dimensional complexity.

4.3.1 Linear models

For linear prediction models, regularised linear regression models are used to predict the one-day ahead IVS given feature matrices (F_t^p through to F_t^s). To train these models, we first create M samples of the feature matrices and the one-day ahead feature matrix using a rolling window on historical data. Let $\mathcal{A} = [F_t^k; \dots; F_{t+M}^k]$ denote the M samples of feature matrices and $\mathcal{B} = [F_{t+1}^k; \dots; F_{t+M+1}^k]$ denote the M samples of one day ahead feature matrix, where $k \in \{p, m, T, s\}$. Then, we perform three different types of linear regressions:

- **Lasso regression:** Lasso regression [59] extends OLS by adding a penalty term on the absolute value of the coefficients of regressions denoted $\mathbf{w}$. Using the l_1 norm results in selecting only the relevant features that most influence the output. The lasso regression problem can be posed as an optimisation problem: $\min_{\mathbf{w}} \left(\frac{1}{2M} \|\mathcal{B} - \mathcal{A}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1 \right)$. The regularisation strength α is computed using hyperparameter optimisation using grid search.
- **Ridge regression:** Ridge regression [70] extends OLS by adding a penalty term on the l_2 norm of the coefficients. Unlike lasso, ridge regression does not eliminate the input features completely, which makes it ideal for cases where only over-fitting and multi-collinearity needs to be overcome. Ridge regression is posed as the optimisation problem: $\min_{\mathbf{w}} \left(\frac{1}{2M} \|\mathcal{B} - \mathcal{A}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_2^2 \right)$. Here again α is found using grid search.
- **Elastic-Net regression:** Elastic-Net regression [60] combines lasso and ridge regression together to benefit from both. This method overcomes over-fitting and multiple-correlated features; as it simultaneously addresses feature selection and multi-collinearity. Elastic-net training can be posed as the optimisation problem: $\min_{\mathbf{w}} \left(\frac{1}{2M} \|\mathcal{B} - \mathcal{A}\mathbf{w}\|_2^2 + \alpha \kappa \|\mathbf{w}\|_1 + \frac{\alpha(1-\kappa)}{2} \|\mathbf{w}\|_2^2 \right)$. The hyperparameters α and κ are found using grid search as before.

Once the regression coefficients ($\mathbf{w}$) are computed. We can predict the one-day ahead feature matrix (F_{r+1}^k) given a single sample input feature matrix F_r^k , where $k \in \{p, m, T, s\}$ and $r > t + M + 1$ some day out-of-samples. Once we have the one-day ahead predicted feature matrix, we can reconstruct the complete IVS using the feature encoders (f) described in section 4.2.

4.3.2 Non-linear models

Non-linear models will also be explored to predict the one-day ahead IVS. Similarly to the linear models, M samples of a feature matrices $\mathcal{A}$ and the one-day ahead feature matrices $\mathcal{B}$ are constructed by a rolling window on the historical data. However, due to hardware constraints some models such as the support vector regression and LSTM models can not be trained for certain lag lengths and feature encoding methods, as indicated by the tables 4.2 and 4.3. Additionally due to the nature of the LSTM and transformer architectures, and the way in which features are encoded for the PCA and CCA encoders, these two predictor models are not compatible with the PCA and CCA encoders. For example, referring to section 4.1.1, PCA and CCA are computed over the flattened lagged single sample input vector $X_t^{(m_i, \tau_j)}$. This results in the lagged sequences being flattened in feature space where the LSTM, and attention model expect sequential data. The use of non-linear models is included to extend upon the work of Wen, Zhai, Wang, *et al.* [16] where they show linear models significantly outperform non-linear models for the point (F_t^p) feature predictions. A major departure from the work in [16] is that all non-linear models do not make use of the m and τ as features. Only lagged features of the IVS are used for training and forecasting. We will show if non-linear models are more capable of capturing information from all feature matrices (F_t^p through to F_t^s) and whether that leads to better overall forecast performance.

- **Random Forest regression**

The random forest regressor from scikit-learn [71] is used to predict the one-day ahead feature matrix. The random forest regressor is an ensemble tree based algorithm, where multiple decision trees are "grown". Each tree is constructed using a subset of the training data and a subset of the features. At each node split in the tree, a random subset of features and values are considered to minimise the mean squared error of the split. There are many hyperparameters that can be tuned to improve the performance of the random forest regressor. ① The number of trees/estimators to grow are considered. Testing showed that increasing the number of estimators from the default scikit-learn value of 100 to 150 improved R^2 and RMSE performance. ② The maximum depth of each tree are set to the default value for scikit-learn of none, which allows the trees to grow until all leaves are pure i.e. all samples in the leaf node belong to the same class or the number of samples is less than the minimum samples required to split an internal node. Only for models using the HAR encoding model, where there were computational constraints for the random forest regressor, did the maximum depth of the trees get set to 10. All other hyperparameters are set to the default values in scikit-learn. This includes the number of features to consider when looking for the best split, which is set to the square root of the number of features.

- **Long Short Term Memory (LSTM)**

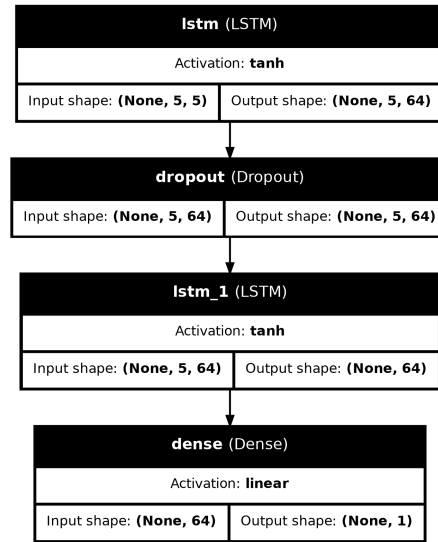


FIGURE 4.1: LSTM architecture

Input shapes shown are for the point feature matrix (F_t^p). Each LSTM layer has a hidden unit size of 64. The final sequence of the second LSTM layer is passed as input to a fully connected dense layer to produce the output feature matrix (F_{t+1}^p).

The LSTM is a type of recurrent neural network that was first introduced by Hochreiter and Schmidhuber [61]. There has been work showing that LSTM models can be used to predict the IVS. Most notably, Wen, Zhai, Wang, *et al.* [16] first use a LSTM model to extract temporal features from the lagged IVS points and then use the extracted features along with the moneyness m and time-to-maturity τ to predict the one-day ahead IVS point. The LSTM model presented in this work instead uses the lagged IVS points as the input feature matrix (F_t^p through to F_t^s) and does not use the moneyness m and time-to-maturity τ as features. The architecture presented is shown in figure 4.1

Unlike the traditional recurrent neural network, the LSTM should be able to more effectively capture long-term dependencies in the data. The lags of the input feature matrix are passed in as a sequence to the LSTM model this allows the LSTM to maintain the temporal structure of the data instead of flattening the input feature matrix into a single vector as done in the linear models.

The LSTM has many hyperparameters to consider. After testing various combinations, the following hyperparameters were selected:

- Number of LSTM layers: 2
- Number of hidden units: 64
- Dropout rate: 0.2
- Learning rate: 0.001
- Batch size: 64
- Number of epochs: 100

- Loss function: Mean Squared Error (MSE)
- Optimizer: Adam

The final output layer of the LSTM is a fully connected layer that maps the hidden state to the output feature matrix. To reduce overfitting, a dropout layer is added after the LSTM layers and an early stopping mechanism is implemented to stop training when validation loss stops improving. We also use a learning rate scheduler to reduce the learning rate on validation loss plateau.

• Attention model

The attention model presented in this work is based on the original transformer architecture introduced by Vaswani, Shazeer, Parmar, *et al.* [65]. The model is an encoder only version of the original transformer architecture and is inspired by the architecture presented in Zerveas, Jayaraman, Patel, *et al.* [72] where an encoder only transformer is used for forecasting and regression tasks on multivariate time series data. Figure 4.2 shows the architecture of the attention model used.

As Zerveas, Jayaraman, Patel, *et al.* point out, only using the encoder means getting rid of half the transformer parameters and reduces the computational complexity and potential overfitting. The encoder only architecture is versatile and can be used for both regression and classification tasks making it suitable for the task at hand.

There are several hyperparameters that can be tuned to improve the performance of the attention model. Computational constraints are a major factor in the choice of hyperparameters. For a static set of hyperparameters, the results show that as the number of input features increased with the different feature vectors (F_t^p through to F_t^s), the performance of the attention model decreased. This is likely due to the high dimensional feature space being compressed into the lower dimensional latent space, which results in the model not being able to learn the relevant features from the input feature matrix. Thus, the hyperparameters were tuned for the surface feature matrix (F_t^s).

The following hyperparameters were selected after testing various combinations:

- Number of attention layers N : 4
- Number of attention heads h : 4
- Hidden size: 64
- Feed forward hidden size: 64
- Dropout rate: 0.1
- Learning rate: 0.0005
- Batch size: 64
- Number of epochs: 100

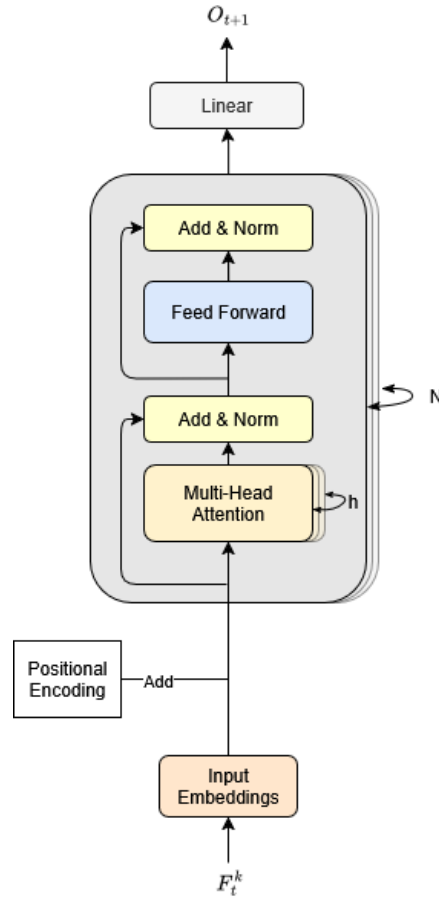


FIGURE 4.2: Attention model architecture

Then input feature matrix F_t^k is for $k \in \{p, m, T, s\}$. The input feature matrix is passed through a multi-head attention layer with h heads (see figure 2.7) and there are N stacked identical attention layers. The output of the attention layers is then passed to a final linear layer to linearly transform the output into the final output feature matrix for the next day F_{t+1}^k . O_t is the output where $O \in \{\sigma_{t+1}^{(m_i, \tau_j)}, IVM_{t+1}^{\tau_i}, IVT_{t+1}^{m_i}, IVS_{t+1}\}$.

- Loss function: Mean Squared Error (MSE)
- Optimizer: Adam

Additionally, a learning rate scheduler is used to reduce the learning rate on validation loss plateau and an early stopping callback is used to stop training when validation loss stops improving. Each of these along with a drop out of 0.1 helps to reduce overfitting and improve the generalisation of the model. For the surface feature matrix, the number of attention heads are increased to 8, the hidden size is increased to 256 and the feed forward hidden size is increased to 256 in an effort to capture the complex relationships in the surface feature matrix. All other aspects of the attention model remain the same as described in section 2.3.2.

- **Support Vector Regression (SVR)**

The last prediction model is the support vector regression (SVR) 2.3.2. The model is implemented using the scikit-learn implementation of the SVR model [71]. The SVR model uses the epsilon-insensitive loss function (2.22), which states that the model should only penalize predictions that fall outside the margin defined by the epsilon parameter ϵ . It also uses an L2 regularisation term that is controlled by the hyperparameter C . The hyperparameter ϵ is set to the default value of 0 which showed the best performance in experimentation. A hyperparameter search is performed to find the best value for the regularisation parameter C . Decreasing the value of C logarithmically down to $1e^{-5}$ showed the best performance.

Once each of the models are trained, the one-day ahead feature matrix (F_{r+1}^k) can be predicted given a single sample input feature matrix F_r^k , where $k \in \{p, m, T, s\}$ and $r > t + M + 1$ some day out-of-samples. Once the one-day ahead predicted feature matrix is obtained, we can reconstruct the complete IVS using the feature encoders (f) described in section 4.2.

Chapter 5

Empirical study for forecasting the IVS

This chapter gives the details of training and out-of-sample testing of the various encoding and feature extraction models described previously. The experimental setup is first described followed by the quantitative measures used to compare forecast accuracy of different feature extraction and encoding techniques. Finally, presenting the quantitative results.

5.1 Experimental setup

The complete dataset from January 2002 — December 2022 is divided into two sets: the training dataset is from 2002–2013. The testing dataset is from 2014–2022. A rolling window based strategy (c.f. Section 4.3) is used to train and test the prediction models. Once the prediction models are trained, the out-of-sample tests are carried out. The task is to predict one day ahead IVS on a rolling basis for the whole 9 years of out-of-sample test dataset. The prediction models are not re-trained when performing out-of-sample forecasts.

Three different lag lengths (l , c.f. section 4.1) 5, 10, and 20 are tested for forecasting. From table 4.1, it can be seen that there are 23 different combinations of feature extraction, encoding, and training techniques. These 23 different techniques work with three different lag lengths, and 3 different linear prediction models (c.f. Section 4.3). The two non-linear models only use the SAM method of feature encoding. Hence, overall there are 225 different forecasts that need to be compared. The input feature matrix construction and encoding is carried out in python using scikit learn [71] and optimizers available in scipy [73].

5.2 Quantitative measures

Three different quantitative measures and associated statistical tests are used to compare the 225 different forecasts. The following quantitative measures and associated statistical significance tests are described herein.

Let $\hat{Y}_t^{\mathcal{M}}$ denote the forecasted out-of-sample IVS for some day t , then $\hat{Y}^{\mathcal{M}}$ is the matrix of IVS forecasts for *all* N days in the nine year (2014–2022) out-of-sample test period.

Here tuple $\mathcal{M} = (\mathcal{F}, \mathcal{E}, \mathcal{P}, \mathcal{L})$ denotes a specific forecast model; and $\mathcal{F} \in \{p, m, T, s\}$ denotes the feature extractions from the lagged IVSs.

Parameter $\mathcal{E} \in \{\text{SAM}, \text{PCA}, \text{CCA}, \text{NS}, \text{ADNS}, \text{SSVI}, \text{HAR}\}$ denotes the different features encoders.

Parameter $\mathcal{P} \in \{\text{Lasso}, \text{Ridge}, \text{ElasticNet}, \text{RandomForest}, \text{LSTM}, \text{Attention}, \text{SVR}\}$ denotes the five different prediction models; and finally $\mathcal{L} \in \{5, 10, 20\}$ denotes the different lag lengths. The notation $\mathcal{M}1$ and $\mathcal{M}2$ is also used to denote two different forecast models. Two forecast models are different, if *any* element of the tuples ($\mathcal{M}1$ and $\mathcal{M}2$) are different.

Furthermore, let Y_t denote the ground-truth IVS during the out-of-sample testing for a given day t and Y denote the matrix of all ground-truth IVSs. Y_t and $Y_t^{\mathcal{M}}$ are 2D matrices of dimension $|I| \times |J|$, where I and J are as defined in section 3.1.2. In the same vein $\hat{Y}^{\mathcal{M}}$ and Y are also 2D matrices of dimension $N \times (|I| * |J|)$. For sake of clarity, the indices for these matrices start at one, and notation $Y[i, j]$ for some $i, j \in \mathbb{Z}^{\geq 1}$ denotes accessing the $(i^{\text{th}}, j^{\text{th}})$ element of the matrix.

5.2.1 The R^2 Statistic

Firstly, the R^2 statistic (Equation (5.1)), for each point on the IVS across the whole out-of-sample set (N), is utilised to assess the predictive power of the different models and their forecasts. The R^2 statistic in equation (5.1) denotes how much variation of the ground-truth has been captured by the predictive forecast model $\mathcal{M}$. The numerator captures the total variance of each point on the IVS. Ideal value of R^2 is 1, which indicates all the variation has been captured. A value of 0, means that the prediction is no better than using the average. The predictive model is deemed arbitrarily worse if the R^2 value is negative. The average R^2 value for all the points on the IVS across the out-of-sample set is given by equation (5.2).

$$R_{\mathcal{M}}^2[j] = 1 - \frac{\sum_{i=1}^N (Y[i, j] - \hat{Y}^{\mathcal{M}}[i, j])^2}{\sum_{i=1}^N (Y[i, j] - \bar{Y}[j])^2}, \forall j \in \{1, \dots, |I| * |J|\} \quad (5.1)$$

$$R_{\mathcal{M}}^2 = \frac{1}{|I| * |J|} \sum_{i=1}^{|I| * |J|} R_{\mathcal{M}}^2[i] \quad (5.2)$$

A higher R^2 statistic is better. However, it is necessary to compare the R^2 values of different forecast models statistically. To make sure that the higher R^2 is not just by chance. A pair-wise model comparison is performed using equation (5.3) as proposed in [24]. Given two models $\mathcal{M}1$ and $\mathcal{M}2$, a positive $R_{\mathcal{M}1, \mathcal{M}2}^2$ means that model $\mathcal{M}2$ outperforms $\mathcal{M}1$ and vice versa. The significance of this output is tested by performing a single-sample one-sided t-test of the statistic ($f_{\mathcal{M}1, \mathcal{M}2}$) in equation (5.4).

$$\begin{aligned}
R_{\mathcal{M}1, \mathcal{M}2}^2 &= 1 - \frac{\sum_{j=1}^{|I|*|J|} \sum_{i=1}^N (\epsilon^{\mathcal{M}2})^2}{\sum_{j=1}^{|I|*|J|} \sum_{i=1}^N (\epsilon^{\mathcal{M}1})^2} \\
\epsilon^{\mathcal{M}1} &= Y - \hat{Y}^{\mathcal{M}1}, \\
\epsilon^{\mathcal{M}2} &= Y - \hat{Y}^{\mathcal{M}2},
\end{aligned} \tag{5.3}$$

$$\begin{aligned}
f_{\mathcal{M}1, \mathcal{M}2} &= \sum_{i=1}^N \sum_{j=1}^{|I|*|J|} (\epsilon^{\mathcal{M}1})^2 - \sum_{i=1}^N \sum_{j=1}^{|I|*|J|} (\epsilon^{\mathcal{M}2} - \epsilon^{\mathcal{M}2, \mathcal{M}1})^2 \\
\epsilon^{\mathcal{M}2, \mathcal{M}1} &= \hat{Y}^{\mathcal{M}2} - \hat{Y}^{\mathcal{M}1},
\end{aligned} \tag{5.4}$$

5.2.2 Root Mean Square Error (RMSE)

The second quantitative measure used to compare forecast accuracy is the *daily* Root Mean Square Error (RMSE) between the predicted IVS and the ground-truth as shown in equation (5.5). The average RMSE for the whole out-of-sample testing period is given in equation (5.6).

$$\text{RMSE}_{\mathcal{M}}[i] = \sqrt{\frac{1}{|I|*|J|} \sum_{j=1}^{|I|*|J|} (Y[i, j] - \hat{Y}^{\mathcal{M}}[i, j])^2}, \forall i \in \{1, \dots, N\} \tag{5.5}$$

$$\text{RMSE}_{\mathcal{M}} = \frac{1}{N} \sum_{i=1}^N \text{RMSE}_{\mathcal{M}}[i] \tag{5.6}$$

In order to compare the forecast accuracy, in terms of daily RMSE, of two different models $\mathcal{M}1$ and $\mathcal{M}2$, the DM [74] test statistic is used like in [21]. The DM test statistic checks if the so-called *loss differential* time series between two error (RMSE) metrics are significantly different to zero. The DM test-statistic is defined in equation (5.7). If the DM test-statistic value in equation (5.7) is statistically significantly greater than zero, then the forecast from model $\mathcal{M}2$ is better, because the $\mathcal{M}1$ forecast error is greater, and vice versa. The significance value (p-value) is obtained by testing for the null hypothesis (H_0) that DM statistic simulates a standard normal distribution (Equation (5.8)).

$$\begin{aligned}
\delta[i] &= \text{RMSE}_{\mathcal{M}1}^2[i] - \text{RMSE}_{\mathcal{M}2}^2[i], \forall i \in \{1, \dots, N\} \\
\delta &= \frac{1}{N} \sum_{i=1}^N \delta[i], \quad \gamma_k = \frac{1}{N} \sum_{i=k+1}^N (\delta[i] - \delta)(\delta[i-k] - \delta) \\
\text{DM} &= \frac{\delta}{\sqrt{\frac{\gamma_0 + 2 \sum_{k=1}^{h-1} \gamma_k}{N}}}, \exists n > k \geq 1 \wedge h = N^{1/3} + 1
\end{aligned} \tag{5.7}$$

$$\text{DM} \sim \mathcal{N}(0, 1) \tag{5.8}$$

5.2.3 Directional accuracy

Elliott and Timmermann [75] argue that statistical measures such as R^2 and RMSE can only give a crude approximation of the volatility traders' objective at hand. Volatility traders usually buy (or sell) volatility depending upon the *expected direction* of the future IVS. Hence, the directional accuracy is also used as a performance criterion to compare forecasting models.

The forecasted direction for each (IV) point on the whole forecasted, one day ahead, IVS surface is given in equation (5.9). If the difference between the one-day ahead forecasted point on the IVS and today's same point on the IVS is greater than some small $\epsilon \in \mathbb{R}^{>0}$, the direction of the IV at that point on the grid $\mathcal{I}$ is considered increasing, if the difference is within some $\epsilon \in \mathbb{R}^{>0}$ value the direction is considered non-changing, and decreasing otherwise.

$$d = \hat{Y}^{\mathcal{M}}[i] - Y[i-1], \forall i \in \{2, \dots, N\}$$

$$\text{sgn}(x) = \begin{cases} 1, & \text{if } x > \epsilon \\ 0, & \text{if } x = |\epsilon| \\ -1, & \text{otherwise} \end{cases}, \forall x \in \mathbb{R}$$

$$\hat{d}_{\mathcal{M}}[i, j] = \text{sgn}(d[i, j]), \forall i \in \{1, \dots, N-1\}, \forall j \in \{1, \dots, |I| * |J|\} \quad (5.9)$$

$$\mathcal{D}_{\mathcal{M}}[j] = \frac{1}{N-1} \sum_{i=1}^{N-1} (\hat{d}_{\mathcal{M}}[i, j] = d_{\mathcal{T}}[i, j]), \forall j \in \{1, \dots, |I| * |J|\} \quad (5.10)$$

Let $d_{\mathcal{T}}$ denote the true direction of each point on the IVS in the out-of-sample test set. Then the average directional accuracy for each point on the IVS for the whole out-of-sample test set is given in equation (5.10). One can check the statistical significance of the forecasted directional accuracy ($d_{\mathcal{M}}$) compared to the ground-truth ($d_{\mathcal{T}}$) using the Pesaran and Timmermann (PT) [76] test for directional accuracy. For some IV point $j \in \{1, \dots, |I| * |J|\}$, let $\hat{d}_{\mathcal{M}}^j = \hat{d}_{\mathcal{M}}[i, j], \forall i \in \{1, \dots, N-1\}$ be the forecasted directional time series. In the same vein we have $d_{\mathcal{T}}^j$, which is the true directional time series for the whole out-of-sample test dataset. Given these two time series, equation (5.11) gives the PT test statistic. The null hypothesis is that forecast direction time series does not predict the true directional time series. This null hypothesis is rejected if the PT statistic is from the upper tail of the standard normal distribution.

$$\begin{aligned} p_y &= \frac{1}{N-1} \sum_{i=1}^{N-1} d_{\mathcal{M}}^j & p_z &= \frac{1}{N-1} \sum_{i=1}^{N-1} d_{\mathcal{T}}^j & p &= p_y p_z + (1 - p_y)(1 - p_z) \\ q_y &= \frac{p_y(1 - p_y)}{N-1} & q_z &= \frac{p_z(1 - p_z)}{N-1} & v &= \frac{p(1 - p)}{N-1} \\ w &= ((2p_y - 1)^2) * q_z + ((2p_z - 1)^2) * q_y + 4q_y q_z \\ \text{PT} &= \frac{(D_{\mathcal{M}}[j] - p)}{\sqrt{v - w}} \end{aligned} \quad (5.11)$$

5.3 Results and Discussion

5.3.1 Out of sample forecasting

This section presents the results comparing different models using the comparison metrics in section 5.2 with the respective statistical significance for the whole 9 years out-of-sample testing period from 2014–2022. All the results are presented for the call option IVS. Robustness checks with the put option IVS are carried out later in section 5.3.3.

The section begins with a comparison of the average R^2 and RMSE statistics for the different feature encoders and predictors for the entire out-of-sample period. This gives an overview of the forecasting performance and a birds-eye comparison can be made of the different feature encoders and predictors. Next, an analysis of the directional accuracy of the models is presented. This is then followed by a comparison of the different lag lengths for the purpose singling out the best performing models for each input feature set. Finally, a comparison of the best performing models is presented.

Tables 5.1– 5.5 present the overall average R^2 , RMSE, and volatility direction accuracy statistics of different feature encoding models via the predictor models for the point-surface input features for all lags. Full comparative results, showing $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ (5.3) and DM (5.7) results comparing each model against the benchmark AR(1) model, can be found in the appendix A.1–A.2. The models for each input feature are compared against the AR(1) model trained on the respective input feature i.e. Point-SAM model is compared against Point-AR(1). The AR(1) model is a standard autoregressive model akin to linear regression as described in section 2.3.2, with one lagged input feature matrix using the SAM method for feature encoding.

For the point input feature, all encoder and predictor model pairs perform the best for a lag of 5 days, except the RF model, which shows lackluster performance compared to all other predictor models. This is evident from the fact that the SAM encoder with SVR predictor at 5 lags is able to capture 0.28% *more* variation of the ground truth IVS than the same for 20 lags. More prominently the R^2 statistic for the SAM encoder with the LSTM predictor drops from 88.49% to 84.58% for 5 lags to 10 lags respectively. This pattern is consistent for all other encoding techniques when using the IV point as an input feature. The RMSE statistic re-enforces this observation. Overall for the point input feature the SAM encoder with the SVR predictor shows a much better R^2 and RMSE performance over every other encoding and predictor technique when using 5 historical lags. As the number of lags increases to 10 and 20, the HAR feature encoder starts outperforming the SAM encoding. For all encoding techniques and predictor models, the results show a trend of decreasing R^2 and increasing RMSE as the number of lags increases for the point input feature. Across all encoders and predictors, the SAM encoder with the SVR predictor outperforms all other models in terms of average R^2 and RMSE statistics. The $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ statistic shows that the SAM encoder with the SVR predictor is significantly better than the AR(1) benchmark model for the point input feature by 23.7%. These results for the point input feature are contradictory to the

findings in Wen, Zhai, Wang, *et al.* [16], where they find for increasing lags predictors tend to perform better

For the Moneyness Skew (MS) input feature, the Sampling (SAM) encoder with the SVR predictor seems to outperform all other feature encoders with an R^2 statistic of 90.09% for a lag of 20. This is only marginally better than the HAR encoder for the same predictor with an R^2 statistic of 89.99% (Table 5.1). Referring to the RMSE statistic 5.3, we see that again the SAM encoder with the SVR predictor outperforms all other models. Again the HAR encoder with the SVR predictor is a close second, with a RMSE statistic of 0.0152 and 0.0151 respectively. These results go to an overall pattern for the moneyness skew input feature where we see marginal improvements in the R^2 statistic as the number of lags increases from 5 to 10 and 20. The same pattern is seen in the comparative $R^2_{M1,M2}$ and DM statistics (Table A.1 and A.2).

For the Term Structure (TS) input feature, the SAM encoder with ridge predictor performs the best, managing to outperform the SVR predictor. The SVR model seems to show a consistent pattern of performing poorly for large numbers of features. Even when applying feature reduction techniques. The HAR encoder is the second best. No other feature encoder comes close to these two in terms of improvement over the benchmark. One should also note that for the TS feature none of the encoding models improve as the number of lags increase for the comparative tests (Table A.1-A.2) or the average R^2 and RMSE statistics (Table 5.1-5.3). Looking at the average R^2 statistics (Table 5.3) one can see that for the TS feature the ridge predictor with the SAM encoder model for 10 lags has the highest R^2 value of 0.8995. This is only 0.01% better than the same model for 5 lags. Further increasing the lags to 20 leads to a decrease in the R^2 value. A similar trend is seen for all other encoding techniques for the ridge predictor.

Finally, when considering the surface input feature set; looking at the R^2 the HAR encoder outperforms every other encoder when used with the ridge predictor, followed by the SAM encoding technique (Table 5.1). The DM statistic (Table A.2) shows that the SAM encoder with ridge predictor outperforms the HAR encoder. However, looking at the average R^2 and RMSE statistics (Tables 5.1 and 5.3), we see that the HAR encoder with ridge predictor has the highest R^2 and RMSE values of 0.8960 and 0.0155 respectively. The surf feature set contrary to the other feature sets shows a pattern of increasing R^2 and decreasing RMSE as the number of lags increase up to a lag length of 10 days. After this point the R^2 and RMSE values tend to decrease for all encoding and predictor techniques.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.8974	0.8926	0.8902	-	-	-	0.8931
		MS	0.8847	0.8581	0.8536	0.8638	-	-	0.8913
		TS	0.8994	0.8832	0.8871	0.8873	-	-	0.8991
		Surf	0.8858	0.8525	0.8609	-	0.7911	0.8467	0.8960
	Enet	Point	-0.7785	-0.7785	0.8902	-	-	-	-0.7739
		MS	-0.7785	-0.7450	0.8302	-2.8305	-	-	-0.7739
		TS	-0.7785	0.7647	0.8867	-0.7795	-	-	-0.7739
		Surf	-0.7785	0.8347	0.8608	-	-0.7642	0.8323	-0.7739
	Lasso	Point	-0.7785	-0.7785	0.8902	-	-	-	-0.7739
		MS	-0.7785	-0.7785	0.8282	-3.3303	-	-	-0.7739
		TS	-0.7785	0.7157	0.8862	-0.7795	-	-	-0.7739
		Surf	-0.7785	0.8204	0.8607	-	-0.7642	0.8323	-0.7739
	RF	Point	0.6466	0.8735	0.7743	-	-	-	0.8598
		MS	0.6491	0.8042	0.8006	-4.3424	-	-	0.8436
		TS	0.6324	0.8510	0.8264	0.7677	-	-	0.8412
		Surf	0.6598	0.6728	0.7608	-	0.1207	0.8456	0.7789
	LSTM	Point	0.8849	-	-	-	-	-	0.8937
		MS	0.8802	-	-	-5.3817	-	-	0.8897
		TS	0.8806	-	-	0.8663	-	-	0.8683
		Surf	-0.6644	-	-	-	-29.6127	-	-0.8029
	Attention	Point	0.9027	-	-	-	-	-	0.8714
		MS	0.8613	-	-	-17.7810	-	-	0.8527
		TS	0.5732	-	-	0.8309	-	-	0.6483
		Surf	0.5045	-	-	-	0.7808	0.8571	-0.9760
	SVR	Point	0.9042	0.9031	0.8797	-	-	-	0.8992
		MS	0.8993	0.8582	0.5023	-22.3135	-	-	0.8999
		TS	0.8863	0.0622	-0.8396	0.7989	-	-	0.8853
		Surf	-8.4230	-0.7699	-0.7778	-	-9.1754	0.7104	-8.4550
10	Ridge	Point	0.8926	0.8773	0.8603	-	-	-	0.8931
		MS	0.8903	0.8518	0.8256	0.8676	-	-	0.8913
		TS	0.8995	0.8811	0.8661	0.8822	-	-	0.8991
		Surf	0.8893	0.8498	0.8465	-	0.7803	0.8489	0.8960
	Enet	Point	-0.7766	-0.7766	0.8603	-	-	-	-0.7739
		MS	-0.7766	-0.5556	0.8061	-3.1275	-	-	-0.7739
		TS	-0.7766	0.7619	0.8657	-0.7776	-	-	-0.7739
		Surf	-0.7766	0.8268	0.8465	-	-0.7623	0.8319	-0.7739
	Lasso	Point	-0.7766	-0.7766	0.8603	-	-	-	-0.7739
		MS	-0.7766	-0.7766	0.7983	-3.4406	-	-	-0.7739
		TS	-0.7766	0.7353	0.8653	-0.7776	-	-	-0.7739
		Surf	-0.7766	0.8073	0.8465	-	-0.7623	0.8319	-0.7739
	RF	Point	0.6471	0.8762	0.7448	-	-	-	0.8598
		MS	0.6455	0.7787	0.7077	-6.5643	-	-	0.8436
		TS	0.6310	0.8152	0.7870	0.7308	-	-	0.8412

20	LSTM	Surf	0.6549	0.6271	0.7336	-	0.0670	0.8487	0.7789
		Point	0.8564	-	-	-	-	-	0.8937
		MS	0.8833	-	-	-2.3391	-	-	0.8897
		TS	0.8867	-	-	0.8428	-	-	0.8683
	Attention	Surf	-0.6807	-	-	-	-185.6314	-	-0.8029
		Point	0.8976	-	-	-	-	-	0.8714
		MS	0.7614	-	-	-85.2427	-	-	0.8527
		TS	0.7197	-	-	0.8904	-	-	0.6483
	SVR	Surf	0.4876	-	-	-	0.6369	0.8561	-0.9760
		Point	0.9025	0.9020	0.8484	-	-	-	0.8992
		MS	0.8994	0.8548	0.4818	-26.9570	-	-	0.8999
		TS	0.8864	0.0713	-0.8374	0.7783	-	-	0.8853
20	Ridge	Surf	-8.4419	-0.7680	-0.7759	-	-9.2082	0.7101	-8.4550
		Point	0.8903	0.8667	0.7990	-	-	-	0.8931
		MS	0.8921	0.8503	0.7616	0.8665	-	-	0.8913
		TS	0.8970	0.8808	0.8317	0.8794	-	-	0.8991
	Enet	Surf	0.8888	0.8489	-	-	0.7910	0.8502	0.8960
		Point	-0.7739	-0.7739	0.7990	-	-	-	-0.7739
		MS	-0.7739	-0.2299	0.7575	-3.7950	-	-	-0.7739
		TS	-0.7739	0.7113	0.8311	-0.7747	-	-	-0.7739
	Lasso	Surf	-0.7739	0.8156	-	-	-0.7591	0.8296	-0.7739
		Point	-0.7739	-0.7739	0.7990	-	-	-	-0.7739
		MS	-0.7739	-0.7477	0.7431	-4.8032	-	-	-0.7739
		TS	-0.7739	0.6976	0.8305	-0.7747	-	-	-0.7739
	RF	Surf	-0.7739	0.7830	-	-	-0.7591	0.8296	-0.7739
		Point	0.6478	0.8713	0.6779	-	-	-	0.8598
		MS	0.6560	0.7465	0.6071	-6.1129	-	-	0.8436
		TS	0.6293	0.7857	0.7462	0.6900	-	-	0.8412
	LSTM	Surf	0.6603	0.4289	-	-	0.0523	0.8490	0.7789
		Point	0.8458	-	-	-	-	-	0.8937
		MS	0.8700	-	-	-5.5629	-	-	0.8897
		TS	0.8811	-	-	0.8545	-	-	0.8683
	Attention	Surf	-0.6445	-	-	-	-852.0242	-	-0.8029
		Point	0.8829	-	-	-	-	-	0.8714
		MS	0.8333	-	-	-53.7186	-	-	0.8527
		TS	0.6873	-	-	0.8648	-	-	0.6483
	SVR	Surf	-0.8484	-	-	-	0.8277	0.8427	-0.9760
		Point	0.9014	0.9011	0.7860	-	-	-	0.8992
		MS	0.9009	0.8545	0.4437	-53.9433	-	-	0.8999
		TS	0.8868	0.0866	-0.8349	0.7731	-	-	0.8853
		Surf	-	-	-	-	-	-	-8.4550

TABLE 5.1: The table present the average R^2 statistic for call options on the out of sample dataset using the Dumas interpolation. The best performing model for each feature set is shown in bold.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.8983	0.8916	0.8897	-	-	-	0.8936
		MS	0.8947	0.8487	0.8765	0.8807	-	-	0.8971
		TS	0.8972	0.8520	0.8867	0.8765	-	-	0.8965
		Surf	0.8501	0.8517	0.8697	-	0.8076	0.8576	0.8557
	Enet	Point	-0.7148	-0.7148	0.8897	-	-	-	-0.7088
		MS	-0.7148	-0.6881	0.8416	-1.3919	-	-	-0.7088
		TS	-0.7148	0.7548	0.8823	-0.7133	-	-	-0.7088
		Surf	-0.7148	0.8342	0.8699	-	-0.7095	0.8478	-0.7088
	Lasso	Point	-0.7148	-0.7148	0.8897	-	-	-	-0.7088
		MS	-0.7148	-0.7148	0.8410	-2.3237	-	-	-0.7088
		TS	-0.7148	0.7130	0.8771	-0.7133	-	-	-0.7088
		Surf	-0.7148	0.8185	0.8700	-	-0.7095	0.8478	-0.7088
	RF	Point	0.8948	0.8145	0.8163	-	-	-	0.8801
		MS	0.8851	0.8329	0.8356	-0.8889	-	-	0.8716
		TS	0.8428	0.8397	0.7861	0.7658	-	-	0.8371
		Surf	0.8005	0.7034	0.3248	-	-0.0922	0.8574	0.7958
	LSTM	Point	0.9034	-	-	-	-	-	0.8967
		MS	0.8937	-	-	-4.4402	-	-	0.9056
		TS	0.8437	-	-	0.8573	-	-	0.8332
		Surf	-0.6894	-	-	-	-159.9678	-	-0.7141
	Attention	Point	-0.5490	-	-	-	-	-	-2.2982
		MS	-0.9551	-	-	-1.6476	-	-	-2.7729
		TS	-2.3094	-	-	0.4582	-	-	-3.8609
		Surf	-3.2899	-	-	-	0.3592	0.4442	-4.5370
	SVR	Point	0.9034	0.8894	0.8806	-	-	-	0.8989
		MS	0.9076	0.8702	0.5453	-22.6290	-	-	0.9066
		TS	0.8990	0.1378	-0.7311	0.8602	-	-	0.8963
		Surf	-7.7388	-0.7068	-0.7140	-	-13.3549	0.7640	-7.6675
10	Ridge	Point	0.8926	0.8740	0.8578	-	-	-	0.8936
		MS	0.8969	0.8469	0.8395	0.8755	-	-	0.8971
		TS	0.8912	0.8441	0.8718	0.8680	-	-	0.8965
		Surf	0.8222	0.8480	0.8564	-	0.8099	0.8527	0.8557
	Enet	Point	-0.7117	-0.7117	0.8578	-	-	-	-0.7088
		MS	-0.7117	-0.4648	0.8130	-1.5623	-	-	-0.7088
		TS	-0.7117	0.7458	0.8688	-0.7101	-	-	-0.7088
		Surf	-0.7117	0.8261	0.8564	-	-0.7063	0.8415	-0.7088
	Lasso	Point	-0.7117	-0.7117	0.8578	-	-	-	-0.7088
		MS	-0.7117	-0.7117	0.8091	-1.7061	-	-	-0.7088
		TS	-0.7117	0.7232	0.8656	-0.7101	-	-	-0.7088
		Surf	-0.7117	0.7997	0.8564	-	-0.7063	0.8415	-0.7088
	RF	Point	0.8893	0.8592	0.7832	-	-	-	0.8801
		MS	0.8766	0.7894	0.7574	-0.9155	-	-	0.8716
		TS	0.8450	0.8335	0.7876	0.7466	-	-	0.8371

		Surf	0.7895	0.6494	0.3506	-	-0.1526	0.8507	0.7958
	LSTM	Point	0.9020	-	-	-	-	-	0.8967
		MS	0.8687	-	-	-9.5191	-	-	0.9056
		TS	0.8345	-	-	0.8280	-	-	0.8332
		Surf	-0.6291	-	-	-	-131.0126	-	-0.7141
	Attention	Point	-1.2937	-	-	-	-	-	-2.2982
		MS	-1.7805	-	-	-5.4754	-	-	-2.7729
		TS	-3.1302	-	-	0.4606	-	-	-3.8609
		Surf	-3.9824	-	-	-	0.4003	0.4346	-4.5370
	SVR	Point	0.9022	0.8914	0.8472	-	-	-	0.8989
		MS	0.9075	0.8565	0.5221	-24.1037	-	-	0.9066
		TS	0.8957	0.1390	-0.7306	0.8478	-	-	0.8963
		Surf	-7.7329	-0.7036	-0.7109	-	-11.9503	0.7575	-7.6675
20	Ridge	Point	0.8907	0.8634	0.7953	-	-	-	0.8936
		MS	0.8980	0.8415	0.7643	0.8729	-	-	0.8971
		TS	0.8815	0.8375	0.7911	0.8638	-	-	0.8965
		Surf	0.7873	0.8454	-	-	0.8162	0.8478	0.8557
	Enet	Point	-0.7088	-0.7088	0.7953	-	-	-	-0.7088
		MS	-0.7088	-0.1039	0.7595	-1.6256	-	-	-0.7088
		TS	-0.7088	0.6928	0.7905	-0.7075	-	-	-0.7088
		Surf	-0.7088	0.8177	-	-	-0.7034	0.8350	-0.7088
	Lasso	Point	-0.7088	-0.7088	0.7953	-	-	-	-0.7088
		MS	-0.7088	-0.6895	0.7509	-1.5357	-	-	-0.7088
		TS	-0.7088	0.6813	0.7896	-0.7075	-	-	-0.7088
		Surf	-0.7088	0.7873	-	-	-0.7034	0.8350	-0.7088
	RF	Point	0.8817	0.8522	0.7141	-	-	-	0.8801
		MS	0.8704	0.7567	0.6566	-0.9233	-	-	0.8716
		TS	0.8430	0.7198	0.7521	0.7148	-	-	0.8371
		Surf	0.7941	0.4268	-	-	-0.1574	0.8454	0.7958
	LSTM	Point	0.9009	-	-	-	-	-	0.8967
		MS	0.8705	-	-	-6.9977	-	-	0.9056
		TS	0.7813	-	-	0.8162	-	-	0.8332
		Surf	-0.6858	-	-	-	-577.0972	-	-0.7141
	Attention	Point	-2.2969	-	-	-	-	-	-2.2982
		MS	-2.7749	-	-	-2.3056	-	-	-2.7729
		TS	-3.8614	-	-	0.4636	-	-	-3.8609
		Surf	-4.5378	-	-	-	0.4640	0.4312	-4.5370
	SVR	Point	0.9019	0.8837	0.7841	-	-	-	0.8989
		MS	0.9078	0.8554	0.4805	-15.6401	-	-	0.9066
		TS	0.8913	0.0868	-0.7334	0.8429	-	-	0.8963
		Surf	-	-	-	-	-	-	-7.6675

TABLE 5.2: The table presents the average overall R^2 statistic for call options on the out of sample dataset using the Nadarya Watson interpolation. The best performing model for each feature set is shown in bold.

It is interesting to note that the elastic-net and lasso predictors do not perform well with any of the feature encoders besides CCA, SSVI and in the cases of TS and surface features PCA also performs well. In these cases the feature reduction techniques, especially CCA and PCA, may be reducing the multicollinearity significantly enough to allow the l1 regularization present in these two predictor models to work effectively. Otherwise, in the presence of multiple predictor variables that are highly correlated such as the lagged features of the IVS, the l1 regularization tends to force many of the weights to zero. During testing, the results showed that the lasso and elastic-net predictors tend to force the weights of many features to zero, thereby removing them as being considered predictive for the one-day ahead forecast. Testing showed lowering the alpha value, which controls the l1 regularization strength, to values below 0.001 did improve results. However, at this point one would just consider using linear regression. The R^2 and RMSE statistic results for the elastic-net and lasso predictors are much worse than the ridge predictor and often show worse performance over the AR(1) benchmark for the $R^2_{\mathcal{M}1, \mathcal{M}2}$ and DM statistics. One should note that the elastic-net and lasso predictors do show consistent gains over the benchmark for the CCA encoding technique. The CCA encoding technique for lasso and elastic-net also shows better $R^2_{\mathcal{M}1, \mathcal{M}2}$ and DM statistics compared to the SAM encoding method for the skew, term structure, and surface features.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.0154	0.0159	0.0160	-	-	-	0.0158
		MS	0.0160	0.0181	0.0181	0.0176	-	-	0.0157
		TS	0.0152	0.0166	0.0162	0.0162	-	-	0.0153
		Surf	0.0160	0.0184	0.0181	-	0.0218	0.0183	0.0155
	Enet	Point	0.0651	0.0651	0.0160	-	-	-	0.0652
		MS	0.0651	0.0644	0.0195	0.0944	-	-	0.0652
		TS	0.0651	0.0235	0.0163	0.0651	-	-	0.0652
		Surf	0.0651	0.0195	0.0181	-	0.0650	0.0190	0.0652
	Lasso	Point	0.0651	0.0651	0.0160	-	-	-	0.0652
		MS	0.0651	0.0651	0.0196	0.1004	-	-	0.0652
		TS	0.0651	0.0259	0.0163	0.0651	-	-	0.0652
		Surf	0.0651	0.0203	0.0181	-	0.0650	0.0190	0.0652
	RF	Point	0.0291	0.0172	0.0232	-	-	-	0.0183
		MS	0.0287	0.0212	0.0216	0.1130	-	-	0.0190
		TS	0.0295	0.0186	0.0200	0.0234	-	-	0.0192
		Surf	0.0281	0.0273	0.0234	-	0.0461	0.0184	0.0221
	LSTM	Point	0.0163	-	-	-	-	-	0.0159
		MS	0.0165	-	-	0.1233	-	-	0.0160
		TS	0.0166	-	-	0.0178	-	-	0.0176
		Surf	0.0630	-	-	-	0.2175	-	0.0657
	Attention	Point	0.0150	-	-	-	-	-	0.0176
		MS	0.0177	-	-	0.2112	-	-	0.0187
		TS	0.0309	-	-	0.0199	-	-	0.0278
		Surf	0.0327	-	-	-	0.0224	0.0177	0.0605
	SVR	Point	0.0149	0.0150	0.0169	-	-	-	0.0154
		MS	0.0152	0.0179	0.0346	0.2321	-	-	0.0152
		TS	0.0161	0.0466	0.0651	0.0216	-	-	0.0163
		Surf	0.1478	0.0650	0.0651	-	0.1548	0.0245	0.1484
10	Ridge	Point	0.0159	0.0171	0.0183	-	-	-	0.0158
		MS	0.0157	0.0184	0.0201	0.0175	-	-	0.0157
		TS	0.0152	0.0168	0.0179	0.0166	-	-	0.0153
		Surf	0.0159	0.0186	0.0191	-	0.0225	0.0182	0.0155
	Enet	Point	0.0652	0.0652	0.0183	-	-	-	0.0652
		MS	0.0652	0.0605	0.0212	0.0980	-	-	0.0652
		TS	0.0652	0.0236	0.0179	0.0652	-	-	0.0652
		Surf	0.0652	0.0201	0.0191	-	0.0650	0.0190	0.0652
	Lasso	Point	0.0652	0.0652	0.0183	-	-	-	0.0652
		MS	0.0652	0.0652	0.0216	0.1021	-	-	0.0652
		TS	0.0652	0.0250	0.0179	0.0652	-	-	0.0652
		Surf	0.0652	0.0212	0.0191	-	0.0650	0.0190	0.0652
	RF	Point	0.0291	0.0171	0.0247	-	-	-	0.0183
		MS	0.0289	0.0226	0.0262	0.1346	-	-	0.0190
		TS	0.0296	0.0208	0.0223	0.0253	-	-	0.0192

		Surf	0.0283	0.0292	0.0249	-	0.0475	0.0182	0.0221
	LSTM	Point	0.0183	-	-	-	-	-	0.0159
		MS	0.0164	-	-	0.0894	-	-	0.0160
		TS	0.0163	-	-	0.0193	-	-	0.0176
		Surf	0.0634	-	-	-	0.5697	-	0.0657
	Attention	Point	0.0155	-	-	-	-	-	0.0176
		MS	0.0235	-	-	0.4505	-	-	0.0187
		TS	0.0250	-	-	0.0160	-	-	0.0278
		Surf	0.0334	-	-	-	0.0289	0.0178	0.0605
	SVR	Point	0.0151	0.0151	0.0191	-	-	-	0.0154
		MS	0.0152	0.0182	0.0354	0.2542	-	-	0.0152
		TS	0.0162	0.0464	0.0651	0.0229	-	-	0.0163
		Surf	0.1481	0.0650	0.0651	-	0.1553	0.0246	0.1484
20	Ridge	Point	0.0161	0.0178	0.0221	-	-	-	0.0158
		MS	0.0157	0.0186	0.0238	0.0177	-	-	0.0157
		TS	0.0154	0.0168	0.0201	0.0169	-	-	0.0153
		Surf	0.0160	0.0188	-	-	0.0221	0.0182	0.0155
	Enet	Point	0.0652	0.0652	0.0221	-	-	-	0.0652
		MS	0.0652	0.0537	0.0240	0.1055	-	-	0.0652
		TS	0.0652	0.0262	0.0201	0.0652	-	-	0.0652
		Surf	0.0652	0.0209	-	-	0.0651	0.0192	0.0652
	Lasso	Point	0.0652	0.0652	0.0221	-	-	-	0.0652
		MS	0.0652	0.0647	0.0247	0.1146	-	-	0.0652
		TS	0.0652	0.0269	0.0201	0.0652	-	-	0.0652
		Surf	0.0652	0.0227	-	-	0.0651	0.0192	0.0652
	RF	Point	0.0291	0.0175	0.0279	-	-	-	0.0183
		MS	0.0285	0.0243	0.0306	0.1307	-	-	0.0190
		TS	0.0297	0.0225	0.0245	0.0272	-	-	0.0192
		Surf	0.0281	0.0362	-	-	0.0479	0.0182	0.0221
	LSTM	Point	0.0192	-	-	-	-	-	0.0159
		MS	0.0174	-	-	0.1254	-	-	0.0160
		TS	0.0167	-	-	0.0185	-	-	0.0176
		Surf	0.0628	-	-	-	1.2056	-	0.0657
	Attention	Point	0.0167	-	-	-	-	-	0.0176
		MS	0.0199	-	-	0.3599	-	-	0.0187
		TS	0.0265	-	-	0.0177	-	-	0.0278
		Surf	0.0586	-	-	-	0.0202	0.0187	0.0605
	SVR	Point	0.0152	0.0152	0.0228	-	-	-	0.0154
		MS	0.0151	0.0183	0.0367	0.3577	-	-	0.0152
		TS	0.0162	0.0462	0.0652	0.0233	-	-	0.0163
		Surf	-	-	-	-	-	-	0.1484

TABLE 5.3: The table presents the average RMSE statistic for call options on the out of sample dataset using the Dumas interpolation. The best performing model for each feature set is shown in bold.

The non-linear predictors, random forest and LSTM, show much worse performance than the linear predictors overall. This is inline with the findings of Wen, Zhai, Wang, *et al.* [16]. However, in contradiction to their findings, the SVR predictor shows much better performance for the point and MS feature sets compared to the ridge regression predictor. The attention predictor also shows very good R^2 performance of 90.27% for the point input and 5 day lag length. Overall, the random forest predictor does not show comparable R^2 and RMSE performance to the linear predictors, when paired with the SAM encoding technique for all feature sets. Only when paired with the PCA and HAR encoding techniques does the random forest predictor show fairly good performance. However, it still falls short of the best performing linear and non-linear predictors for all feature sets by a minimum of 3.16% for the R^2 statistic. The LSTM predictor does show some improvement over the AR(1) benchmark model for the point, MS, and TS features sets (Table A.1-A.2). We would expect the LSTM predictor to improve as the number of lags increase due to its ability to more effectively capture long-term dependencies. However, for the point feature set R^2 and RMSE performance worsen for increasing lags. For the MS and TS features set, there is marginal improvement in the R^2 and DM statistic from 5 to 10 lags. Any more than 10 lags leads to a decrease in the R^2 and RMSE statistic. For the surface feature set, the LSTM predictor consistently shows negative R^2 performance, with only marginal improvement from 5 to 20 lags. This result is expected due to the large number of features in the surface feature set. The LSTM architecture is fairly small with only 2 LSTM layers and 64 hidden units each, this limits its capacity to learn complex patterns in high-dimensional data, especially for the surface feature set which has 10237 features for the entire surface. Similarly, the attention model shows poor performance compared to the linear predictors and again lack luster performance for the surface feature input can be attributed to the fairly small number of parameters compared to the number of features.

The results in table 5.1 show that all the best models capture approximately 89% of the ground truth variation on average irrespective of the lags. However, the average RMSE and R^2 for the SVR predictor and SAM encoder on the point and MS feature sets are clearly better than the ridge predictor and SAM encoder using the TS and the ridge predictor using the HAR encoder for the surface feature set. These results suggest that the non-linear SVR predictor using the SAM encoding technique trained on the point and MS feature sets are best suited for forecasting the call option IVS for the nine year out-of-sample period. However, this needs to be statistically shown. In the same vein the average R^2 and RMSE values suggest that adding extra lags for most encoding techniques does not have any improvement in forecasting performance. The exceptions being the MS feature set for every predictor and encoding technique, where we consistently see better performance for increasing lag lengths. However, this too needs to be statistically investigated.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.0158	0.0164	0.0165	-	-	-	0.0162
		MS	0.0159	0.0192	0.0174	0.0171	-	-	0.0158
		TS	0.0157	0.0190	0.0166	0.0174	-	-	0.0159
		Surf	0.0190	0.0188	0.0177	-	0.0218	0.0185	0.0188
	Enet	Point	0.0650	0.0650	0.0165	-	-	-	0.0651
		MS	0.0650	0.0645	0.0197	0.0752	-	-	0.0651
		TS	0.0650	0.0243	0.0169	0.0650	-	-	0.0651
		Surf	0.0650	0.0199	0.0177	-	0.0650	0.0190	0.0651
	Lasso	Point	0.0650	0.0650	0.0165	-	-	-	0.0651
		MS	0.0650	0.0650	0.0197	0.0867	-	-	0.0651
		TS	0.0650	0.0265	0.0173	0.0650	-	-	0.0651
		Surf	0.0650	0.0209	0.0177	-	0.0650	0.0190	0.0651
	RF	Point	0.0160	0.0216	0.0215	-	-	-	0.0173
		MS	0.0166	0.0203	0.0201	0.0684	-	-	0.0176
		TS	0.0195	0.0196	0.0219	0.0234	-	-	0.0199
		Surf	0.0215	0.0264	0.0396	-	0.0522	0.0184	0.0218
	LSTM	Point	0.0153	-	-	-	-	-	0.0160
		MS	0.0160	-	-	0.1159	-	-	0.0152
		TS	0.0196	-	-	0.0187	-	-	0.0202
		Surf	0.0646	-	-	-	0.5375	-	0.0652
	Attention	Point	0.0812	-	-	-	-	-	0.1187
		MS	0.0908	-	-	0.1031	-	-	0.1267
		TS	0.1181	-	-	0.0480	-	-	0.1435
		Surf	0.1338	-	-	-	0.0521	0.0485	0.1530
	SVR	Point	0.0153	0.0165	0.0172	-	-	-	0.0157
		MS	0.0150	0.0179	0.0337	0.2411	-	-	0.0151
		TS	0.0157	0.0456	0.0643	0.0186	-	-	0.0160
		Surf	0.1459	0.0649	0.0650	-	0.1849	0.0235	0.1457
10	Ridge	Point	0.0163	0.0177	0.0188	-	-	-	0.0162
		MS	0.0158	0.0194	0.0200	0.0175	-	-	0.0158
		TS	0.0162	0.0197	0.0176	0.0180	-	-	0.0159
		Surf	0.0208	0.0191	0.0187	-	0.0218	0.0188	0.0188
	Enet	Point	0.0650	0.0650	0.0188	-	-	-	0.0651
		MS	0.0650	0.0600	0.0215	0.0778	-	-	0.0651
		TS	0.0650	0.0249	0.0179	0.0650	-	-	0.0651
		Surf	0.0650	0.0206	0.0187	-	0.0650	0.0194	0.0651
	Lasso	Point	0.0650	0.0650	0.0188	-	-	-	0.0651
		MS	0.0650	0.0650	0.0217	0.0798	-	-	0.0651
		TS	0.0650	0.0260	0.0181	0.0650	-	-	0.0651
		Surf	0.0650	0.0221	0.0187	-	0.0650	0.0194	0.0651
	RF	Point	0.0165	0.0187	0.0234	-	-	-	0.0173
		MS	0.0171	0.0227	0.0246	0.0689	-	-	0.0176
		TS	0.0194	0.0201	0.0224	0.0247	-	-	0.0199

20	LSTM	Surf	0.0221	0.0285	0.0388	-	0.0537	0.0189	0.0218
		Point	0.0155	-	-	-	-	-	0.0160
		MS	0.0178	-	-	0.1609	-	-	0.0152
		TS	0.0202	-	-	0.0206	-	-	0.0202
	Attention	Surf	0.0635	-	-	-	0.4844	-	0.0652
		Point	0.0989	-	-	-	-	-	0.1187
		MS	0.1085	-	-	0.1570	-	-	0.1267
		TS	0.1320	-	-	0.0479	-	-	0.1435
	SVR	Surf	0.1447	-	-	-	0.0506	0.0490	0.1530
		Point	0.0154	0.0164	0.0196	-	-	-	0.0157
		MS	0.0150	0.0188	0.0346	0.2489	-	-	0.0151
		TS	0.0160	0.0456	0.0644	0.0195	-	-	0.0160
20	Ridge	Surf	0.1460	0.0649	0.0650	-	0.1768	0.0238	0.1457
		Point	0.0165	0.0185	0.0227	-	-	-	0.0162
		MS	0.0158	0.0198	0.0242	0.0178	-	-	0.0158
		TS	0.0169	0.0201	0.0228	0.0184	-	-	0.0159
	Enet	Surf	0.0228	0.0193	-	-	0.0215	0.0192	0.0188
		Point	0.0651	0.0651	0.0227	-	-	-	0.0651
		MS	0.0651	0.0520	0.0245	0.0788	-	-	0.0651
		TS	0.0651	0.0276	0.0228	0.0651	-	-	0.0651
	Lasso	Surf	0.0651	0.0212	-	-	0.0650	0.0198	0.0651
		Point	0.0651	0.0651	0.0227	-	-	-	0.0651
		MS	0.0651	0.0647	0.0250	0.0776	-	-	0.0651
		TS	0.0651	0.0281	0.0229	0.0651	-	-	0.0651
	RF	Surf	0.0651	0.0230	-	-	0.0650	0.0198	0.0651
		Point	0.0171	0.0192	0.0269	-	-	-	0.0173
		MS	0.0176	0.0244	0.0292	0.0692	-	-	0.0176
		TS	0.0195	0.0256	0.0247	0.0265	-	-	0.0199
	LSTM	Surf	0.0219	0.0366	-	-	0.0540	0.0193	0.0218
		Point	0.0156	-	-	-	-	-	0.0160
		MS	0.0177	-	-	0.1398	-	-	0.0152
		TS	0.0233	-	-	0.0213	-	-	0.0202
	Attention	Surf	0.0647	-	-	-	1.0181	-	0.0652
		Point	0.1187	-	-	-	-	-	0.1187
		MS	0.1267	-	-	0.1168	-	-	0.1267
		TS	0.1436	-	-	0.0479	-	-	0.1435
	SVR	Surf	0.1530	-	-	-	0.0480	0.0492	0.1530
		Point	0.0155	0.0170	0.0233	-	-	-	0.0157
		MS	0.0150	0.0190	0.0361	0.2030	-	-	0.0151
		TS	0.0163	0.0474	0.0646	0.0198	-	-	0.0160
		Surf	-	-	-	-	-	-	0.1457

TABLE 5.4: The table reports the average RMSE for call options on the out of sample dataset using the Nadarya Watson interpolation. The best performing model for each feature set is shown in bold.

The results for the same out of sample testing period but using the Nadaraya Watson interpolation techniques paint a similar picture to the results for the Dumas et al. interpolation technique. For both the R^2 and RMSE results the SVR predictor with the SAM encoder shows a dominant performance over all other models for the point, MS and TS input feature sets. The linear ridge predictor still shows comparable performance to the SVR predictor for the point input feature set. However, the SVR predictor excels when applied to the SAM and HAR encoded point, MS, and TS input feature sets, with all models regardless of lag lengths showing close to or above 90% R^2 results. Interestingly the results show that all models show slightly better forecasting performance especially the random forest predictors and the LSTM predictors for all encoding techniques. The only exception is the attention predictor which shows negative R^2 results for almost all encoding techniques. The same patterns are still observed for the Nadaraya Watson results. ① Increasing lags for the point, TS and surface input feature sets leads to overall worse R^2 and RMSE results. The comparative $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ and DM statistics support this observation. ② The MS input feature set shows a consistent improvement in the R^2 and RMSE results as the number of lags increase up to 10 lags, after which the performance of the models start to decrease. ③ The SVR predictor with the SAM encoding technique shows the best performance for the point and MS input feature sets. ④ The encoding technique that consistently shows the best performance for all input feature sets except the surface feature set is the SAM encoding technique. The HAR encoding technique shows the best performance for the surface feature set.

Directionality Analysis

Most volatility trading desks use the direction of volatility in addition to the variation of the IVS. Hence, it is important to check that the forecasting models learn the direction of the volatility forecast correctly in addition to the overall RMSE and R^2 performance. In this section we compare the average directional forecast (Equation (5.10)) of all points on the IVS as well as the average direction forecast for each point and their statistical significance using the Pesaran-Timmermann statistic (Equation (5.11)).

The average directional accuracy (Equation (5.10)) for all points in the IVS over the nine year out-of-sample testing period is shown in figure 5.5. The very first thing we note is that the top performing models are similar to the R^2 and RMSE results as shown by the bolded values. The results show that overall the SAM encoding method does not perform as well in the directional accuracy measurement with the HAR method of feature encoding outperforming SAM for the TS feature set and showing the best performance for the surface feature set. This can be explained by the HAR method of encoding filtering out the noise into long-term, medium-term, and short-term feature representations that better capture the macro-level volatility features. The best performing model for both the MS and point feature sets however, is the non-linear SVR predictor with the SAM encoding technique. This is expected as the SVR predictor has shown to be the best performing predictor for both the R^2 and RMSE results for the point and MS feature sets. Overall compared to the R^2 and RMSE results, the

directional accuracy results leave a lot to be desired. The best performing model only achieves 61% directional accuracy.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.5939	0.5949	0.6035	-	-	-	0.5869
		MS	0.5830	0.5709	0.5800	0.5714	-	-	0.5908
		TS	0.5953	0.5892	0.5969	0.5900	-	-	0.5991
		Surf	0.5818	0.5751	0.5781	-	0.5575	0.5712	0.5913
	Enet	Point	0.5122	0.5122	0.6035	-	-	-	0.5119
		MS	0.5122	0.5122	0.5731	0.5161	-	-	0.5119
		TS	0.5122	0.5488	0.5964	0.5122	-	-	0.5119
		Surf	0.5122	0.5735	0.5781	-	0.5120	0.5690	0.5119
	Lasso	Point	0.5122	0.5122	0.6035	-	-	-	0.5119
		MS	0.5122	0.5122	0.5730	0.5098	-	-	0.5119
		TS	0.5122	0.5403	0.5963	0.5122	-	-	0.5119
		Surf	0.5122	0.5727	0.5781	-	0.5120	0.5690	0.5119
	RF	Point	0.5189	0.5788	0.5663	-	-	-	0.5769
		MS	0.5451	0.5618	0.5700	0.5127	-	-	0.5695
		TS	0.5303	0.5731	0.5739	0.5583	-	-	0.5740
		Surf	0.5537	0.5547	0.5651	-	0.5278	0.5699	0.5643
	LSTM	Point	0.5939	-	-	-	-	-	0.5948
		MS	0.5770	-	-	0.5149	-	-	0.5806
		TS	0.5876	-	-	0.5639	-	-	0.5616
		Surf	0.5132	-	-	-	0.4959	-	0.5113
	Attention	Point	0.6027	-	-	-	-	-	0.5891
		MS	0.5682	-	-	0.5099	-	-	0.5698
		TS	0.5168	-	-	0.5322	-	-	0.5219
		Surf	0.5285	-	-	-	0.5604	0.5688	0.5198
	SVR	Point	0.6119	0.6058	0.5801	-	-	-	0.6098
		MS	0.5890	0.5708	0.5295	0.5071	-	-	0.5928
		TS	0.5870	0.5235	0.5171	0.5606	-	-	0.5888
		Surf	0.5120	0.5122	0.5122	-	0.5120	0.5344	0.5123
10	Ridge	Point	0.6014	0.6000	0.6011	-	-	-	0.5869
		MS	0.5894	0.5763	0.5812	0.5777	-	-	0.5908
		TS	0.5966	0.5915	0.5951	0.5923	-	-	0.5991
		Surf	0.5838	0.5764	0.5774	-	0.5610	0.5722	0.5913
	Enet	Point	0.5119	0.5119	0.6011	-	-	-	0.5119
		MS	0.5119	0.5129	0.5769	0.5134	-	-	0.5119
		TS	0.5119	0.5593	0.5948	0.5119	-	-	0.5119
		Surf	0.5119	0.5760	0.5774	-	0.5117	0.5695	0.5119
	Lasso	Point	0.5119	0.5119	0.6011	-	-	-	0.5119
		MS	0.5119	0.5119	0.5753	0.5066	-	-	0.5119
		TS	0.5119	0.5504	0.5944	0.5119	-	-	0.5119
		Surf	0.5119	0.5731	0.5774	-	0.5117	0.5695	0.5119
	RF	Point	0.5188	0.5884	0.5688	-	-	-	0.5769

		MS	0.5448	0.5649	0.5683	0.5106	-	-	0.5695
		TS	0.5301	0.5700	0.5702	0.5556	-	-	0.5740
		Surf	0.5530	0.5532	0.5631	-	0.5257	0.5710	0.5643
	LSTM	Point	0.5666	-	-	-	-	-	0.5948
		MS	0.5863	-	-	0.5137	-	-	0.5806
		TS	0.5972	-	-	0.5507	-	-	0.5616
		Surf	0.5128	-	-	-	0.5202	-	0.5113
	Attention	Point	0.6058	-	-	-	-	-	0.5891
		MS	0.5532	-	-	0.4925	-	-	0.5698
		TS	0.5327	-	-	0.5830	-	-	0.5219
		Surf	0.5302	-	-	-	0.5599	0.5678	0.5198
	SVR	Point	0.6128	0.6079	0.5794	-	-	-	0.6098
		MS	0.5917	0.5730	0.5303	0.5113	-	-	0.5928
		TS	0.5884	0.5238	0.5170	0.5561	-	-	0.5888
		Surf	0.5123	0.5119	0.5119	-	0.5123	0.5347	0.5123
20	Ridge	Point	0.6026	0.5999	0.5888	-	-	-	0.5869
		MS	0.5901	0.5773	0.5742	0.5760	-	-	0.5908
		TS	0.5938	0.5914	0.5880	0.5925	-	-	0.5991
		Surf	0.5845	0.5762	-	-	0.5629	0.5723	0.5913
	Enet	Point	0.5119	0.5119	0.5888	-	-	-	0.5119
		MS	0.5119	0.5160	0.5739	0.5127	-	-	0.5119
		TS	0.5119	0.5611	0.5877	0.5120	-	-	0.5119
		Surf	0.5119	0.5748	-	-	0.5117	0.5703	0.5119
	Lasso	Point	0.5119	0.5119	0.5888	-	-	-	0.5119
		MS	0.5119	0.5120	0.5723	0.5059	-	-	0.5119
		TS	0.5119	0.5543	0.5874	0.5120	-	-	0.5119
		Surf	0.5119	0.5710	-	-	0.5117	0.5703	0.5119
	RF	Point	0.5187	0.5945	0.5677	-	-	-	0.5769
		MS	0.5455	0.5666	0.5644	0.5114	-	-	0.5695
		TS	0.5298	0.5688	0.5660	0.5556	-	-	0.5740
		Surf	0.5536	0.5417	-	-	0.5231	0.5739	0.5643
	LSTM	Point	0.5705	-	-	-	-	-	0.5948
		MS	0.5817	-	-	0.5078	-	-	0.5806
		TS	0.5926	-	-	0.5725	-	-	0.5616
		Surf	0.5132	-	-	-	0.4925	-	0.5113
	Attention	Point	0.5995	-	-	-	-	-	0.5891
		MS	0.5731	-	-	0.5055	-	-	0.5698
		TS	0.5250	-	-	0.5684	-	-	0.5219
		Surf	0.5204	-	-	-	0.5600	0.5639	0.5198
	SVR	Point	0.6134	0.6057	0.5734	-	-	-	0.6098
		MS	0.5931	0.5718	0.5312	0.5094	-	-	0.5928
		TS	0.5906	0.5241	0.5171	0.5582	-	-	0.5888
		Surf	-	-	-	-	-	-	0.5123

TABLE 5.5: The table presents the average forecast accuracy of the directional change in volatility for call options on the out of sample dataset using the Dumas interpolation. The best models for each feature set are shown in bold.

The average directional accuracy (Equation (5.10)) of the best performing models for each point in the IVS over the nine year out-of-sample testing period is shown in appendix B. The Pesaran-Timmermann statistic confirms the statistical significance of the directional accuracy results for all models at the 1% level. Furthermore, every points directional accuracy is greater than 50% for most models. Hence, most of the models, on average, have learnt the correct direction of future volatility. The darker regions indicate higher directional accuracy. For the best performing model in directional accuracy, the Point-SVR model for the 20 day lagged feature input (Figure B.3c) has a more uniform directional accuracy across all moneyness and maturity levels with its highest performing region being the long term and medium term in the money (ITM) regions. A similar pattern can be seen with the best performing models for directional accuracy. TS-HAR (Figure B.3d), Surface-HAR (figure B.3a) and MS-SVR (figure B.2a) models only have small regions of lower directional accuracy, but on average perform well, with the best performing regions being ITM.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.5924	0.5937	0.5985	-	-	-	0.5860
		MS	0.5906	0.5708	0.5877	0.5825	-	-	0.5943
		TS	0.5962	0.5830	0.5923	0.5903	-	-	0.5969
		Surf	0.5684	0.5747	0.5795	-	0.5627	0.5723	0.5748
	Enet	Point	0.5094	0.5094	0.5985	-	-	-	0.5094
		MS	0.5094	0.5095	0.5737	0.5070	-	-	0.5094
		TS	0.5094	0.5497	0.5925	0.5094	-	-	0.5094
		Surf	0.5094	0.5722	0.5798	-	0.5093	0.5677	0.5094
	Lasso	Point	0.5094	0.5094	0.5985	-	-	-	0.5094
		MS	0.5094	0.5094	0.5736	0.5032	-	-	0.5094
		TS	0.5094	0.5407	0.5925	0.5094	-	-	0.5094
		Surf	0.5094	0.5710	0.5800	-	0.5093	0.5677	0.5094
	RF	Point	0.5850	0.5684	0.5674	-	-	-	0.5957
		MS	0.5927	0.5687	0.5791	0.5093	-	-	0.5913
		TS	0.5838	0.5784	0.5712	0.5719	-	-	0.5818
		Surf	0.5704	0.5629	0.5407	-	0.5183	0.5725	0.5686
	LSTM	Point	0.5975	-	-	-	-	-	0.6060
		MS	0.6015	-	-	0.5156	-	-	0.6090
		TS	0.5584	-	-	0.5782	-	-	0.5578
		Surf	0.5095	-	-	-	0.5119	-	0.5090
	Attention	Point	0.5602	-	-	-	-	-	0.5508
		MS	0.5567	-	-	0.5238	-	-	0.5461
		TS	0.5365	-	-	0.5877	-	-	0.5278
		Surf	0.5246	-	-	-	0.5628	0.5757	0.5277
	SVR	Point	0.6057	0.5986	0.5794	-	-	-	0.6043
		MS	0.6111	0.5775	0.5263	0.5156	-	-	0.6133
		TS	0.5963	0.5236	0.5149	0.5797	-	-	0.5984
		Surf	0.5169	0.5094	0.5094	-	0.5169	0.5449	0.5170
10	Ridge	Point	0.5964	0.5955	0.5949	-	-	-	0.5860
		MS	0.5950	0.5716	0.5789	0.5852	-	-	0.5943
		TS	0.5937	0.5816	0.5927	0.5910	-	-	0.5969
		Surf	0.5634	0.5751	0.5778	-	0.5655	0.5727	0.5748
	Enet	Point	0.5092	0.5092	0.5949	-	-	-	0.5094
		MS	0.5092	0.5104	0.5744	0.5061	-	-	0.5094
		TS	0.5092	0.5577	0.5923	0.5091	-	-	0.5094
		Surf	0.5092	0.5743	0.5779	-	0.5091	0.5681	0.5094
	Lasso	Point	0.5092	0.5092	0.5949	-	-	-	0.5094
		MS	0.5092	0.5092	0.5740	0.5051	-	-	0.5094
		TS	0.5092	0.5504	0.5915	0.5091	-	-	0.5094
		Surf	0.5092	0.5712	0.5780	-	0.5091	0.5681	0.5094
	RF	Point	0.5960	0.5833	0.5691	-	-	-	0.5957
		MS	0.5958	0.5672	0.5699	0.5092	-	-	0.5913
		TS	0.5833	0.5820	0.5712	0.5707	-	-	0.5818

	LSTM	Surf	0.5700	0.5600	0.5354	-	0.5183	0.5722	0.5686
		Point	0.5988	-	-	-	-	-	0.6060
		MS	0.6012	-	-	0.5111	-	-	0.6090
		TS	0.5625	-	-	0.5772	-	-	0.5578
	Attention	Surf	0.5099	-	-	-	0.5152	-	0.5090
		Point	0.5657	-	-	-	-	-	0.5508
		MS	0.5537	-	-	0.5197	-	-	0.5461
		TS	0.5308	-	-	0.5919	-	-	0.5278
	SVR	Surf	0.5273	-	-	-	0.5640	0.5748	0.5277
		Point	0.6058	0.5996	0.5786	-	-	-	0.6043
		MS	0.6134	0.5796	0.5272	0.5161	-	-	0.6133
		TS	0.5955	0.5240	0.5148	0.5784	-	-	0.5984
20	Ridge	Surf	0.5172	0.5092	0.5092	-	0.5172	0.5448	0.5170
		Point	0.5974	0.5961	0.5869	-	-	-	0.5860
		MS	0.5960	0.5713	0.5692	0.5850	-	-	0.5943
		TS	0.5905	0.5798	0.5816	0.5912	-	-	0.5969
	Enet	Surf	0.5615	0.5728	-	-	0.5707	0.5713	0.5748
		Point	0.5094	0.5094	0.5869	-	-	-	0.5094
		MS	0.5094	0.5146	0.5735	0.5057	-	-	0.5094
		TS	0.5094	0.5622	0.5816	0.5094	-	-	0.5094
	Lasso	Surf	0.5094	0.5723	-	-	0.5093	0.5665	0.5094
		Point	0.5094	0.5094	0.5869	-	-	-	0.5094
		MS	0.5094	0.5095	0.5733	0.5104	-	-	0.5094
		TS	0.5094	0.5555	0.5815	0.5094	-	-	0.5094
	RF	Surf	0.5094	0.5705	-	-	0.5093	0.5665	0.5094
		Point	0.5967	0.5877	0.5678	-	-	-	0.5957
		MS	0.5962	0.5670	0.5691	0.5090	-	-	0.5913
		TS	0.5826	0.5643	0.5710	0.5694	-	-	0.5818
	LSTM	Surf	0.5708	0.5439	-	-	0.5189	0.5715	0.5686
		Point	0.6043	-	-	-	-	-	0.6060
		MS	0.5959	-	-	0.5097	-	-	0.6090
		TS	0.5440	-	-	0.5693	-	-	0.5578
	Attention	Surf	0.5095	-	-	-	0.5063	-	0.5090
		Point	0.5535	-	-	-	-	-	0.5508
		MS	0.5437	-	-	0.5186	-	-	0.5461
		TS	0.5278	-	-	0.5935	-	-	0.5278
	SVR	Surf	0.5278	-	-	-	0.5788	0.5708	0.5277
		Point	0.6069	0.5987	0.5729	-	-	-	0.6043
		MS	0.6145	0.5797	0.5296	0.5173	-	-	0.6133
		TS	0.5931	0.5240	0.5153	0.5794	-	-	0.5984
		Surf	-	-	-	-	-	-	0.5170

TABLE 5.6: The table reports the average forecast accuracy of directional change in volatility for call options on the out of sample dataset using the Nadarya Watson interpolation. The best models for each feature set are shown in bold.

Moving forward, we compare the best performing models for each input feature set. For the point input feature, the best performing model proved to be the SAM encoder with the SVR predictor. The SAM encoder with SVR predictor is also the best performing for the MS input feature set. The SAM encoder with the ridge model is the best performing for the TS input feature set. The surface input feature set is more ambiguous. The HAR encoder with ridge predictor shows consistently better R^2 and RMSE results. However, the SAM encoder is slightly better in the DM statistic but the HAR encoder performs better in the $R^2_{\mathcal{M}1, \mathcal{M}2}$ statistic. A decision was made to use the HAR encoder with ridge predictor due to the consistently better average R^2 and RMSE results. Despite the linear predictors not performing well compared to the non-linear SVR predictor, the best performing linear predictors (SAM encoder with ridge predictor for both TS and MS) will still be compared for each lag length so that the best performing lag length can be determined for these models. This helps facilitate the comparison of feature importance later in section 5.3.2.

Forecast comparison across varying lags

In order to understand the effect of lags and determine the best performing lag length. We perform a comparative analysis of the best performing models for each input feature using the $R^2_{\mathcal{M}1, \mathcal{M}2}$ and DM statistics from equations (5.3) and (5.7), respectively. The results of this comparison are shown in tables 5.7–5.10. The HAR feature encoder uses all information up to 20 lags (c.f. Section 4.2.7) to make forecasts. Hence, there is no affect of changing the lags on the HAR encoder. However, the SAM encoders (c.f. Section 4.2.1) do get affected by the lags. Hence, we only compare the Point-SAM, Point-SVR, TS-SAM and MS-SVR models with varying lag lengths. For the tables lag lengths in the columns are compared against the lag lengths in the rows.

(A) DM statistic			(B) R^2 test statistic		
	Lag 10	Lag 5		Lag 10	Lag 5
Lag 20	5.9280 ^a	4.6741 ^a	Lag 20	0.0212 ^a	0.0627 ^a
Lag 10		4.0965 ^a	Lag 10		0.0429 ^a

TABLE 5.7: Comparative test across lags for the SAM encoder and Ridge predictor using **Point** IVS feature inputs. Positive value indicates the column beats the row. *a, b, c* denote significance at the 1%, 5%, 10% level, respectively.

(A) DM statistic			(B) R^2 test statistic		
	Lag 10	Lag 5		Lag 10	Lag 5
Lag 20	4.3329 ^a	3.2270 ^a	Lag 20	0.0130 ^a	0.0301 ^a
Lag 10		2.6431 ^a	Lag 10		0.0176 ^a

TABLE 5.8: Comparative test across lags for the SAM encoder and SVR predictor using **Point** IVS feature inputs. Positive value indicates the column beats the row. *a, b, c* denote significance at the 1%, 5%, 10% level, respectively.

For the point input feature matrix; the results in table 5.7 and table 5.8 show for decreasing lag lengths the R^2 and RMSE performance improves. The DM statistic comparing the RMSE of a lag length of 20 day to 10 and 5 days is positive. A positive loss differential (Equation (5.7)) indicates that the RMSE for lags 10 and 5 is lower than 20. Similarly, the RMSE for lag 5 is lower than lag 10, making the 5 lag model the best performing. The $R^2_{\mathcal{M}1\mathcal{M}2}$ statistic shows a similar pattern (Table 5.7 panel B). This result is in contrast to the work in [16], which showed that when predicting the IVS surface point by point, increasing the lag length improves the forecast performance.

(A) DM statistic			(B) R^2 test statistic		
	Lag 10	Lag 5		Lag 10	Lag 5
Lag 20	-4.4687 ^a	-11.9279 ^a	Lag 20	-0.0136 ^c	-0.0676
Lag 10		-10.9488 ^a	Lag 10		-0.0530 ^c

TABLE 5.9: Comparative test across lags for the SAM encoder and Ridge predictor using **Skew** IVS feature inputs. Negative value means row beats column. *a, b, c* denote significance at the 1%, 5%, 10% level, respectively.

The MS input feature matrix shows a different pattern. As shown in table 5.9 and table 5.10, the R^2 and RMSE performance improves for increasing lag lengths. The DM statistic result of -10.9488 in table 5.9 clearly shows that using a lag length of 10 days leads to better RMSE performance over using a lag length of 5 days. Comparing 5 lag length to 20 we see even better performance over 5 lags. When comparing the lag lengths of 10 and 20 days there is even further improvement. The $R^2_{\mathcal{M}1\mathcal{M}2}$ statistic shows similar results. However, the statistical significance of the $R^2_{\mathcal{M}1\mathcal{M}2}$ statistic is not as strong as the DM statistic. The DM statistic shows with statistical significance that a lag length of 20 days shows the best RMSE performance for the out of sample test period for both the ridge and SVR predictors.

(A) DM statistic			(B) R^2 test statistic		
	Lag 10	Lag 5		Lag 10	Lag 5
Lag 20	-11.4725 ^a	-2.8767 ^a	Lag 20	-0.0132	-0.0138
Lag 10		-0.1072	Lag 10		-0.0004 ^b

TABLE 5.10: Comparative test across lags for the SAM encoder and SVR predictor using **Skew** IVS feature inputs. Positive value indicates the column beats the row. *a, b, c* denote significance at the 1%, 5%, 10% level, respectively.

Comparison of the TS-SAM model with the ridge predictor and varying lags shows similar performance to the Point-SAM model. Both the $R^2_{\mathcal{M}1\mathcal{M}2}$ statistic (Table 5.11 in panel B) and DM statistic (Table 5.11 in panel A) show that 5 lag outperforms 10 and 20 lags. This is in contrast to the R^2 results in table 5.1 which shows that the TS-SAM model with 10 lags has the highest R^2 value of 0.8995. However, it follows the benchmark comparison results (Tables A.1–A.2) and the RMSE results (Table 5.3). Therefore,

statistically, using the $R^2_{\mathcal{M}1\mathcal{M}2}$ statistic we determine the lag lengths of 10 is the best performing for the term structure feature representation.

(A) DM statistic			(B) R^2 test statistic		
	Lag 10	Lag 5		Lag 10	Lag 5
Lag 20	8.8663 ^a	3.0353 ^a	Lag 20	0.0243 ^a	0.0252 ^a
Lag 10		0.1769	Lag 10		0.0012 ^a

TABLE 5.11: Comparative test across lags for the SAM encoder and Ridge predictor using **Term structure** IVS feature inputs. a, b, c denote significance at the 1%, 5%, 10% level, respectively.

Comparing best feature input models

Overall, the four best models for the four input feature matrices are: ① the SVR predictor and SAM encoder trained on 5 lag lengths (Point-SVR), ② the SVR predictor and SAM encoder trained on 20 lag lengths (MS-SVR), ③ the ridge regression predictor with SAM encoder trained on 5 lag lengths (TS-SAM), and ④ the ridge regression predictor with HAR encoder trained on 20 lag lengths (Surface-HAR). The comparison between these four models; using the $R^2_{\mathcal{M}1\mathcal{M}2}$ and DM statistic is shown in tables 5.12.

	MS-SVR	TS-SAM	Surface-HAR
Point-SVR	-0.040238 ^a	-0.052183 ^a	-0.095062 ^a
MS-SVR		-0.11483 ^a	-0.052704 ^a
TS-SAM			-0.040753 ^a

(A) Comparative R^2 results, comparing the best models for each feature input

	MS-SVR	TS-SAM	Surface-HAR
Point-SVR	-5.3797 ^a	-9.7223 ^a	-11.1843 ^a
MS-SVR		-1.1308	-6.7347 ^a
TS-SAM			-5.5685 ^a

(B) Comparative DM RMSE results, comparing the best models for each feature input

TABLE 5.12: Comparative DM RMSE and R^2 results, comparing across features for the best models including: Point-SVR, Moneyness MS-SVR, Term Structure (TS)-SAM, and Surface-HAR.

From table 5.12a, one can see that the Point-SVR model outperforms every other model, in terms of variation of the ground truth captured in the forecast. The Point-SVR model outperforms the MS-SVR model which is the next best performing model by 4.0238% with a significance level at the 1% level. If one were to use R^2 as the sole comparative measure; then for the ridge predictor, the models would be ranked as: ① Point-SVR, ② MS-SVR, ③ TS-SAM, and finally ④ Surface-HAR. The DM statistic further supports this ranking. These results contradict the general observation made

	MS-SVR	TS-SAM	Surface-HAR
Point-SVR	0.053690 ^a	-0.061876 ^a	-0.049106 ^a
MS-SVR		-0.122123 ^a	-0.108628 ^a
TS-SAM			0.012026 ^a

(A) Comparative R^2 results, comparing the best models for each feature input

	MS-SVR	TS-SAM	Surface-HAR
Point-SVR	9.1938 ^a	-10.5831 ^a	-5.8281 ^a
MS-SVR		-19.8588 ^a	-18.4575 ^a
TS-SAM			2.0608 ^b

(B) Comparative DM RMSE results, comparing the best models for each feature input

TABLE 5.13: Comparative DM RMSE and R^2 results, comparing across features for the best models including: Point-SVR, Moneyness MS-SVR, Term Structure (TS)-SAM, and Surface-HAR. For the put options

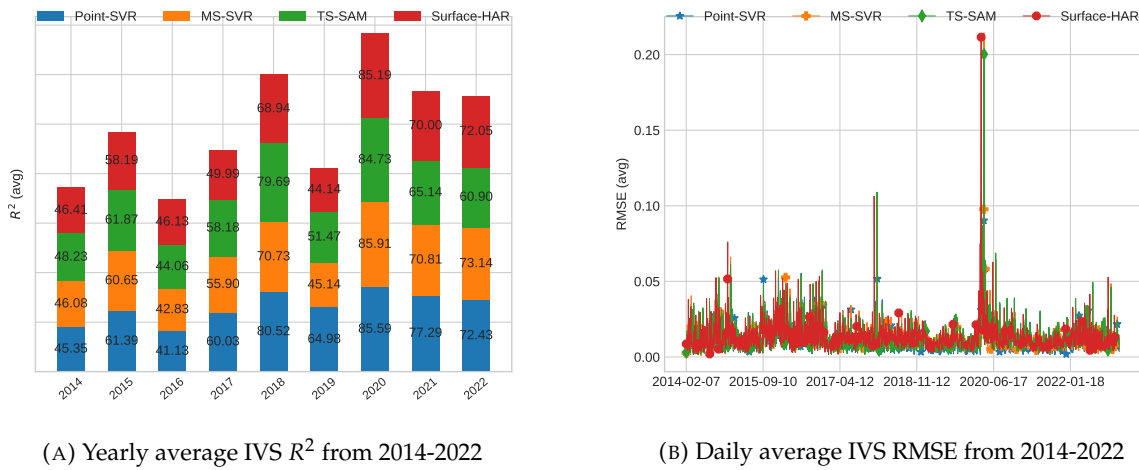
in Wen, Zhai, Wang, *et al.* [16] that linear models outperform non-linear models when predicting the IVS.

A similar comparative analysis can be performed for the lasso and elastic-net predictors. The comparative analysis is also shown in the tables A.1–A.2. The R^2 and RMSE is shown in the appendix in tables 5.1–5.3 however, the R^2 and RMSE values forecast from these predictors are poor. We can clearly see that the ridge predictor significantly outperforms the lasso and elastic-net predictors. In many cases, the R^2 results for the forecasts from the lasso and elastic-net predictors are negative. This is inline with the observation made in [16], when predicting single point on the IVS using the lasso and elastic-net predictors. Unlike lasso and elastic-net, ridge predictors retain all features. Ridge’s superior predictions indicate that most lagged features are informative.

Now that we have determined the best performing models; in the upcoming sections we will investigate: ① why are the predictions different, and ② how the lagged features influence the predictions for these models.

Forecast performance across years

In order to further investigate the performance differences between the best models for every input feature set, we consider their individual yearly forecast over the nine years in the out-of-sample testing set. Figure 5.1a shows the yearly average R^2 for the four different best performing models. All the best models show very similar yearly average R^2 performance. However, in some years (e.g., 2019) the MS-SVR and Surface-HAR models give a lower R^2 score. While the TS-SAM and Point-SVR models have a higher R^2 score, with the Point-SVR model showing a much higher R^2 by 13 points compared to the TS-SAM model. TS-SAM and Point-SVR models also underperform in some years, e.g., 2022 and 2016, respectively.

FIGURE 5.1: Average RMSE and R^2 over time from 2014-2022

Average forecast performance for entire test period from 2014-2022. Shown are the best performing models for each feature input for the ridge predictor.

Notably we see that during period of high volatility, such as 2018 and 2020 all models show much higher R^2 . The TS-SAM and Point-SVR models show more consistent performance across the years, with both models having greater than 50% R^2 for most years. The daily RMSE comparison for the four models are shown in figure 5.1b. Here to the RMSE magnitudes are similar. The Surface-HAR model and the TS-SAM model show higher sensitivity to spikes in the RMSE during periods of high volatility. Notably, the MS-SVR and Point-SVR models show much lower peaks and consistently lower RMSE values across the years.

Moneyneess and Time to maturity forecast comparisons

Models	MS-SVR			TS-SAM			Surface-HAR			
		ST	MT	LT	ST	MT	LT	ST	MT	LT
Point-SVR	ITM	-8.4574 ^a	-14.0978 ^a	-2.8234 ^a	-1.0988 ^a	-2.1633 ^b	-16.0159 ^b	-23.9045 ^a	-19.4063 ^a	-4.0825 ^a
	ATM	-1.3508 ^a	-0.0625 ^a	1.7436 ^a	3.8053 ^a	2.7886 ^a	-8.2821 ^b	-7.0899 ^a	-0.7826 ^a	-1.8223 ^a
	OTM	-2.7077	-1.7604 ^c	1.8207 ^a	3.7509 ^a	1.5789 ^a	-0.2516 ^a	-3.3024 ^b	-2.4312 ^a	-4.7244 ^a
MS-SVR	ITM				6.7453 ^a	10.4075 ^a	-12.9365 ^a	-14.2425 ^a	-4.6526	-1.2245 ^a
	ATM				5.0714 ^a	2.7942 ^a	-10.2504 ^b	-5.6627	-0.7196 ^a	-3.6292 ^b
	OTM				6.2605 ^a	3.2171 ^a	-2.1148 ^b	-0.5790 ^a	-0.6592 ^a	-6.6665 ^c
TS-SAM	ITM							-22.5059 ^a	-16.8096 ^a	10.3704 ^a
	ATM							-11.3075 ^b	-3.6148 ^b	6.0057 ^a
	OTM							-7.2963	-4.0052	-4.4575 ^a

TABLE 5.14: Table of comparative R^2 results in (%), comparing the best models performance for every term structure and moneyneess pair forecasting IVS out-of-sample. Negative values indicate the row outperforms the column. The inner column headers ST, MT, and LT are short-term, medium-term, and long-term maturities respectively. ^a, ^b, ^c denote significance at the 1%, 5%, 10% level, respectively.

The IVS can be differentiated into nine different regions depending upon the moneyneess and time to maturity. The moneyneess of the IVS is divided into: ① In-The-Money (ITM), ② At-The-Money (ATM), and ③ Out-The-Money (OTM) (c.f. Section 3.1.1).

For *each* of these moneyness regions, there are the time to maturity regions defined as: ① short term maturity, ② medium term maturity, and ③ long term maturity as defined in section 3.1. This section investigates the IVS region that is best forecasted by the four best models. The $R^2_{\mathcal{M}_1\mathcal{M}_2}$ statistic and its statistical significance (Equations (5.3) and (5.4)) are used to compare the four best forecast models. The results of comparisons are shown in table 5.14.

Firstly, one notes that the TS-SAM model shows a dominant performance for the short term and medium term to maturity options irrespective of the moneyness. The only exception being the Point-SVR model for ITM options. The $R^2_{\mathcal{M}_1\mathcal{M}_2}$ statistic shows this for both Point-SVR and MS-SVR at a 1% significance level. Compared to the Surface-HAR model, the statistical significance drops to 5% for the ATM and no statistical significance for the OTM options.

Notably, the Surface-HAR model outperforms the TS-SAM model for the both ITM and ATM options. Compared to the Point-SVR and the MS-SVR models, the Surface-HAR model is outperformed for all moneyness and time to maturity regions. Although, the results for the ATM and OTM options do not show any statistical significance for both the Point-SVR and MS-SVR comparisons. Overall, the MS-SVR model shows varying performance improvements across moneyness and time to maturity compared to the other models. It shows greater R^2 performance than the Point-SVR model for Long Term (LT), ATM and OTM options, although with statistical significance outside the 10% threshold. However, compared to the Surface-HAR model, the MS-SVR model shows superior performance for all regions of the IVS.

Models		MS-SVR			TS-SAM			Surface-HAR		
		ST	MT	LT	ST	MT	LT	ST	MT	LT
Point-SVR	ITM	-3.3008 ^a	-5.1490 ^a	-1.3146	-0.9810	-1.6021	-9.9627 ^a	-4.8280 ^a	-4.8074 ^a	-1.8279 ^c
	ATM	-0.2916	-0.0140	0.3545	0.8820	0.4973	-1.8260 ^c	-0.6391	-0.1034	-0.4524
	OTM	-0.6680	-0.4291	0.9559	1.0240	0.4707	-0.0702	-0.6143	-0.6303	-1.6426
MS-SVR	ITM				3.3660 ^a	4.5402 ^a	-5.0616 ^a	-4.7682 ^a	-2.2339 ^b	-0.9803
	ATM				0.9252	0.4247	-1.2188	-0.5908	-0.1128	-0.5708
	OTM				1.0500	0.5330	-0.4130	-0.0949	-0.1779	-1.7040 ^c
TS-SAM	ITM							-5.1744 ^a	-4.8710 ^a	5.1272 ^a
	ATM							-1.2510	-0.6671	1.7165 ^c
	OTM							-2.1370 ^b	-1.0315	-2.3434 ^b

TABLE 5.15: Table of comparative *RMSE* results, comparing the best models performance for every term structure and moneyness pair forecasting IVS out-of-sample. Negative values indicate the row outperforms the column. The inner column headers ST, MT, and LT are short-term, medium-term, and long-term maturities respectively. *a, b, c* denote significance at the 1%, 5%, 10% level, respectively.

The Point-SVR model shows the best performance for Short Term (ST), Medium Term (MT) and LT ITM options, outperforming all other models. The TS-SAM model however, shows higher $R^2_{\mathcal{M}_1\mathcal{M}_2}$ results for ST and MT ATM and OTM options. Given these results, a trading desk might prefer using the TS-SAM model for predicting and trading the liquid (short term, ATM) call options as substantially more variation of the IVS would be correctly predicted by the TS-SAM model. In the same vein someone

hedging or trading long term to maturity options should prefer the MS-SVR or Point-SVR model. The DM statistic results in table 5.15 reinforce these findings however, fewer values are statistically significant.

These results show that there is no single model best suited for predicting every region of the IVS. They also reinforce the findings in [24], and show that there is information in the whole IVS surface that needs to be considered when forecasting future IVS.

5.3.2 Function of features

In this section we study the effect of *individual* components of the feature matrices for the four best performing linear models. The ridge predictor assigns a weight value ($\mathbf{w}$) to each individual component of the feature matrix (c.f. Section 4.3). The higher the value of the weight for each individual feature, the more prominent its contribution to the forecast — the weight values give the feature importance of each individual feature from the feature set. For each of the four models, the worst and best performing IVS points are taken from figure B, and their individual feature importance is studied.

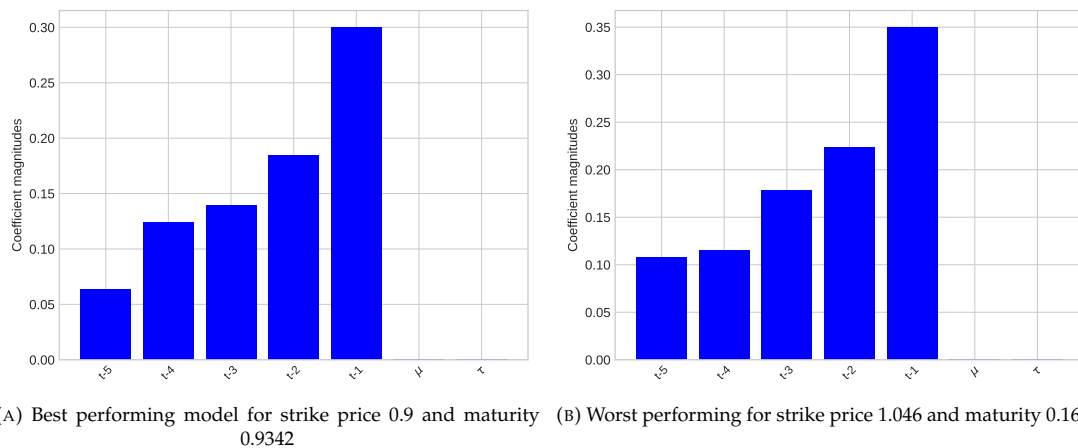


FIGURE 5.2: Ridge regression prediction model coefficient magnitude using the points of IVS as feature input and the SAM method of feature encoding.

The feature importance for Point-SAM model is shown in figure 5.2. The exogenous input features — moneyness and term structure do not play any role in the forecast, seen from the zero weight given to each of these features of the input set (Equation (4.1)). The most recent lagged IVS feature has the largest influence in determining the one-day-ahead forecast. The role of other features reduces as the lags start to increase. This is as expected for both the best case and the worst case scenario.

The feature importance for the MS-SAM model shows a much more interesting picture. The feature influence in this case is not linear. Figure 5.3a gives the feature importance of the lagged feature set (Equation (4.2)) for forecast at maturity (0.9973, ITM). The one-day prior feature vector, around this moneyness, has the largest contribution. However, note that this ITM forecast is also affected significantly by ATM (near 1.0) and

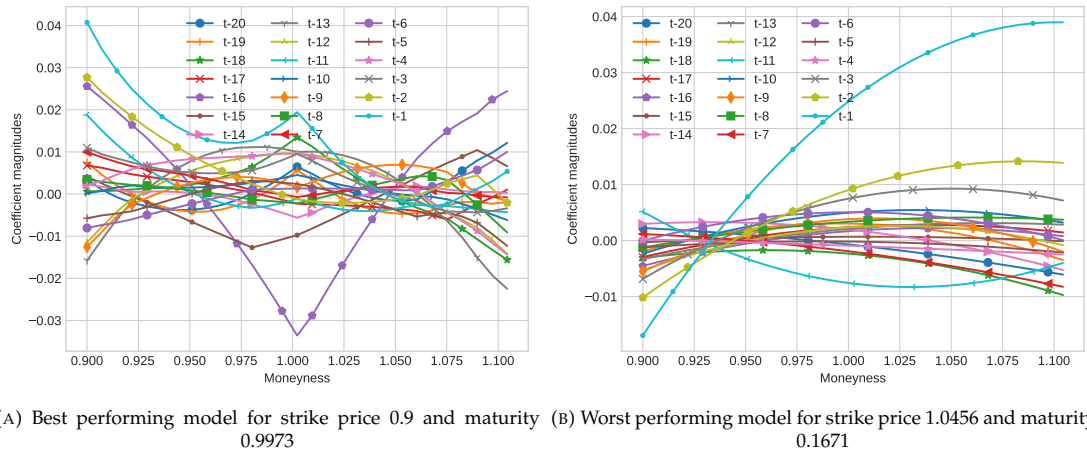


FIGURE 5.3: Ridge regression prediction model coefficient magnitude using the moneyness skew of the IVS as feature input and the HAR method of feature encoding.

OTM (near 1.1) lagged moneyness skew slices. This is evidence for one of our main research questions (c.f. Section 1.2, (2)) that different moneyness values encode different information and the whole moneyness skew is needed for improved prediction of future moneyness skew.

Firstly, comparing the influence of the lagged features. The point showing the best out of sample performance, figure 5.3a, has a non-linear distribution of feature importance across the lags. Most notably, for the ITM region (moneyness 0.9), the one-day prior lag has the largest influence, followed by the two day. We would then expect the three-day prior lag to have the next largest weighting, similar to the Point-SAM model. However, instead we see the 16-day prior lag has the next largest weighting, followed by the 10-day prior lag. This shows that information as far as 16 days prior can be useful for increasing the forecast accuracy of the IVS. The trend continues for other regions of the moneyness skew, such as the ATM region (moneyness 1.0) and the OTM region (moneyness 1.1), where the 16-day prior lag has the largest influence, compared to the other lags for that region. This is in contrast to the worst performing case, figure 5.3b, where we see a much more linear distribution of feature importance across the lags. As one would expect, the largest influence is from the one-day prior lag, followed by the two-day prior lag, and then the three-day prior lag. For lags further than three days, the weighting values drop below 0.01 and most lags have little to no influence on the forecast.

Now looking at the weight distribution across the moneyness skew. The best performing case clearly shows that information from the entire skew is needed for improved forecasting accuracy. The best performing point, which is ITM (moneyness 0.9), gives the largest weighting to the correction ITM point of 0.9. However, it also gives significant weightings to the ATM and OTM regions, with several lags showing spikes at the ITM region. Most notably a large negative spike at the 16-day prior lagged feature vector with almost as much influence as the 1-day prior ITM feature. The worst performing point is OTM (moneyness 1.0456) and much like the lags, shows a much

more linear distribution of feature importance across the moneyness skew. The largest weighting is given to the OTM region, but it does not quite correctly assign the largest weighting to the 1.0456 moneyness. Unlike the best performing case, the rest of the moneyness skew is not given much importance, especially for longer lags. This shows further evidence that the moneyness skew encodes different information and that the entire skew is needed for improved forecasting accuracy. This also opens further scope for improving the weight assignment optimisation problem (c.f. Section 4.3) in the future.

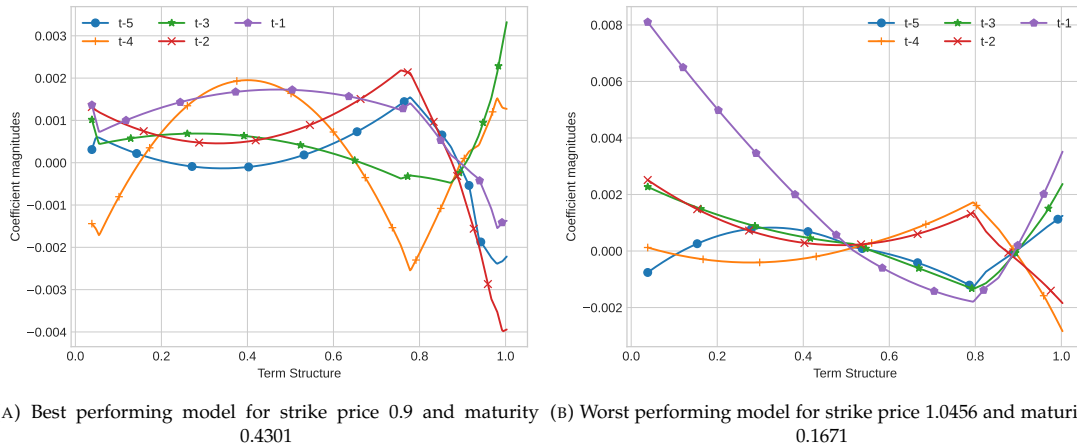


FIGURE 5.4: Ridge regression prediction model coefficient magnitude using the term structure of the IVS as feature input and the SAM method of feature encoding.

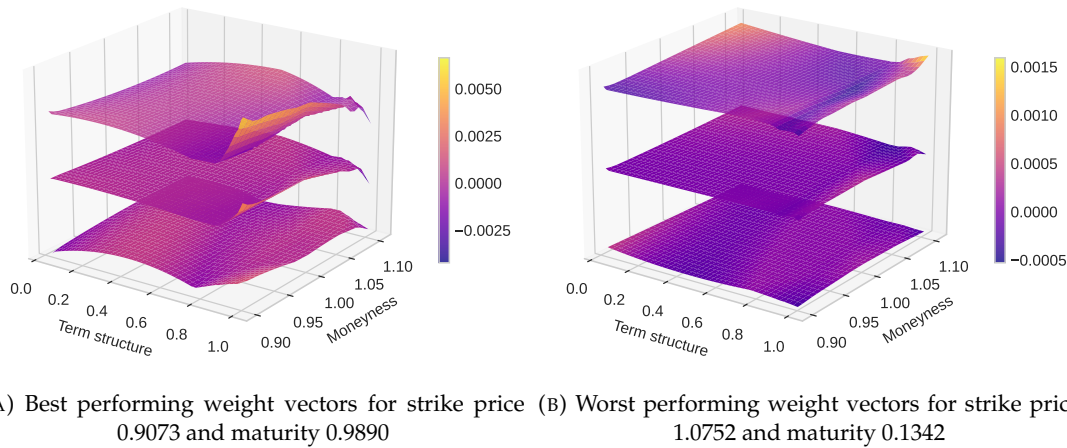


FIGURE 5.5: Ridge regression prediction model coefficient magnitude using the surface of the IVS as feature input and HAR method of feature encoding. Surfaces are stacked for better visibility and implied volatility values are shown via the heat map.

Any HAR model encodes the lagged feature set into a three feature vector (equation (4.15)); the one-day prior lagged value, the average of the 5-day prior lagged values, and the average of the 20-day prior lagged values. The ridge predictor then assigns a weight to each of these features for forecast. The importance of the three encoded features for the Surface-HAR model is shown in figure 5.5. It is much more difficult to

interpret the feature importance for the Surface-HAR model. Paying attention to the magnitudes of the colour bar for each plot, we see a similar distribution of weights as for the TS-SAM and MS-SAM models. The best performing point, figure 5.5a, correctly assigns the largest weighting to the in-the-money long term maturity region (strike 0.9073, maturity 0.9890). There is also a large portion of the medium term maturity region with a significant weighting. The largest weighting for the worst performing point, figure 5.5b, is approximately 0.0012 and the largest weighting for the best performing point is approximately 0.005. There is also an order of magnitude more influence given to negative weights for the best performing point compared to the worst performing point, much like the MS-SAM example.

The TS-SAM and Surface-HAR best and worst case feature importances; Figures 5.4 and 5.5, respectively, follow a similar pattern to the MS-HAR model. ① The feature importances are non-linear, ② the one-day prior feature vector has the largest impact, and ③ the ridge predictor assigns the largest weights for the correct maturity and moneyness for the best forecast outcome and leaves a lot to be desired for the worst forecast outcome. It is interesting to note that for MS-SAM, TS-SAM and Surface-HAR, the best performing points have significant negative weighting in areas further from the corresponding strike/maturity being forecasted. Although, in the worst performing cases the regressor may correctly identify the strike/maturity and assign greater weighting. Comparatively, it sets the weights of the rest of the lags and term-structure/moneyness close to zero, thereby leading to worse overall performance. We also note that the best performing cases distribute the weight more uniformly across the entire term structure which shows further evidence, as suggested by Guo, Han, and Lin [24], that the entire term structure contains information for forecasting future IVS. This can even be extended to previous lags, as can be seen in 5.3a feature vectors as far as 20 lags prior having a significant weighting and the feature vector for lag 4 having even higher weighting for the corresponding maturity than the feature vector for lag of 1.

5.3.3 Robustness Check

The quantitative comparison until section 5.3.2 has been between different forecast models using the call option IVS. In this section we perform robustness checks by carrying out the comparison of the forecast models for the put option IVS. Three performance metrics are shown: ① R^2 statistic (Equation (5.2)), ② RMSE statistic (Equation (5.6)), and ③ directional accuracy. The comparative $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ and RMSE statistics are shown in the appendix in tables 5.16–5.17. For the purposes of the robustness check only the results for the Dumas interpolation technique are shown.

The comparisons for all forecast models are shown in table 5.16–5.18, respectively. These results are obtained by performing one-day ahead IVS forecast for the nine-year test period from 2014–2022 for the put options IVS using the Dumas et al. interpolation method. As we can see from table 5.16, the HAR encoder with ridge predictor outperforms every other model in terms of the R^2 statistic for the surface and TS features. It is able to capture more of the variation of the true IVS verses any of the other forecast

models. For the point and MS feature, the SAM encoder with SVR predictor manages to outperform the HAR encoder with a very small margin. The RMSE results show the same results.

The results follow from the call options IVS results, the only difference being that the HAR encoding method outperforms the SAM encoding method for the ridge predictor model. For the point feature the SAM model with SVR predictor is the best performing model for the R^2 statistic, and the best performing models for the RMSE statistic. The same diminishing returns for increased lags is also observed for the Point-SAM ridge model. In fact for most features and encoders, the 5 lag length is the best performing for the R^2 and RMSE statistic. The best performing model for the MS features is the SAM encoder with SVR predictor at 20 lags and the best performing model for the TS features is the HAR encoder with ridge predictor at 20 lags. The overall trend is similar for the put options IVS. However, the HAR encoding technique seems to perform better for the put options IVS than the call options IVS compared to the other encoding techniques.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.8755	0.8666	0.8652	-	-	-	0.8757
		MS	0.8633	0.7604	0.8210	0.8391	-	-	0.8712
		TS	0.8787	0.8454	0.8588	0.8651	-	-	0.8817
		Surf	0.8636	0.7477	0.8041	-	0.7562	0.8315	0.8819
	Enet	Point	-0.5795	-0.5795	0.8652	-	-	-	-0.5832
		MS	-0.5795	-0.5538	0.7553	-3.2361	-	-	-0.5832
		TS	-0.5795	0.7706	0.8584	-0.5828	-	-	-0.5832
		Surf	-0.5795	0.7393	0.8041	-	-0.5676	0.7956	-0.5832
	Lasso	Point	-0.5795	-0.5795	0.8652	-	-	-	-0.5832
		MS	-0.5795	-0.5795	0.7547	-4.4513	-	-	-0.5832
		TS	-0.5795	0.7418	0.8575	-0.5828	-	-	-0.5832
		Surf	-0.5795	0.7369	0.8041	-	-0.5676	0.7956	-0.5832
	RF	Point	0.6248	0.8542	0.7745	-	-	-	0.8714
		MS	0.6109	0.7736	0.8127	-2.4904	-	-	0.8288
		TS	0.6274	0.8178	0.7783	0.6884	-	-	0.8557
		Surf	0.6368	0.7001	0.6866	-	-1.3633	0.8248	0.6540
	LSTM	Point	0.8614	-	-	-	-	-	0.8851
		MS	0.8736	-	-	-3.3978	-	-	0.8847
		TS	0.8724	-	-	0.0001	-	-	0.8560
		Surf	-0.3283	-	-	-	-658.5656	-	-0.5557
	Attention	Point	0.8834	-	-	-	-	-	0.8612
		MS	0.8523	-	-	-69.4151	-	-	0.8549
		TS	0.7833	-	-	0.8679	-	-	0.8398
		Surf	-28.6152	-	-	-	0.7586	0.8317	-1.6445
	SVR	Point	0.8877	0.8842	0.8601	-	-	-	0.8845
		MS	0.8888	0.7986	0.5628	-81.9202	-	-	0.8934
		TS	0.8762	0.1129	-0.6425	0.8173	-	-	0.8718

		Surf	-9.5873	-0.5734	-0.5789	-	-10.0051	0.5742	-9.5491
10	Ridge	Point	0.8701	0.8414	0.8274	-	-	-	0.8757
		MS	0.8616	0.7420	0.7767	0.8278	-	-	0.8712
		TS	0.8713	0.8429	0.8264	0.8555	-	-	0.8817
		Surf	0.8501	0.7446	0.7763	-	0.7541	0.8252	0.8819
	Enet	Point	-0.5799	-0.5799	0.8274	-	-	-	-0.5832
		MS	-0.5799	-0.3584	0.7211	-5.1879	-	-	-0.5832
		TS	-0.5799	0.7503	0.8263	-0.5833	-	-	-0.5832
		Surf	-0.5799	0.7247	0.7762	-	-0.5679	0.7881	-0.5832
	Lasso	Point	-0.5799	-0.5799	0.8274	-	-	-	-0.5832
		MS	-0.5799	-0.5799	0.7165	-5.6120	-	-	-0.5832
		TS	-0.5799	0.7356	0.8261	-0.5833	-	-	-0.5832
		Surf	-0.5799	0.7121	0.7762	-	-0.5679	0.7881	-0.5832
	RF	Point	0.6247	0.8536	0.7281	-	-	-	0.8714
		MS	0.6105	0.7665	0.7675	-2.5753	-	-	0.8288
		TS	0.6282	0.7846	0.7503	0.6457	-	-	0.8557
		Surf	0.6354	0.5484	0.6734	-	-1.5299	0.8168	0.6540
	LSTM	Point	0.8395	-	-	-	-	-	0.8851
		MS	0.8800	-	-	-2.5973	-	-	0.8847
		TS	0.8774	-	-	0.0373	-	-	0.8560
		Surf	-0.3564	-	-	-	-406.7992	-	-0.5557
	Attention	Point	0.8812	-	-	-	-	-	0.8612
		MS	0.7368	-	-	-56.3769	-	-	0.8549
		TS	0.7864	-	-	0.8530	-	-	0.8398
		Surf	-6.8038	-	-	-	0.3388	0.8242	-1.6445
	SVR	Point	0.8870	0.8848	0.8210	-	-	-	0.8845
		MS	0.8918	0.7941	0.5343	-125.0652	-	-	0.8934
		TS	0.8748	0.1013	-0.6441	0.7935	-	-	0.8718
		Surf	-9.6219	-0.5746	-0.5793	-	-8.4295	0.5665	-9.5491
20	Ridge	Point	0.8654	0.8121	0.7564	-	-	-	0.8757
		MS	0.8655	0.7398	0.7103	0.8271	-	-	0.8712
		TS	0.8669	0.8413	0.7944	0.8496	-	-	0.8817
		Surf	0.8377	0.7433	-	-	0.7523	0.8240	0.8819
	Enet	Point	-0.5832	-0.5832	0.7564	-	-	-	-0.5832
		MS	-0.5832	0.0474	0.6715	-6.4358	-	-	-0.5832
		TS	-0.5832	0.6914	0.7939	-0.5865	-	-	-0.5832
		Surf	-0.5832	0.7236	-	-	-0.5709	0.7865	-0.5832
	Lasso	Point	-0.5832	-0.5832	0.7564	-	-	-	-0.5832
		MS	-0.5832	-0.5662	0.6479	-7.4536	-	-	-0.5832
		TS	-0.5832	0.6847	0.7933	-0.5865	-	-	-0.5832
		Surf	-0.5832	0.7047	-	-	-0.5709	0.7865	-0.5832
	RF	Point	0.6258	0.8474	0.6515	-	-	-	0.8714
		MS	0.6093	0.7488	0.6755	-2.4448	-	-	0.8288
		TS	0.6244	0.7271	0.7205	0.6016	-	-	0.8557
		Surf	0.6280	0.5422	-	-	-1.6806	0.8165	0.6540
	LSTM	Point	0.8213	-	-	-	-	-	0.8851

	MS	0.8577	-	-	-2.7826	-	-	0.8847
	TS	0.8655	-	-	0.0108	-	-	0.8560
	Surf	-0.3433	-	-	-	-677.2849	-	-0.5557
Attention	Point	0.8651	-	-	-	-	-	0.8612
	MS	0.8375	-	-	-166.8052	-	-	0.8549
	TS	0.7289	-	-	0.8750	-	-	0.8398
	Surf	-0.9697	-	-	-	0.6162	0.8209	-1.6445
SVR	Point	0.8868	0.8807	0.7491	-	-	-	0.8845
	MS	0.8940	0.7917	0.4859	-144.0717	-	-	0.8934
	TS	0.8734	0.1021	-0.6502	0.7735	-	-	0.8718
	Surf	-	-	-	-	-	-	-9.5491

TABLE 5.16: The table presents the average R^2 statistic for put contracts on the out of sample dataset using the Dumas interpolation. The best models for each feature set are shown in bold.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.0176	0.0184	0.0185	-	-	-	0.0177
		MS	0.0181	0.0237	0.0208	0.0200	-	-	0.0177
		TS	0.0172	0.0199	0.0189	0.0183	-	-	0.0171
		Surf	0.0182	0.0245	0.0217	-	0.0243	0.0196	0.0171
	Enet	Point	0.0648	0.0648	0.0185	-	-	-	0.0650
		MS	0.0648	0.0641	0.0240	0.1055	-	-	0.0650
		TS	0.0648	0.0242	0.0189	0.0649	-	-	0.0650
		Surf	0.0648	0.0250	0.0217	-	0.0647	0.0215	0.0650
	Lasso	Point	0.0648	0.0648	0.0185	-	-	-	0.0650
		MS	0.0648	0.0648	0.0240	0.1173	-	-	0.0650
		TS	0.0648	0.0259	0.0190	0.0649	-	-	0.0650
		Surf	0.0648	0.0252	0.0217	-	0.0647	0.0215	0.0650
	RF	Point	0.0311	0.0192	0.0242	-	-	-	0.0181
		MS	0.0314	0.0237	0.0219	0.0960	-	-	0.0208
		TS	0.0309	0.0212	0.0233	0.0270	-	-	0.0187
		Surf	0.0295	0.0269	0.0276	-	0.0787	0.0200	0.0275
	LSTM	Point	0.0185	-	-	-	-	-	0.0170
		MS	0.0175	-	-	0.1073	-	-	0.0169
		TS	0.0179	-	-	0.0503	-	-	0.0189
		Surf	0.0595	-	-	-	0.9816	-	0.0645
	Attention	Point	0.0170	-	-	-	-	-	0.0189
		MS	0.0189	-	-	0.4254	-	-	0.0189
		TS	0.0228	-	-	0.0180	-	-	0.0200
		Surf	0.2358	-	-	-	0.0247	0.0196	0.0745
	SVR	Point	0.0166	0.0169	0.0189	-	-	-	0.0170
		MS	0.0165	0.0222	0.0343	0.4575	-	-	0.0162
		TS	0.0175	0.0482	0.0655	0.0216	-	-	0.0179
		Surf	0.1585	0.0647	0.0648	-	0.1616	0.0310	0.1585
10	Ridge	Point	0.0181	0.0204	0.0213	-	-	-	0.0177

		MS	0.0181	0.0248	0.0237	0.0207	-	-	0.0177
		TS	0.0176	0.0201	0.0211	0.0191	-	-	0.0171
		Surf	0.0188	0.0246	0.0236	-	0.0245	0.0200	0.0171
	Enet	Point	0.0649	0.0649	0.0213	-	-	-	0.0650
		MS	0.0649	0.0591	0.0262	0.1285	-	-	0.0650
		TS	0.0649	0.0255	0.0212	0.0649	-	-	0.0650
		Surf	0.0649	0.0258	0.0236	-	0.0647	0.0220	0.0650
	Lasso	Point	0.0649	0.0649	0.0213	-	-	-	0.0650
		MS	0.0649	0.0649	0.0264	0.1288	-	-	0.0650
		TS	0.0649	0.0263	0.0212	0.0649	-	-	0.0650
		Surf	0.0649	0.0266	0.0236	-	0.0647	0.0220	0.0650
	RF	Point	0.0312	0.0193	0.0268	-	-	-	0.0181
		MS	0.0314	0.0242	0.0247	0.0972	-	-	0.0208
		TS	0.0309	0.0232	0.0251	0.0288	-	-	0.0187
		Surf	0.0296	0.0333	0.0283	-	0.0814	0.0205	0.0275
	LSTM	Point	0.0201	-	-	-	-	-	0.0170
		MS	0.0172	-	-	0.0976	-	-	0.0169
		TS	0.0175	-	-	0.0493	-	-	0.0189
		Surf	0.0601	-	-	-	0.7222	-	0.0645
	Attention	Point	0.0173	-	-	-	-	-	0.0189
		MS	0.0258	-	-	0.3825	-	-	0.0189
		TS	0.0226	-	-	0.0191	-	-	0.0200
		Surf	0.1236	-	-	-	0.0412	0.0200	0.0745
	SVR	Point	0.0167	0.0169	0.0217	-	-	-	0.0170
		MS	0.0163	0.0226	0.0354	0.5644	-	-	0.0162
		TS	0.0177	0.0486	0.0655	0.0231	-	-	0.0179
		Surf	0.1588	0.0648	0.0649	-	0.1497	0.0313	0.1585
20	Ridge	Point	0.0185	0.0222	0.0256	-	-	-	0.0177
		MS	0.0180	0.0249	0.0276	0.0209	-	-	0.0177
		TS	0.0179	0.0202	0.0232	0.0195	-	-	0.0171
		Surf	0.0196	0.0248	-	-	0.0248	0.0201	0.0171
	Enet	Point	0.0650	0.0650	0.0256	-	-	-	0.0650
		MS	0.0650	0.0494	0.0291	0.1407	-	-	0.0650
		TS	0.0650	0.0286	0.0232	0.0651	-	-	0.0650
		Surf	0.0650	0.0260	-	-	0.0649	0.0221	0.0650
	Lasso	Point	0.0650	0.0650	0.0256	-	-	-	0.0650
		MS	0.0650	0.0646	0.0301	0.1476	-	-	0.0650
		TS	0.0650	0.0290	0.0232	0.0651	-	-	0.0650
		Surf	0.0650	0.0271	-	-	0.0649	0.0221	0.0650
	RF	Point	0.0312	0.0199	0.0306	-	-	-	0.0181
		MS	0.0315	0.0253	0.0295	0.0956	-	-	0.0208
		TS	0.0311	0.0263	0.0268	0.0308	-	-	0.0187
		Surf	0.0299	0.0338	-	-	0.0840	0.0206	0.0275
	LSTM	Point	0.0215	-	-	-	-	-	0.0170
		MS	0.0190	-	-	0.1003	-	-	0.0169
		TS	0.0184	-	-	0.0499	-	-	0.0189
		Surf	0.0599	-	-	-	1.0730	-	0.0645

Attention	Point	0.0187	-	-	-	-	-	0.0189
	MS	0.0201	-	-	0.6580	-	-	0.0189
	TS	0.0255	-	-	0.0176	-	-	0.0200
	Surf	0.0652	-	-	-	0.0317	0.0202	0.0745
SVR	Point	0.0168	0.0173	0.0260	-	-	-	0.0170
	MS	0.0162	0.0228	0.0373	0.6065	-	-	0.0162
	TS	0.0178	0.0486	0.0658	0.0243	-	-	0.0179
	Surf	-	-	-	-	-	-	0.1585

TABLE 5.17: The table presents the average RMSE statistic for put contracts on the out of sample dataset using the Dumas interpolation. The best models for each feature set are shown in bold.

Directional accuracy results in figure 5.18 reinforce the results in table 5.5 for call options. The SAM encoder method with SVR predictor outperforms in the point feature for directional accuracy. For the best performing model on the MS feature, the HAR encoder with the SVR predictor very narrowly outperforms the same predictor with the SAM encoder for 20 lags. The results of the PT test (Equation 5.11) can be viewed in the appendix (Figures B.4–B.5). They tally with the results in table 5.18. The darker areas, in figure B.6a, clearly show that the Surface-HAR predictor gets a larger area of the IVS correct directionally, on average, than the other predictors. However, there is more variation, the lighter area shows a much lower directional accuracy score compared to other models.

Overall, the results of the robustness check show similar trends to the call options IVS results. The best performing models for R^2 and RMSE are the same as for the call options IVS besides the TS feature where the HAR encoder with ridge predictor outperforms the SAM encoder with ridge predictor. The results do however, show a lot more variation in the directional accuracy results for the put options IVS where more lackluster predictor models such as the LSTM showed the best directional accuracy for the TS feature. As for the other features similar results are observed to the call options IVS results.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.6040	0.6042	0.6057	-	-	-	0.5994
		MS	0.5818	0.5701	0.5881	0.5666	-	-	0.5897
		TS	0.5910	0.5916	0.5998	0.5963	-	-	0.5973
		Surf	0.5748	0.5687	0.5807	-	0.5619	0.5783	0.5933
	Enet	Point	0.5219	0.5219	0.6057	-	-	-	0.5218
		MS	0.5219	0.5220	0.5712	0.5074	-	-	0.5218
		TS	0.5219	0.5691	0.6000	0.5218	-	-	0.5218
		Surf	0.5219	0.5704	0.5808	-	0.5225	0.5685	0.5218
	Lasso	Point	0.5219	0.5219	0.6057	-	-	-	0.5218
		MS	0.5219	0.5219	0.5714	0.5049	-	-	0.5218
		TS	0.5219	0.5601	0.5995	0.5218	-	-	0.5218
		Surf	0.5219	0.5701	0.5808	-	0.5225	0.5685	0.5218

	RF	Point	0.5176	0.5796	0.5739	-	-	-	0.6049
		MS	0.5324	0.5566	0.5767	0.5103	-	-	0.5709
		TS	0.5368	0.5786	0.5768	0.5703	-	-	0.5890
		Surf	0.5478	0.5614	0.5626	-	0.5151	0.5758	0.5551
	LSTM	Point	0.5901	-	-	-	-	-	0.6062
		MS	0.5892	-	-	0.5231	-	-	0.5906
		TS	0.5989	-	-	0.5220	-	-	0.5785
		Surf	0.5288	-	-	-	0.5089	-	0.5224
	Attention	Point	0.5976	-	-	-	-	-	0.5995
		MS	0.5585	-	-	0.5055	-	-	0.5569
		TS	0.5256	-	-	0.5716	-	-	0.5276
		Surf	0.5044	-	-	-	0.5324	0.5787	0.5167
	SVR	Point	0.6136	0.6088	0.6027	-	-	-	0.6144
		MS	0.5941	0.5641	0.5422	0.5064	-	-	0.6122
		TS	0.5809	0.5329	0.5251	0.5797	-	-	0.5817
		Surf	0.5131	0.5219	0.5219	-	0.5146	0.5335	0.5132
10	Ridge	Point	0.6049	0.6022	0.6011	-	-	-	0.5994
		MS	0.5814	0.5723	0.5878	0.5666	-	-	0.5897
		TS	0.5901	0.5933	0.5934	0.5989	-	-	0.5973
		Surf	0.5761	0.5670	0.5813	-	0.5673	0.5797	0.5933
	Enet	Point	0.5217	0.5217	0.6011	-	-	-	0.5218
		MS	0.5217	0.5233	0.5753	0.5014	-	-	0.5218
		TS	0.5217	0.5765	0.5941	0.5217	-	-	0.5218
		Surf	0.5217	0.5717	0.5813	-	0.5224	0.5694	0.5218
	Lasso	Point	0.5217	0.5217	0.6011	-	-	-	0.5218
		MS	0.5217	0.5217	0.5748	0.5018	-	-	0.5218
		TS	0.5217	0.5715	0.5944	0.5217	-	-	0.5218
		Surf	0.5217	0.5718	0.5813	-	0.5224	0.5694	0.5218
	RF	Point	0.5176	0.5868	0.5760	-	-	-	0.6049
		MS	0.5324	0.5657	0.5797	0.5090	-	-	0.5709
		TS	0.5367	0.5866	0.5797	0.5703	-	-	0.5890
		Surf	0.5474	0.5456	0.5668	-	0.5145	0.5770	0.5551
	LSTM	Point	0.5598	-	-	-	-	-	0.6062
		MS	0.6017	-	-	0.5096	-	-	0.5906
		TS	0.6094	-	-	0.5243	-	-	0.5785
		Surf	0.5280	-	-	-	0.4881	-	0.5224
	Attention	Point	0.5953	-	-	-	-	-	0.5995
		MS	0.5579	-	-	0.5140	-	-	0.5569
		TS	0.5342	-	-	0.5598	-	-	0.5276
		Surf	0.5111	-	-	-	0.5256	0.5778	0.5167
	SVR	Point	0.6166	0.6113	0.5994	-	-	-	0.6144
		MS	0.6045	0.5671	0.5448	0.5047	-	-	0.6122
		TS	0.5845	0.5338	0.5251	0.5781	-	-	0.5817
		Surf	0.5133	0.5218	0.5217	-	0.5146	0.5349	0.5132
20	Ridge	Point	0.6040	0.6017	0.5972	-	-	-	0.5994
		MS	0.5824	0.5707	0.5791	0.5706	-	-	0.5897

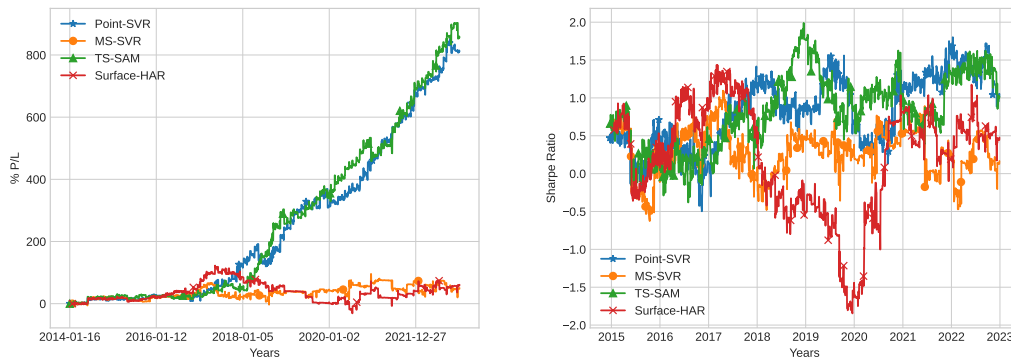
	TS	0.5854	0.5921	0.5906	0.5975	-	-	0.5973
	Surf	0.5786	0.5686	-	-	0.5729	0.5777	0.5933
Enet	Point	0.5218	0.5218	0.5972	-	-	-	0.5218
	MS	0.5218	0.5292	0.5786	0.4973	-	-	0.5218
	TS	0.5218	0.5790	0.5910	0.5218	-	-	0.5218
	Surf	0.5218	0.5706	-	-	0.5224	0.5667	0.5218
Lasso	Point	0.5218	0.5218	0.5972	-	-	-	0.5218
	MS	0.5218	0.5219	0.5759	0.4980	-	-	0.5218
	TS	0.5218	0.5751	0.5912	0.5218	-	-	0.5218
	Surf	0.5218	0.5715	-	-	0.5224	0.5667	0.5218
RF	Point	0.5177	0.5932	0.5755	-	-	-	0.6049
	MS	0.5320	0.5691	0.5773	0.5110	-	-	0.5709
	TS	0.5365	0.5780	0.5745	0.5700	-	-	0.5890
	Surf	0.5456	0.5490	-	-	0.5134	0.5757	0.5551
LSTM	Point	0.5723	-	-	-	-	-	0.6062
	MS	0.5867	-	-	0.5138	-	-	0.5906
	TS	0.5925	-	-	0.5208	-	-	0.5785
	Surf	0.5283	-	-	-	0.5039	-	0.5224
Attention	Point	0.5998	-	-	-	-	-	0.5995
	MS	0.5669	-	-	0.4938	-	-	0.5569
	TS	0.5286	-	-	0.5709	-	-	0.5276
	Surf	0.5199	-	-	-	0.5305	0.5753	0.5167
SVR	Point	0.6189	0.6085	0.5909	-	-	-	0.6144
	MS	0.6122	0.5688	0.5474	0.5112	-	-	0.6122
	TS	0.5838	0.5339	0.5251	0.5763	-	-	0.5817
	Surf	-	-	-	-	-	-	0.5132

TABLE 5.18: The table presents the average forecast accuracy of the directional change in volatility for put contracts using the Dumas interpolation. The best models for each feature set are shown in bold.

Chapter 6

Economic Significance

Results in chapter 5 show that different IV prediction models are necessary to forecast the call option IVS, and the Point-SVR model is, in general, the best predictor for the call option IVS. In this chapter the forecast models are *applied* to evaluate them from a trading desk's perspective. Economic measures are used to test if the models can generate abnormal profits. A simple algorithm is also developed to trade the *call* option IVS. We purposely choose call option IVS as a test, to see if the statistically over-performing Point-SVR model, also over-performs for economic activity.



(A) Portfolio growth over the nine year out-of-sample test period for the call option IVS trading strategy using different forecast models.

(B) Annualized rolling Sharpe ratio over out of sample test period for the call option IVS trading strategy using different forecast

FIGURE 6.1: Trading results — Growth of portfolio and rolling Sharpe ratio for volatility trading on the call option IVS for out-of-sample test period: 2014–2022.

The trading strategy is as follows: ① using the best models, predict the one-day ahead IVS. ② Take the *single* IVS point, with moneyness m_i and time to maturity τ_j , with the greatest absolute change compared to today's IVS for each model. ③a If the largest change is greater than 1% and positive, then buy (go long) a volatility contract — buy 1 call option and 1 put option at moneyness m_i and maturity τ_j — a so-called straddle. ③b If the largest change is greater than 1% and negative, then sell (go short) a volatility contract — sell 1 call option and 1 put option at moneyness m_i and maturity τ_j . ④ If the largest change is less than 1%, then do not trade. ⑤ The straddle position is held until maturity or until the position is held for a predefined maximum number of days, provided there are no abnormal profits greater than 5% while holding the position. Profit is harvested as soon as the straddle profit crosses the 5% mark, else profit

or loss (P/L) is harvested at maturity or the maximum number of days. ⑥ We consider 1.5% of the option price as the transaction cost for each option buy/sell [77].

Models	Point-SVR	MS-SVR	TS-SAM	Surface-HAR	Point-AR(1)	MS-AR(1)	TS-AR(1)	Surface-AR(1)
# Trades	690	694	621	686	678	706	553	667
Longs Pos	438	613	434	603	640	608	442	566
Shorts Pos	252	81	187	83	38	98	111	101
Long Wins (%)	73.2	67.2	69.3	68.3	70.6	67.2	69.0	64.6
Short Wins (%)	74.2	58.0	82.3	59.0	84.2	46.9	89.1	51.4
Wins (%)	73.6	66.1	73.2	67.2	71.3	64.4	73.0	62.6
Max. win (\$)	27,915.27	18,520.90	24,851.60	20,083.89	26,903.21	21,603.76	30,178.39	17,028.10
Max. loss (\$)	-22,124.24	-29,923.74	-28,922.34	-33,268.26	-35,847.88	-37,893.26	-23,061.62	-19,479.02
Avg. win (\$)	1,170.81	72.77	1,348.77	88.78	160.26	-55.60	1,303.08	-85.63
Median win (\$)	367.85	243.78	395.89	243.05	397.90	240.92	414.99	203.66
std (\$)	4567.1	4007.7	5422.9	4461.6	5768.2	4597.6	5347.9	2399.4
Alpha (%)	22.9	-0.4	26.9	1.5	11.3	-inf	20.6	-inf
Beta	0.242	0.281	-0.502	0.014	-1.432	-1.659	0.488	0.489
CAGR (%)	27.8	4.6	28.5	5.4	8.8	-5.3	26.6	-8.9
Avg. SR	0.826	0.282	0.824	0.210	0.377	0.138	0.911	-0.510
Alpha gain on AR(1)	11.54	-	6.32	-	-	-	-	-

TABLE 6.1: Trading statistics

Trade algorithm statistics. The volatility trading algorithm on the call option IVS for nine years out-of-sample test period. A 1.5% of the option price transaction cost is considered for each option buy/sell. SR stands for Sharpe-Ratio. Gain on alpha refers to the models gain on the benchmark AR(1) model. Benchmark models are shown bellow the midrule

The results of applying this simple trading strategy are shown in figure 6.1 and table 6.1. Over the nine years out-of-sample test period, each forecast model results in a different number of trades (Table 6.1, row-1). This is expected, due to the difference in the forecast performance of each model. Out of these trades, the number of wins, after accounting for trading costs, is largest for the Point-SVR and TS-SAM model. This is also expected, because the Point-SVR model has one of the best directional accuracy results (Table 5.5) for the call option IVS and the TS-SAM model shows the best out-of-sample forecast performance for short term and medium term at the money options. The win percentage is calculated as the number of straddle positions that are closed with a positive profit and loss (P/L) divided by the total number of trades. One will notice that although the TS-SAM, PM-SVR and Surface-HAR models have similar win percentages, their alpha percentages are quite different. This is likely due to the nature of the trading strategy employed. Large movements in the Implied Volatility can either result in large profits in the case of a long straddle positions, or large losses in the case of short straddle positions. The answer to the difference in alpha is the average win profit shown in table 6.1. One can see that both the MS-SVR and Surface-HAR average win profit is very low compared to the TS-SAM and PM-SVR models. This indicates that these two models are more capable of capturing the large movements in the IVS and hence are able to generate larger profits and greater alpha values. Looking at the average directional accuracy results (Table 5.5), we would expect that the Surface-HAR would have some of the greatest excess profits over other models. However, notice

the Peseran-Timmermann heat map (Figure B.1) shows that the Surface-HAR model performs best for the long term to maturity In-The-Money options which are not as sensitive to volatility shocks in the underlying market as shorter dated options which the TS-SAM model captures more effectively.

The alpha value of the trading strategy shows the *excess* profit a strategy can generate over the S&P 500 buy-hold strategy. A higher value of alpha is better. We see that TS-SAM model gives the highest alpha and outperforms all the other models. The high alpha indicates that the TS-SAM model along with the applied trading strategy produces the highest excess returns. This is further emphasized with the TS-SAM model showing one of the highest win ratios, sharp ratio and the greatest Compounded Annual Growth Rate (CAGR) of 73.2%, 0.82 and 28.5% respectively. This indicates that the TS-SAM model is giving the most consistent excess profit.

Sharpe-Ratio captures the excess profit of the strategy verses the risk-free rate (10 year bond yield) accounting for the risk taken. The calculated Sharpe-Ratio is annualized over 252 trading days. A Sharpe-ratio of greater than 1 is considered good, above 2 outstanding, and below 1 average. The rolling Sharpe-ratio for the different forecast models applied to the trading strategy are shown in figure 6.1b. A similar trend can be seen for the two best performing models, Point-SVR and TS-SAM. At the beginning of the out of sample period both models struggle dropping close to or below 0 Sharpe-ratio. This is expected as the yearly R^2 forecast results shown in figure 5.1a show significantly worse performance for the first 3 years compared to the last 6 years. The Surface-HAR model for the given trading strategy seems to struggle during high volatility periods from 2018 to 2020 where Sharpe-ratio drops as low as -1.75. On the other hand the TS-SAM and Point-SVR model show consistent positive Sharpe-ratio after the first three years. The economic benefits agree with the statistical results. Hence, the best predictors, TS-SAM or Point-SVR, should be used for trading the call option IVS. Results for the put options are shown in the appendix C. The put trading results are similar in that the best performing models are the TS-SAM and Point-SVR models. Interestingly, all models show a positive alpha value. Overall the win ratios are lower for the put options compared to the call options, and all models show fairly high average win profits. One would expect due to put-call parity that the models would perform similarly for both call and put options.

Chapter 7

Conclusion

In this thesis, several different models are compared for forecasting the options Implied Volatility Surface (IVS). Different features of the IVS are extracted and encoded with well known data-driven and non-parametric models. New encoding and feature reduction techniques are also applied: Canonical Covariance Analysis (CCA), Surface Stochastic Volatility Inspired (SSVI), and Heterogenous Autoregressive (HAR) to the extracted IVS features. A comparison of different forecast models is performed using three different statistical measures: R^2 , RMSE, and directional accuracy. Similarly to Guo, Han, and Lin [24], the R^2_{M1M2} is used for statistical comparisons between models and between the models and a benchmark model. The Diebold and Mariano statistic is also used for the same purpose. The out-of-sample tests show that, for S&P index options; different regions of call option IVS require different forecast models. The same can be said for different regions of the put options IVS.

Several dimensions of the IVS are investigated in this thesis. ① A comparative analysis of 7 different encoding techniques and 7 different predictors is performed, introducing methods not previously applied in this context: Canonical Covariance Analysis (CCA), Surface Stochastic Volatility Inspired (SSVI), and Heterogenous Autoregressive (HAR). The results are in contrast to the findings of Wen, Zhai, Wang, *et al.* [16], that simple linear models tend to outperform non-linear models, with the support vector regression (SVR) model being the best performing predictor overall and the attention model also showing strong performance compared to the simple linear models. Out of the non-linear models, the SVR model is the best performing, showing much better performance than the ridge regression model for the point and moneyness skew feature set. Additionally, the results show that simply sampling the IVS leads to the best performance overall. Notably, the HAR encoding technique which is a fairly parsimonious technique that effectively captures short, medium and long term temporal information, shows comparable performance to the SAM encoding technique.

② The work of Wen, Zhai, Wang, *et al.* [16] is extended in this thesis by showing that the past dependence of the Implied Volatility Surface is not only applicable to point by point prediction, but also to predicting entire cross-sections of the IVS in both moneyness and time to maturity as well as the entire surface. The results contradict the findings of Wen, Zhai, Wang, *et al.*, for the out-of-sample period, in that increasing lags past 5 days does not improve forecasting accuracy for point by point prediction. However, when predicting the skew and term structure, increasing lags up to 20 and 10 days respectively, can improve performance. The results are further supported by

the robustness tests, that shows the HAR encoding technique with the ridge regression predictor, is the best performing for the term structure and surface feature sets. While the SVR model is the best performing for the point and moneyness skew feature sets.

③ A complex relationship among all points of the IVS is shown by analysing the feature importance of the best performing ridge regression model. The results extend upon the findings of Guo, Han, and Lin [24], by showing that not only is information in the term structure of the IVS important but also the skew. This indicates that the entire IVS surface should be considered when making predictions. A comparison of the forecast performance for each region of the IVS is performed, which shows that different regions of the IVS require different forecast models.

④ Finally the best performing models are evaluated using a simple volatility trading strategy. The results show that the predictive forecast models developed in this work are applicable to significant economic activity. Various economic measures indicate that excess profits can be generated using the developed predictive forecast models over the benchmark S&P index.

7.1 Future work

It is clear that the IVS has complex relationships between all points of the surface. Future work should focus on the development of more complex models that can capture these relationships. The results show that non-linear models such as attention based models and support vector machines can show superior performance. Recent work has also shown that ensemble neural networks with increased complexity can show high out of sample performance [78]. Future work should consider the use of more complex models such as ensemble neural networks that can capture the complex relationships between all points of the IVS surface.

The results of this paper are limited to the S&P index options and show that the simple models developed in this work are applicable to significant economic activity. Future work should look at whether these results are generalizable to other derivative options markets, such as forex, and single stock options.

Additionally, results for directional accuracy are not as strong as the R^2 and RMSE measures. For the purpose of trading, the directional accuracy is far more important, and with most models struggling to achieve more than 60% directional accuracy, there is a lot more to be desired in this area. With additional work on specifically forecasting the change in the IVS and not the absolute value, it is possible that the directional accuracy can be improved.

Appendix A

Comparative out of sample forecasting results

The following tables contain the supplementary results of the $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ (5.3) and DM (5.7) statistics for the out-of-sample forecasting of the models described in Chapter 4.3. The results are presented for both call and put options contracts where each model is compared against the corresponding benchmark AR(1) model for the feature set. In other words, the models trained on the point features are compared against the AR(1) model trained on the point features, and the models trained on the skew features are compared against the AR(1) model trained on the skew features etc.

A.1 Call Options Contracts

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.186 ^a	0.147 ^a	0.127 ^a	-	-	-	0.151 ^a
		MS	0.965	0.957	0.957	0.959	-	-	0.968
		TS	0.257^a	0.140 ^a	0.172 ^a	0.171 ^a	-	-	0.254 ^a
		Surf	0.206 ^a	-0.032 ^a	0.049 ^a	-	-0.377 ^a	-0.120 ^a	0.279^a
	Enet	Point	-12.802 ^a	-12.802 ^a	0.127 ^a	-	-	-	-12.782 ^a
		MS	0.481	0.495	0.949	-0.075	-	-	0.483
		TS	-11.771 ^b	-0.798	0.167 ^a	-11.772 ^b	-	-	-11.747 ^b
		Surf	-10.894 ^a	-0.157 ^a	0.048 ^a	-	-10.814 ^a	-0.232 ^b	-10.903 ^a
	Lasso	Point	-12.802 ^a	-12.802 ^a	0.127 ^a	-	-	-	-12.782 ^a
		MS	0.481	0.481	0.949	-0.276	-	-	0.483
		TS	-11.771 ^b	-1.151	0.163 ^a	-11.772 ^b	-	-	-11.747 ^b
		Surf	-10.894 ^a	-0.263 ^a	0.047 ^a	-	-10.814 ^a	-0.232 ^b	-10.903 ^a
	RF	Point	-1.776 ^a	-0.001 ^a	-0.741 ^a	-	-	-	-0.113 ^a
		MS	0.893	0.940	0.940	-0.533	-	-	0.952
		TS	-1.659 ^b	-0.105 ^a	-0.279 ^a	-0.687 ^a	-	-	-0.189 ^a
		Surf	-1.393 ^a	-1.292 ^a	-0.680	-	-4.854	-0.130 ^a	-0.572 ^b
	LSTM	Point	0.080 ^a	-	-	-	-	-	0.153 ^a
		MS	0.964	-	-	-0.812	-	-	0.967
		TS	0.123 ^a	-	-	0.029 ^a	-	-	0.031 ^a
		Surf	-10.165 ^a	-	-	-	-174.712	-	-11.078 ^a

10	Attention	Point	0.223 ^a	-	-	-	-	-	-0.022 ^a
		MS	0.958	-	-	-4.285	-	-	0.956
		TS	-1.988 ^a	-	-	-0.190 ^a	-	-	-1.522 ^a
		Surf	-2.371 ^a	-	-	-	-0.422 ^a	-0.015 ^a	-12.240 ^a
	SVR	Point	0.237^a	0.228 ^a	0.049 ^a	-	-	-	0.196 ^a
		MS	0.970	0.957	0.853	-5.405	-	-	0.970
		TS	0.162 ^a	-5.525	-11.826 ^b	-0.452 ^a	-	-	0.152 ^a
		Surf	-59.866 ^a	-10.836 ^a	-10.889 ^a	-	-64.678 ^a	-0.924 ^a	-60.147 ^a
	Ridge	Point	0.150 ^a	0.028 ^a	-0.112 ^a	-	-	-	0.151 ^a
		MS	0.967	0.955	0.949	0.961	-	-	0.968
		TS	0.257 ^a	0.121 ^a	0.024 ^a	0.135 ^a	-	-	0.254 ^a
		Surf	0.233 ^a	-0.052 ^a	-0.042 ^a	-	-0.446 ^a	-0.097 ^a	0.279^a
	Enet	Point	-12.782 ^a	-12.782 ^a	-0.112 ^a	-	-	-	-12.782 ^a
		MS	0.482	0.562	0.942	-0.144	-	-	0.483
		TS	-11.751 ^b	-0.820	0.020 ^a	-11.752 ^b	-	-	-11.747 ^b
		Surf	-10.882 ^a	-0.211 ^a	-0.042 ^a	-	-10.801 ^a	-0.234 ^b	-10.903 ^a
	Lasso	Point	-12.782 ^a	-12.782 ^a	-0.112 ^a	-	-	-	-12.782 ^a
		MS	0.482	0.482	0.940	-0.265	-	-	0.483
		TS	-11.751 ^b	-1.013	0.017 ^a	-11.752 ^b	-	-	-11.747 ^b
		Surf	-10.882 ^a	-0.348 ^a	-0.042 ^a	-	-10.801 ^a	-0.234 ^b	-10.903 ^a
	RF	Point	-1.771 ^a	0.018 ^a	-0.980 ^a	-	-	-	-0.113 ^a
		MS	0.892	0.931	0.909	-1.171	-	-	0.952
		TS	-1.669 ^b	-0.374 ^a	-0.565 ^a	-0.950 ^a	-	-	-0.189 ^a
		Surf	-1.429 ^a	-1.615 ^a	-0.853	-	-5.215 ^c	-0.105 ^a	-0.572 ^b
	LSTM	Point	-0.140 ^a	-	-	-	-	-	0.153 ^a
		MS	0.965	-	-	0.041	-	-	0.967
		TS	0.166 ^a	-	-	-0.134 ^a	-	-	0.031 ^a
		Surf	-10.258 ^a	-	-	-	-1120.667 ^c	-	-11.078 ^a
20	Attention	Point	0.182 ^a	-	-	-	-	-	-0.022 ^a
		MS	0.927	-	-	-22.968	-	-	0.956
		TS	-0.982 ^a	-	-	0.191 ^a	-	-	-1.522 ^a
		Surf	-2.519 ^a	-	-	-	-1.322 ^a	-0.014 ^a	-12.240 ^a
	SVR	Point	0.223 ^a	0.219 ^a	-0.200 ^a	-	-	-	0.196 ^a
		MS	0.970	0.956	0.847	-6.669	-	-	0.970
		TS	0.161 ^a	-5.469	-11.802 ^c	-0.599 ^a	-	-	0.152 ^a
		Surf	-59.899 ^a	-10.824 ^a	-10.877 ^a	-	-64.844 ^a	-0.924 ^a	-60.147 ^a
	Ridge	Point	0.132 ^a	-0.050 ^a	-0.604 ^a	-	-	-	0.151 ^a
		MS	0.968	0.955	0.930	0.960	-	-	0.968
		TS	0.239 ^a	0.118 ^a	-0.213 ^a	0.113 ^a	-	-	0.254 ^a
		Surf	0.228 ^a	-0.056 ^a	-	-	-0.381 ^a	-0.084 ^a	0.279^a
	Enet	Point	-12.782 ^a	-12.782 ^a	-0.604 ^a	-	-	-	-12.782 ^a
		MS	0.483	0.658	0.928	-0.321	-	-	0.483
		TS	-11.747 ^b	-1.199	-0.218 ^a	-11.747 ^b	-	-	-11.747 ^b
		Surf	-10.903 ^a	-0.288 ^a	-	-	-10.820 ^a	-0.256 ^b	-10.903 ^a
	Lasso	Point	-12.782 ^a	-12.782 ^a	-0.604 ^a	-	-	-	-12.782 ^a
		MS	0.483	0.494	0.923	-0.576	-	-	0.483
		TS	-11.747 ^b	-1.299	-0.223 ^a	-11.747 ^b	-	-	-11.747 ^b

RF	Surf	-10.903 ^a	-0.507 ^a	-	-	-10.820 ^a	-0.256 ^b	-10.903 ^a
	Point	-1.769 ^a	-0.031 ^a	-1.514 ^a	-	-	-	-0.113 ^a
	MS	0.895	0.921	0.879	-1.046	-	-	0.952
	TS	-1.683 ^a	-0.605 ^a	-0.876 ^a	-1.247 ^a	-	-	-0.189 ^a
LSTM	Surf	-1.397 ^a	-2.973 ^a	-	-	-5.346 ^b	-0.106 ^a	-0.572 ^b
	Point	-0.234 ^a	-	-	-	-	-	0.153 ^a
	MS	0.961	-	-	-0.859	-	-	0.967
	TS	0.119 ^a	-	-	-0.069 ^a	-	-	0.031 ^a
Attention	Surf	-10.058 ^a	-	-	-	-5431.872 ^b	-	-11.078 ^a
	Point	0.065 ^a	-	-	-	-	-	-0.022 ^a
	MS	0.949	-	-	-14.230	-	-	0.956
	TS	-1.214 ^a	-	-	0.001 ^a	-	-	-1.522 ^a
SVR	Surf	-11.389 ^a	-	-	-	-0.156 ^a	-0.091 ^a	-12.240 ^a
	Point	0.214 ^a	0.211 ^a	-0.701 ^a	-	-	-	0.196 ^a
	MS	0.970	0.956	0.836	-14.093	-	-	0.970
	TS	0.163 ^a	-5.370	-11.803 ^c	-0.635 ^a	-	-	0.152 ^a
	Surf	-	-	-	-	-	-	-60.147 ^a

TABLE A.1: The table reports the $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ statistic for the out of sample dataset on *call options* using the *Dumas interpolation*. *a, b, c* denote statistical significance at the 1%, 5%, and 10% levels respectively.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	32.910 ^a	23.444 ^a	21.612 ^a	-	-	-	26.400 ^a
		MS	2.000 ^b	1.982 ^b	1.983 ^b	1.987 ^b	-	-	1.997 ^b
		TS	40.238^a	19.751 ^a	24.614 ^a	27.830 ^a	-	-	36.859 ^a
		Surf	30.055^a	-7.038 ^a	5.164 ^a	-	-21.451 ^a	-8.322 ^a	28.032 ^a
	Enet	Point	-114.093 ^a	-114.093 ^a	21.612 ^a	-	-	-	-113.171 ^a
		MS	0.996	1.024	1.966 ^b	-0.155	-	-	0.996
		TS	-111.927 ^a	-44.430 ^a	24.115 ^a	-111.889 ^a	-	-	-111.031 ^a
		Surf	-122.542 ^a	-38.098 ^a	5.105 ^a	-	-125.962 ^a	-12.165 ^a	-121.507 ^a
	Lasso	Point	-114.093 ^a	-114.093 ^a	21.612 ^a	-	-	-	-113.171 ^a
		MS	0.996	0.996	1.965 ^b	-0.574	-	-	0.996
		TS	-111.927 ^a	-53.765 ^a	23.421 ^a	-111.889 ^a	-	-	-111.031 ^a
		Surf	-122.542 ^a	-53.670 ^a	5.061 ^a	-	-125.962 ^a	-12.165 ^a	-121.507 ^a
	RF	Point	-71.780 ^a	-0.260	-143.935 ^a	-	-	-	-18.023 ^a
		MS	1.849 ^c	1.947 ^c	1.947 ^c	-1.103	-	-	1.964 ^b
		TS	-108.494 ^a	-19.184 ^a	-50.225 ^a	-81.676 ^a	-	-	-31.285 ^a
		Surf	-65.396 ^a	-51.936 ^a	-63.881 ^a	-	-181.584 ^a	-8.715 ^a	-45.329 ^a
	LSTM	Point	21.798 ^a	-	-	-	-	-	23.027 ^a
		MS	1.996 ^b	-	-	-1.681 ^c	-	-	1.995 ^b
		TS	21.933 ^a	-	-	3.960 ^a	-	-	4.619 ^a
		Surf	-116.091 ^a	-	-	-	-43.244 ^a	-	-125.074 ^a
	Attention	Point	37.117 ^a	-	-	-	-	-	-2.873 ^a
		MS	1.984 ^b	-	-	-8.874 ^a	-	-	1.972 ^b

		TS	-50.815 ^a	-	-	-19.280 ^a	-	-	-37.849 ^a
		Surf	-40.665 ^a	-	-	-	-28.737 ^a	-1.389	-34.305 ^a
	SVR	Point	40.759^a	39.270 ^a	7.203 ^a	-	-	-	32.321 ^a
		MS	2.008^b	1.983 ^b	1.767 ^c	-11.214 ^a	-	-	2.002 ^b
		TS	28.960 ^a	-98.575 ^a	-101.247 ^a	-64.055 ^a	-	-	25.070 ^a
		Surf	-138.696 ^a	-122.635 ^a	-122.564 ^a	-	-231.813 ^a	-36.967 ^a	-142.924 ^a
10	Ridge	Point	24.244 ^a	3.232 ^a	-14.877 ^a	-	-	-	26.400 ^a
		MS	2.001 ^b	1.975 ^b	1.964 ^b	1.987 ^b	-	-	1.997 ^b
		TS	37.488 ^a	15.540 ^a	2.849 ^a	19.346 ^a	-	-	36.859 ^a
		Surf	27.712 ^a	-12.498 ^a	-3.756 ^a	-	-25.210 ^a	-7.414 ^a	28.032 ^a
	Enet	Point	-114.186 ^a	-114.186 ^a	-14.877 ^a	-	-	-	-113.171 ^a
		MS	0.996	1.162	1.950 ^c	-0.297	-	-	0.996
		TS	-112.011 ^a	-46.248 ^a	2.422 ^b	-111.968 ^a	-	-	-111.031 ^a
		Surf	-122.671 ^a	-45.607 ^a	-3.777 ^a	-	-126.187 ^a	-12.410 ^a	-121.507 ^a
	Lasso	Point	-114.186 ^a	-114.186 ^a	-14.877 ^a	-	-	-	-113.171 ^a
		MS	0.996	0.996	1.944 ^c	-0.548	-	-	0.996
		TS	-112.011 ^a	-51.669 ^a	2.063 ^b	-111.968 ^a	-	-	-111.031 ^a
		Surf	-122.671 ^a	-66.725 ^a	-3.785 ^a	-	-126.187 ^a	-12.410 ^a	-121.507 ^a
	RF	Point	-72.053 ^a	3.075 ^a	-160.390 ^a	-	-	-	-18.023 ^a
		MS	1.845 ^c	1.927 ^c	1.881 ^c	-2.419 ^b	-	-	1.964 ^b
		TS	-109.528 ^a	-40.116 ^a	-83.439 ^a	-80.191 ^a	-	-	-31.285 ^a
		Surf	-65.288 ^a	-52.062 ^a	-93.842 ^a	-	-174.752 ^a	-7.564 ^a	-45.329 ^a
	LSTM	Point	-38.184 ^a	-	-	-	-	-	23.027 ^a
		MS	1.996 ^b	-	-	0.084	-	-	1.995 ^b
		TS	25.359 ^a	-	-	-20.661 ^a	-	-	4.619 ^a
		Surf	-120.792 ^a	-	-	-	-55.690 ^a	-	-125.074 ^a
	Attention	Point	29.181 ^a	-	-	-	-	-	-2.873 ^a
		MS	1.917 ^c	-	-	-47.635 ^a	-	-	1.972 ^b
		TS	-33.147 ^a	-	-	24.658 ^a	-	-	-37.849 ^a
		Surf	-42.040 ^a	-	-	-	-64.264 ^a	-1.322	-34.305 ^a
	SVR	Point	36.752 ^a	35.955 ^a	-24.767 ^a	-	-	-	32.321 ^a
		MS	2.006 ^b	1.978 ^b	1.753 ^c	-13.826 ^a	-	-	2.002 ^b
		TS	27.159 ^a	-98.095 ^a	-101.358 ^a	-93.242 ^a	-	-	25.070 ^a
		Surf	-143.188 ^a	-122.727 ^a	-122.693 ^a	-	-263.262 ^a	-37.861 ^a	-142.924 ^a
20	Ridge	Point	21.777 ^a	-4.804 ^a	-52.184 ^a	-	-	-	26.400 ^a
		MS	1.997 ^b	1.970 ^b	1.919 ^c	1.982 ^b	-	-	1.997 ^b
		TS	34.810 ^a	15.717 ^a	-21.774 ^a	16.219 ^a	-	-	36.859 ^a
		Surf	24.187 ^a	-9.984 ^a	-	-	-23.333 ^a	-6.859 ^a	28.032 ^a
	Enet	Point	-113.171 ^a	-113.171 ^a	-52.184 ^a	-	-	-	-113.171 ^a
		MS	0.996	1.358	1.916 ^c	-0.662	-	-	0.996
		TS	-111.031 ^a	-56.342 ^a	-22.746 ^a	-110.933 ^a	-	-	-111.031 ^a
		Surf	-121.507 ^a	-52.361 ^a	-	-	-125.012 ^a	-12.981 ^a	-121.507 ^a
	Lasso	Point	-113.171 ^a	-113.171 ^a	-52.184 ^a	-	-	-	-113.171 ^a
		MS	0.996	1.019	1.905 ^c	-1.188	-	-	0.996
		TS	-111.031 ^a	-58.113 ^a	-23.625 ^a	-110.933 ^a	-	-	-111.031 ^a
		Surf	-121.507 ^a	-72.518 ^a	-	-	-125.012 ^a	-12.981 ^a	-121.507 ^a
	RF	Point	-71.492 ^a	-4.339 ^a	-163.709 ^a	-	-	-	-18.023 ^a

	MS	1.846 ^c	1.901 ^c	1.814 ^c	-2.156 ^b	-	-	1.964 ^b
	TS	-107.849 ^a	-44.116 ^a	-74.809 ^a	-96.164 ^a	-	-	-31.285 ^a
	Surf	-65.253 ^a	-57.733 ^a	-	-	-161.857 ^a	-7.580 ^a	-45.329 ^a
LSTM	Point	-39.368 ^a	-	-	-	-	-	23.027 ^a
	MS	1.982 ^b	-	-	-1.772 ^c	-	-	1.995 ^b
	TS	18.635 ^a	-	-	-13.341 ^a	-	-	4.619 ^a
	Surf	-119.454 ^a	-	-	-	-43.252 ^a	-	-125.074 ^a
Attention	Point	9.180 ^a	-	-	-	-	-	-2.873 ^a
	MS	1.959 ^c	-	-	-29.415 ^a	-	-	1.972 ^b
	TS	-40.137 ^a	-	-	0.179	-	-	-37.849 ^a
	Surf	-34.578 ^a	-	-	-	-12.056 ^a	-6.917 ^a	-34.305 ^a
SVR	Point	34.687 ^a	34.267 ^a	-59.213 ^a	-	-	-	32.321 ^a
	MS	2.002 ^b	1.974 ^b	1.726 ^c	-29.159 ^a	-	-	2.002 ^b
	TS	26.791 ^a	-97.899 ^a	-101.220 ^a	-103.249 ^a	-	-	25.070 ^a
	Surf	-	-	-	-	-	-	-142.924 ^a

TABLE A.2: The table reports the Diebold and Mariano (DM) statistic for the out of sample dataset on *call options* using the *Dumas interpolation*. *a, b, c* denote statistical significance at the 1%, 5%, and 10% levels respectively.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.169 ^a	0.117 ^a	0.099 ^a	-	-	-	0.129 ^a
		MS	0.240 ^a	-0.056 ^a	0.139 ^a	0.144 ^a	-	-	0.262 ^a
		TS	0.939 ^a	0.915 ^a	0.934 ^a	0.927 ^a	-	-	0.939 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
	Enet	Point	-12.256 ^a	-12.256 ^a	0.099 ^a	-	-	-	-12.239 ^a
		MS	-10.531 ^a	-10.278 ^a	-0.111 ^a	-14.394 ^a	-	-	-10.513 ^a
		TS	0.056 ^a	0.855 ^a	0.932 ^a	0.056 ^a	-	-	0.058 ^a
		Surf	0.998 ^a	1.000 ^a	1.000^a	-	0.998 ^a	1.000 ^a	0.998 ^a
	Lasso	Point	-12.256 ^a	-12.256 ^a	0.099 ^a	-	-	-	-12.239 ^a
		MS	-10.531 ^a	-10.531 ^a	-0.115 ^a	-21.560 ^a	-	-	-10.513 ^a
		TS	0.056 ^a	0.832 ^a	0.929 ^a	0.056 ^a	-	-	0.058 ^a
		Surf	0.998 ^a	1.000 ^a	1.000^a	-	0.998 ^a	1.000 ^a	0.998 ^a
	RF	Point	0.137 ^a	-0.469 ^a	-0.455 ^a	-	-	-	0.027 ^a
		MS	0.173 ^a	-0.168 ^a	-0.170 ^a	-11.515	-	-	0.069 ^a
		TS	0.906 ^a	0.906 ^a	0.874 ^a	0.862 ^a	-	-	0.902 ^a
		Surf	1.000 ^a	1.000 ^a	0.999 ^a	-	0.999 ^a	1.000 ^a	1.000 ^a
	LSTM	Point	0.206^a	-	-	-	-	-	0.154 ^a
		MS	0.235 ^a	-	-	-35.271 ^a	-	-	0.325 ^a
		TS	0.909 ^a	-	-	0.917 ^a	-	-	0.902 ^a
		Surf	0.998 ^a	-	-	-	0.840 ^a	-	0.998 ^a
	Attention	Point	-19.814 ^a	-	-	-	-	-	-42.609 ^a
		MS	-21.743 ^a	-	-	-22.075 ^b	-	-	-42.341 ^a
		TS	-2.056 ^a	-	-	0.928 ^a	-	-	-3.499 ^a
		Surf	0.992 ^a	-	-	-	1.000 ^a	1.000 ^a	0.990 ^a

10	SVR	Point	0.206 ^a	0.097 ^a	0.032 ^a	-	-	-	0.169 ^a
		MS	0.341 ^a	0.090 ^a	-2.089 ^a	-154.539 ^a	-	-	0.335 ^a
		TS	0.941^a	0.538 ^a	0.076 ^a	0.918 ^a	-	-	0.939 ^a
		Surf	0.991 ^a	0.998 ^a	0.998 ^a	-	0.985 ^a	1.000 ^a	0.991 ^a
	Ridge	Point	0.125 ^a	-0.013 ^a	-0.154 ^a	-	-	-	0.129 ^a
		MS	0.259 ^a	-0.075 ^a	-0.099 ^a	0.116 ^a	-	-	0.262 ^a
		TS	0.936 ^a	0.911 ^a	0.926 ^a	0.922 ^a	-	-	0.939 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
	Enet	Point	-12.231 ^b	-12.231 ^b	-0.154 ^a	-	-	-	-12.239 ^a
		MS	-10.510 ^a	-8.595 ^a	-0.305 ^a	-15.563 ^a	-	-	-10.513 ^a
		TS	0.058 ^a	0.850 ^a	0.924 ^a	0.058 ^a	-	-	0.058 ^a
		Surf	0.998 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	0.998 ^a
	Lasso	Point	-12.231 ^b	-12.231 ^b	-0.154 ^a	-	-	-	-12.239 ^a
		MS	-10.510 ^a	-10.510 ^a	-0.337 ^a	-17.078 ^a	-	-	-10.513 ^a
		TS	0.058 ^a	0.837 ^a	0.923 ^a	0.058 ^a	-	-	0.058 ^a
		Surf	0.998 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	0.998 ^a
	RF	Point	0.086 ^a	-0.132 ^a	-0.716 ^a	-	-	-	0.027 ^a
		MS	0.104 ^a	-0.491 ^a	-0.718 ^a	-11.701	-	-	0.069 ^a
		TS	0.907 ^a	0.903 ^a	0.876 ^a	0.853 ^a	-	-	0.902 ^a
		Surf	1.000 ^a	1.000 ^a	0.999 ^a	-	0.999 ^a	1.000 ^a	1.000 ^a
	LSTM	Point	0.196 ^a	-	-	-	-	-	0.154 ^a
		MS	0.054 ^a	-	-	-69.552 ^a	-	-	0.325 ^a
		TS	0.904 ^a	-	-	0.900 ^a	-	-	0.902 ^a
		Surf	0.998 ^a	-	-	-	0.869 ^a	-	0.998 ^a
	Attention	Point	-29.530 ^a	-	-	-	-	-	-42.609 ^a
		MS	-31.095 ^a	-	-	-61.890 ^a	-	-	-42.341 ^a
		TS	-2.814 ^a	-	-	0.928 ^a	-	-	-3.499 ^a
		Surf	0.991 ^a	-	-	-	1.000 ^a	1.000 ^a	0.990 ^a
	SVR	Point	0.197 ^a	0.113 ^a	-0.231 ^a	-	-	-	0.169 ^a
		MS	0.342 ^a	-0.014 ^a	-2.243 ^a	-164.824 ^a	-	-	0.335 ^a
		TS	0.939 ^a	0.539 ^a	0.077 ^a	0.911 ^a	-	-	0.939 ^a
		Surf	0.991 ^a	0.998 ^a	0.998 ^a	-	0.986 ^a	1.000 ^a	0.991 ^a
	20 Ridge	Point	0.109 ^a	-0.089 ^a	-0.651 ^a	-	-	-	0.129 ^a
		MS	0.269 ^a	-0.117 ^a	-0.608 ^a	0.103 ^a	-	-	0.262 ^a
		TS	0.930 ^a	0.907 ^a	0.878 ^a	0.920 ^a	-	-	0.939 ^a
		Surf	1.000 ^a	1.000 ^a	-	-	1.000 ^a	1.000 ^a	1.000 ^a
	Enet	Point	-12.239 ^a	-12.239 ^a	-0.651 ^a	-	-	-	-12.239 ^a
		MS	-10.513 ^a	-6.170	-0.658 ^a	-16.099 ^a	-	-	-10.513 ^a
		TS	0.058 ^a	0.819 ^a	0.878 ^a	0.058 ^a	-	-	0.058 ^a
		Surf	0.998 ^a	1.000 ^a	-	-	0.998 ^a	1.000 ^a	0.998 ^a
	Lasso	Point	-12.239 ^a	-12.239 ^a	-0.651 ^a	-	-	-	-12.239 ^a
		MS	-10.513 ^a	-10.327 ^a	-0.739 ^a	-16.486	-	-	-10.513 ^a
		TS	0.058 ^a	0.813 ^a	0.878 ^a	0.058 ^a	-	-	0.058 ^a
		Surf	0.998 ^a	1.000 ^a	-	-	0.998 ^a	1.000 ^a	0.998 ^a
	RF	Point	0.024 ^a	-0.201 ^a	-1.264 ^a	-	-	-	0.027 ^a
		MS	0.057 ^a	-0.734 ^a	-1.430 ^a	-11.773 ^c	-	-	0.069 ^a
		TS	0.907 ^a	0.834 ^a	0.855 ^a	0.835 ^a	-	-	0.902 ^a

	Surf	1.000 ^a	0.999 ^a	-	-	0.999 ^a	1.000 ^a	1.000 ^a
LSTM	Point	0.187 ^a	-	-	-	-	-	0.154 ^a
	MS	0.074 ^a	-	-	-51.343 ^b	-	-	0.325 ^a
	TS	0.875 ^a	-	-	0.891 ^a	-	-	0.902 ^a
	Surf	0.998 ^a	-	-	-	0.412 ^a	-	0.998 ^a
Attention	Point	-42.592 ^a	-	-	-	-	-	-42.609 ^a
	MS	-42.361 ^a	-	-	-27.720	-	-	-42.341 ^a
	TS	-3.500 ^a	-	-	0.926 ^a	-	-	-3.499 ^a
	Surf	0.990 ^a	-	-	-	1.000 ^a	1.000 ^a	0.990 ^a
SVR	Point	0.193 ^a	0.051 ^a	-0.733 ^a	-	-	-	0.169 ^a
	MS	0.343^a	-0.019 ^a	-2.536 ^b	-108.649 ^a	-	-	0.335 ^a
	TS	0.937 ^a	0.503 ^a	0.073 ^a	0.909 ^a	-	-	0.939 ^a
	Surf	-	-	-	-	-	-	0.991 ^a

TABLE A.3: The table reports the $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ statistic for the out of sample dataset on *call options* for the *Nadaraya Watson interpolation*. *a, b, c* denote statistical significance at the 1%, 5%, and 10% levels respectively.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	32.175 ^a	20.192 ^a	19.232 ^a	-	-	-	26.356 ^a
		MS	30.872^a	-5.965 ^a	13.695 ^a	20.715 ^a	-	-	27.901 ^a
		TS	32.589 ^a	31.806 ^a	32.397 ^a	32.134 ^a	-	-	32.416 ^a
		Surf	49.089 ^a	49.089 ^a	49.089 ^a	-	49.086 ^a	49.090 ^a	48.992 ^a
	Enet	Point	-120.900 ^a	-120.900 ^a	19.232 ^a	-	-	-	-119.828 ^a
		MS	-127.580 ^a	-140.402 ^a	-15.156 ^a	-122.901 ^a	-	-	-126.401 ^a
		TS	2.088 ^b	30.241 ^a	32.316 ^a	2.105 ^b	-	-	2.165 ^b
		Surf	49.015 ^a	49.089 ^a	49.089 ^a	-	49.012 ^a	49.090 ^a	48.918 ^a
	Lasso	Point	-120.900 ^a	-120.900 ^a	19.232 ^a	-	-	-	-119.828 ^a
		MS	-127.580 ^a	-127.580 ^a	-16.235 ^a	-52.526 ^a	-	-	-126.401 ^a
		TS	2.088 ^b	29.520 ^a	32.231 ^a	2.105 ^b	-	-	2.165 ^b
		Surf	49.015 ^a	49.089 ^a	49.089 ^a	-	49.012 ^a	49.090 ^a	48.918 ^a
	RF	Point	28.588 ^a	-94.403 ^a	-91.441 ^a	-	-	-	4.971 ^a
		MS	20.899 ^a	-20.119 ^a	-27.015 ^a	-218.983 ^a	-	-	12.300 ^a
		TS	31.522 ^a	31.639 ^a	31.002 ^a	30.405 ^a	-	-	31.274 ^a
		Surf	49.090 ^a	49.084 ^a	49.066 ^a	-	49.041 ^a	49.090^a	48.993 ^a
	LSTM	Point	39.917 ^a	-	-	-	-	-	29.542 ^a
		MS	27.546 ^a	-	-	-148.034 ^a	-	-	28.922 ^a
		TS	31.490 ^a	-	-	31.803 ^a	-	-	31.195 ^a
		Surf	49.016 ^a	-	-	-	41.606 ^a	-	48.918 ^a
	Attention	Point	-113.890 ^a	-	-	-	-	-	-149.110 ^a
		MS	-109.303 ^a	-	-	-118.305 ^a	-	-	-141.475 ^a
		TS	-64.786 ^a	-	-	32.208 ^a	-	-	-96.349 ^a
		Surf	48.763 ^a	-	-	-	49.086 ^a	49.090 ^a	48.569 ^a
	SVR	Point	42.443^a	18.559 ^a	5.620 ^a	-	-	-	35.976 ^a
		MS	30.010 ^a	8.353 ^a	-144.512 ^a	-182.089 ^a	-	-	28.888 ^a

10	Ridge	TS	32.615 ^a	18.540 ^a	2.580 ^a	31.818 ^a	-	-	32.421 ^a
		Surf	48.714 ^a	49.015 ^a	49.015 ^a	-	48.438 ^a	49.085 ^a	48.618 ^a
		Point	23.009 ^a	-1.738 ^c	-25.600 ^a	-	-	-	26.356 ^a
		MS	29.674 ^a	-9.276 ^a	-8.136 ^a	13.771 ^a	-	-	27.901 ^a
	Enet	TS	32.450 ^a	31.607 ^a	32.098 ^a	31.940 ^a	-	-	32.416 ^a
		Surf	49.036 ^a	49.036 ^a	49.036 ^a	-	49.033 ^a	49.037 ^a	48.992 ^a
		Point	-120.751 ^a	-120.751 ^a	-25.600 ^a	-	-	-	-119.828 ^a
		MS	-127.431 ^a	-257.912 ^a	-39.795 ^a	-107.525 ^a	-	-	-126.401 ^a
	Lasso	TS	2.156 ^b	30.030 ^a	32.037 ^a	2.173 ^b	-	-	2.165 ^b
		Surf	48.962 ^a	49.036 ^a	49.036 ^a	-	48.960 ^a	49.037 ^a	48.918 ^a
		Point	-120.751 ^a	-120.751 ^a	-25.600 ^a	-	-	-	-119.828 ^a
		MS	-127.431 ^a	-127.431 ^a	-47.896 ^a	-74.828 ^a	-	-	-126.401 ^a
	RF	TS	2.156 ^b	29.650 ^a	31.985 ^a	2.173 ^b	-	-	2.165 ^b
		Surf	48.962 ^a	49.035 ^a	49.036 ^a	-	48.960 ^a	49.037 ^a	48.918 ^a
		Point	13.898 ^a	-24.430 ^a	-133.406 ^a	-	-	-	4.971 ^a
		MS	18.819 ^a	-63.358 ^a	-86.338 ^a	-206.777 ^a	-	-	12.300 ^a
	LSTM	TS	31.553 ^a	31.454 ^a	30.774 ^a	30.045 ^a	-	-	31.274 ^a
		Surf	49.037 ^a	49.030 ^a	49.015 ^a	-	48.985 ^a	49.037 ^a	48.993 ^a
		Point	36.349 ^a	-	-	-	-	-	29.542 ^a
		MS	8.870 ^a	-	-	-122.492 ^a	-	-	28.922 ^a
	Attention	TS	31.259 ^a	-	-	31.171 ^a	-	-	31.195 ^a
		Surf	48.966 ^a	-	-	-	43.025 ^a	-	48.918 ^a
		Point	-130.533 ^a	-	-	-	-	-	-149.110 ^a
		MS	-124.416 ^a	-	-	-60.853 ^a	-	-	-141.475 ^a
	SVR	TS	-83.407 ^a	-	-	32.134 ^a	-	-	-96.349 ^a
		Surf	48.658 ^a	-	-	-	49.031 ^a	49.037 ^a	48.569 ^a
		Point	39.072 ^a	20.955 ^a	-35.514 ^a	-	-	-	35.976 ^a
		MS	29.658 ^a	-1.633	-144.371 ^a	-168.563 ^a	-	-	28.888 ^a
20	Ridge	TS	32.523 ^a	18.550 ^a	2.590 ^a	31.560 ^a	-	-	32.421 ^a
		Surf	48.661 ^a	48.963 ^a	48.962 ^a	-	48.447 ^a	49.032 ^a	48.618 ^a
		Point	20.442 ^a	-9.619 ^a	-69.800 ^a	-	-	-	26.356 ^a
		MS	27.897 ^a	-14.511 ^a	-37.371 ^a	10.918 ^a	-	-	27.901 ^a
	Enet	TS	32.179 ^a	31.442 ^a	30.284 ^a	31.749 ^a	-	-	32.416 ^a
		Surf	48.991 ^a	48.992 ^a	-	-	48.989 ^a	48.992 ^a	48.992 ^a
		Point	-119.828 ^a	-119.828 ^a	-69.800 ^a	-	-	-	-119.828 ^a
		MS	-126.401 ^a	-394.036 ^a	-50.718 ^a	-98.090 ^a	-	-	-126.401 ^a
	Lasso	TS	2.165 ^b	28.994 ^a	30.272 ^a	2.179 ^b	-	-	2.165 ^b
		Surf	48.918 ^a	48.991 ^a	-	-	48.915 ^a	48.993 ^a	48.918 ^a
		Point	-119.828 ^a	-119.828 ^a	-69.800 ^a	-	-	-	-119.828 ^a
		MS	-126.401 ^a	-135.869 ^a	-76.182 ^a	-58.776 ^a	-	-	-126.401 ^a
	RF	TS	2.165 ^b	28.807 ^a	30.256 ^a	2.179 ^b	-	-	2.165 ^b
		Surf	48.918 ^a	48.990 ^a	-	-	48.915 ^a	48.993 ^a	48.918 ^a
		Point	3.315 ^a	-32.369 ^a	-154.342 ^a	-	-	-	4.971 ^a
		MS	12.292 ^a	-66.137 ^a	-84.899 ^a	-203.771 ^a	-	-	12.300 ^a
	LSTM	TS	31.423 ^a	29.844 ^a	29.929 ^a	29.323 ^a	-	-	31.274 ^a
		Surf	48.993 ^a	48.976 ^a	-	-	48.941 ^a	48.993 ^a	48.993 ^a
		Point	34.275 ^a	-	-	-	-	-	29.542 ^a

	MS	9.240 ^a	-	-	-169.812 ^a	-	-	28.922 ^a
	TS	30.125 ^a	-	-	30.841 ^a	-	-	31.195 ^a
	Surf	48.919 ^a	-	-	-	19.175 ^a	-	48.918 ^a
Attention	Point	-149.024 ^a	-	-	-	-	-	-149.110 ^a
	MS	-141.684 ^a	-	-	-205.131 ^a	-	-	-141.475 ^a
	TS	-96.380 ^a	-	-	31.993 ^a	-	-	-96.349 ^a
	Surf	48.569 ^a	-	-	-	48.989 ^a	48.992 ^a	48.569 ^a
SVR	Point	38.266 ^a	9.287 ^a	-75.986 ^a	-	-	-	35.976 ^a
	MS	29.321 ^a	-2.019 ^b	-135.551 ^a	-196.456 ^a	-	-	28.888 ^a
	TS	32.336 ^a	17.604 ^a	2.471 ^b	31.359 ^a	-	-	32.421 ^a
	Surf	-	-	-	-	-	-	48.618 ^a

TABLE A.4: The table reports the DM statistic for the out of sample dataset on *call options* using *Nadaraya Watson interpolation*. *a, b, c* denote statistical significance at the 1%, 5%, and 10% levels respectively.

A.2 Put Options Contracts

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.101 ^a	0.028 ^a	0.014 ^a	-	-	-	0.098 ^a
		MS	0.980 ^a	0.966 ^a	0.973 ^a	0.976 ^a	-	-	0.981 ^a
		TS	0.989 ^a	0.986 ^a	0.988 ^a	0.988 ^a	-	-	0.990^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000^a
	Enet	Point	-10.270 ^a	-10.270 ^a	0.014 ^a	-	-	-	-10.313 ^a
		MS	0.768 ^a	0.775 ^a	0.965 ^a	0.399 ^a	-	-	0.768 ^a
		TS	0.862 ^a	0.979 ^a	0.988 ^a	0.862 ^a	-	-	0.862 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
	Lasso	Point	-10.270 ^a	-10.270 ^a	0.014 ^a	-	-	-	-10.313 ^a
		MS	0.768 ^a	0.768 ^a	0.965 ^a	0.250 ^a	-	-	0.768 ^a
		TS	0.862 ^a	0.976 ^a	0.988 ^a	0.862 ^a	-	-	0.862 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
	RF	Point	-1.813 ^a	-0.064 ^a	-0.586 ^a	-	-	-	0.041 ^a
		MS	0.941 ^a	0.965 ^a	0.972 ^a	0.505 ^a	-	-	0.973 ^a
		TS	0.966 ^a	0.984 ^a	0.981 ^a	0.972 ^a	-	-	0.987 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
	LSTM	Point	-0.019 ^a	-	-	-	-	-	0.156 ^a
		MS	0.981 ^a	-	-	0.385 ^a	-	-	0.983 ^a
		TS	0.989 ^a	-	-	0.916 ^a	-	-	0.987 ^a
		Surf	1.000 ^a	-	-	-	0.900 ^a	-	1.000 ^a
	Attention	Point	0.133 ^a	-	-	-	-	-	-0.037 ^a
		MS	0.978 ^a	-	-	-8.585 ^a	-	-	0.978 ^a
		TS	0.981 ^a	-	-	0.988 ^a	-	-	0.986 ^a
		Surf	0.994 ^a	-	-	-	1.000 ^a	1.000 ^a	1.000 ^a
	SVR	Point	0.180^a	0.154 ^a	-0.023 ^a	-	-	-	0.154 ^a
		MS	0.983 ^a	0.969 ^a	0.935 ^a	-10.091 ^a	-	-	0.984 ^a
		TS	0.989 ^a	0.921 ^a	0.853 ^a	0.984 ^a	-	-	0.989 ^a
		Surf	0.998 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	0.998 ^a
10	Ridge	Point	0.065 ^a	-0.162 ^a	-0.273 ^a	-	-	-	0.098 ^a
		MS	0.980 ^a	0.963 ^a	0.967 ^a	0.975 ^a	-	-	0.981 ^a
		TS	0.989 ^a	0.986 ^a	0.985 ^a	0.987 ^a	-	-	0.990^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000^a
	Enet	Point	-10.272 ^a	-10.272 ^a	-0.273 ^a	-	-	-	-10.313 ^a
		MS	0.769 ^a	0.811 ^a	0.959 ^a	0.083 ^a	-	-	0.768 ^a
		TS	0.862 ^a	0.977 ^a	0.985 ^a	0.862 ^a	-	-	0.862 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
	Lasso	Point	-10.272 ^a	-10.272 ^a	-0.273 ^a	-	-	-	-10.313 ^a
		MS	0.769 ^a	0.769 ^a	0.959 ^a	0.106 ^a	-	-	0.768 ^a
		TS	0.862 ^a	0.976 ^a	0.985 ^a	0.862 ^a	-	-	0.862 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
	RF	Point	-1.813 ^a	-0.095 ^a	-0.921 ^a	-	-	-	0.041 ^a
		MS	0.941 ^a	0.964 ^a	0.965 ^a	0.493 ^a	-	-	0.973 ^a

20	LSTM	TS	0.966 ^a	0.980 ^a	0.978 ^a	0.968 ^a	-	-	0.987 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
		Point	-0.181 ^a	-	-	-	-	-	0.156 ^a
		MS	0.982 ^a	-	-	0.489 ^a	-	-	0.983 ^a
	Attention	TS	0.989 ^a	-	-	0.919 ^a	-	-	0.987 ^a
		Surf	1.000 ^a	-	-	-	0.939 ^a	-	1.000 ^a
		Point	0.116 ^a	-	-	-	-	-	-0.037 ^a
		MS	0.962 ^a	-	-	-6.721 ^a	-	-	0.978 ^a
	SVR	TS	0.981 ^a	-	-	0.987 ^a	-	-	0.986 ^a
		Surf	0.999 ^a	-	-	-	1.000 ^a	1.000 ^a	1.000 ^a
		Point	0.174 ^a	0.157 ^a	-0.318 ^a	-	-	-	0.154 ^a
		MS	0.984 ^a	0.969 ^a	0.931 ^a	-15.840 ^a	-	-	0.984 ^a
	Ridge	TS	0.989 ^a	0.920 ^a	0.853 ^a	0.982 ^a	-	-	0.989 ^a
		Surf	0.998 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	0.998 ^a
		Point	0.034 ^a	-0.356 ^a	-0.817 ^a	-	-	-	0.098 ^a
		MS	0.980 ^a	0.963 ^a	0.956 ^a	0.975 ^a	-	-	0.981 ^a
	Enet	TS	0.989 ^a	0.986 ^a	0.982 ^a	0.987 ^a	-	-	0.990^a
		Surf	1.000 ^a	1.000 ^a	-	-	1.000 ^a	1.000 ^a	1.000^a
		Point	-10.313 ^a	-10.313 ^a	-0.817 ^a	-	-	-	-10.313 ^a
		MS	0.768 ^a	0.866 ^a	0.951 ^a	-0.073 ^a	-	-	0.768 ^a
	Lasso	TS	0.862 ^a	0.972 ^a	0.982 ^a	0.862 ^a	-	-	0.862 ^a
		Surf	1.000 ^a	1.000 ^a	-	-	1.000 ^a	1.000 ^a	1.000 ^a
		Point	-10.313 ^a	-10.313 ^a	-0.817 ^a	-	-	-	-10.313 ^a
		MS	0.768 ^a	0.773 ^a	0.948 ^a	-0.159 ^a	-	-	0.768 ^a
	RF	TS	0.862 ^a	0.971 ^a	0.982 ^a	0.862 ^a	-	-	0.862 ^a
		Surf	1.000 ^a	1.000 ^a	-	-	1.000 ^a	1.000 ^a	1.000 ^a
		Point	-1.812 ^a	-0.152 ^a	-1.500 ^a	-	-	-	0.041 ^a
		MS	0.941 ^a	0.961 ^a	0.951 ^a	0.511 ^a	-	-	0.973 ^a
	LSTM	TS	0.966 ^a	0.975 ^a	0.975 ^a	0.964 ^a	-	-	0.987 ^a
		Surf	1.000 ^a	1.000 ^a	-	-	0.999 ^a	1.000 ^a	1.000 ^a
		Point	-0.317 ^a	-	-	-	-	-	0.156 ^a
		MS	0.978 ^a	-	-	0.463 ^a	-	-	0.983 ^a
	Attention	TS	0.988 ^a	-	-	0.917 ^a	-	-	0.987 ^a
		Surf	1.000 ^a	-	-	-	0.891 ^a	-	1.000 ^a
		Point	-0.023 ^a	-	-	-	-	-	-0.037 ^a
		MS	0.976 ^a	-	-	-21.774 ^a	-	-	0.978 ^a
	SVR	TS	0.975 ^a	-	-	0.989 ^a	-	-	0.986 ^a
		Surf	1.000 ^a	-	-	-	1.000 ^a	1.000 ^a	1.000 ^a
		Point	0.171 ^a	0.126 ^a	-0.867 ^a	-	-	-	0.154 ^a
		MS	0.984^a	0.969 ^a	0.923 ^a	-18.353 ^a	-	-	0.984 ^a
		TS	0.989 ^a	0.920 ^a	0.853 ^a	0.981 ^a	-	-	0.989 ^a
		Surf	-	-	-	-	-	-	0.998 ^a

TABLE A.5: The table reports the $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ statistic for the out of sample dataset on *put options* for the *Dumas interpolation*. a, b, c denote statistical significance at the 1%, 5%, and 10% levels respectively.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	17.012 ^a	4.119 ^a	2.139 ^b	-	-	-	15.346 ^a
		MS	2.647 ^a	2.609 ^a	2.630 ^a	2.637 ^a	-	-	2.641 ^a
		TS	5.947^a	5.929 ^a	5.937 ^a	5.940 ^a	-	-	5.928 ^a
		Surf	35.906^a	35.905 ^a	35.905 ^a	-	35.905 ^a	35.905 ^a	35.786 ^a
	Enet	Point	-122.805 ^a	-122.805 ^a	2.139 ^b	-	-	-	-122.332 ^a
		MS	2.076 ^b	2.093 ^b	2.606 ^a	1.079	-	-	2.069 ^b
		TS	5.189 ^a	5.886 ^a	5.936 ^a	5.189 ^a	-	-	5.172 ^a
		Surf	35.897 ^a	35.905 ^a	35.905 ^a	-	35.897 ^a	35.905 ^a	35.778 ^a
	Lasso	Point	-122.805 ^a	-122.805 ^a	2.139 ^b	-	-	-	-122.332 ^a
		MS	2.076 ^b	2.076 ^b	2.606 ^a	0.675	-	-	2.069 ^b
		TS	5.189 ^a	5.870 ^a	5.936 ^a	5.189 ^a	-	-	5.172 ^a
		Surf	35.897 ^a	35.905 ^a	35.905 ^a	-	35.897 ^a	35.905 ^a	35.778 ^a
	RF	Point	-46.820 ^a	-12.329 ^a	-74.767 ^a	-	-	-	5.892 ^a
		MS	2.542 ^b	2.608 ^a	2.626 ^a	1.363	-	-	2.620 ^a
		TS	5.814 ^a	5.915 ^a	5.897 ^a	5.849 ^a	-	-	5.914 ^a
		Surf	35.904 ^a	35.905 ^a	35.904 ^a	-	35.892 ^a	35.905 ^a	35.785 ^a
	LSTM	Point	-6.333 ^a	-	-	-	-	-	23.133 ^a
		MS	2.650 ^a	-	-	1.041	-	-	2.646 ^a
		TS	5.942 ^a	-	-	5.515 ^a	-	-	5.914 ^a
		Surf	35.899 ^a	-	-	-	31.566 ^a	-	35.778 ^a
	Attention	Point	21.859 ^a	-	-	-	-	-	-4.875 ^a
		MS	2.642 ^a	-	-	-23.166 ^a	-	-	2.634 ^a
		TS	5.897 ^a	-	-	5.941 ^a	-	-	5.908 ^a
		Surf	35.728 ^a	-	-	-	35.905 ^a	35.905 ^a	35.771 ^a
	SVR	Point	30.397^a	26.057 ^a	-3.391 ^a	-	-	-	25.431 ^a
		MS	2.656^a	2.619 ^a	2.526 ^b	-27.268 ^a	-	-	2.649 ^a
		TS	5.945 ^a	5.547 ^a	5.150 ^a	5.917 ^a	-	-	5.923 ^a
		Surf	35.835 ^a	35.897 ^a	35.897 ^a	-	35.833 ^a	35.904 ^a	35.716 ^a
10	Ridge	Point	9.895 ^a	-15.832 ^a	-30.008 ^a	-	-	-	15.346 ^a
		MS	2.644 ^a	2.599 ^a	2.609 ^a	2.631 ^a	-	-	2.641 ^a
		TS	5.937 ^a	5.921 ^a	5.914 ^a	5.928 ^a	-	-	5.928 ^a
		Surf	35.866 ^a	35.865 ^a	35.865 ^a	-	35.865 ^a	35.865 ^a	35.786 ^a
	Enet	Point	-122.495 ^a	-122.495 ^a	-30.008 ^a	-	-	-	-122.332 ^a
		MS	2.074 ^b	2.188 ^b	2.589 ^a	0.225	-	-	2.069 ^b
		TS	5.184 ^a	5.869 ^a	5.914 ^a	5.184 ^a	-	-	5.172 ^a
		Surf	35.857 ^a	35.865 ^a	35.865 ^a	-	35.858 ^a	35.865 ^a	35.778 ^a
	Lasso	Point	-122.495 ^a	-122.495 ^a	-30.008 ^a	-	-	-	-122.332 ^a
		MS	2.074 ^b	2.074 ^b	2.587 ^a	0.285	-	-	2.069 ^b
		TS	5.184 ^a	5.860 ^a	5.914 ^a	5.184 ^a	-	-	5.172 ^a
		Surf	35.857 ^a	35.865 ^a	35.865 ^a	-	35.858 ^a	35.865 ^a	35.778 ^a
	RF	Point	-46.952 ^a	-12.633 ^a	-86.390 ^a	-	-	-	5.892 ^a
		MS	2.540 ^b	2.602 ^a	2.604 ^a	1.331	-	-	2.620 ^a
		TS	5.808 ^a	5.890 ^a	5.875 ^a	5.820 ^a	-	-	5.914 ^a
		Surf	35.864 ^a	35.864 ^a	35.864 ^a	-	35.851 ^a	35.865 ^a	35.785 ^a

20	LSTM	Point	-33.344 ^a	-	-	-	-	-	23.133 ^a
		MS	2.649 ^a	-	-	1.321	-	-	2.646 ^a
		TS	5.939 ^a	-	-	5.530 ^a	-	-	5.914 ^a
		Surf	35.859 ^a	-	-	-	33.244 ^a	-	35.778 ^a
	Attention	Point	17.779 ^a	-	-	-	-	-	-4.875 ^a
		MS	2.597 ^a	-	-	-18.130 ^a	-	-	2.634 ^a
		TS	5.894 ^a	-	-	5.926 ^a	-	-	5.908 ^a
		Surf	35.818 ^a	-	-	-	35.863 ^a	35.865 ^a	35.771 ^a
	SVR	Point	27.959 ^a	25.419 ^a	-34.429 ^a	-	-	-	25.431 ^a
		MS	2.654 ^a	2.614 ^a	2.512 ^b	-42.717 ^a	-	-	2.649 ^a
		TS	5.938 ^a	5.535 ^a	5.144 ^a	5.898 ^a	-	-	5.923 ^a
		Surf	35.795 ^a	35.857 ^a	35.857 ^a	-	35.804 ^a	35.864 ^a	35.716 ^a
	Ridge	Point	5.539 ^a	-27.193 ^a	-54.400 ^a	-	-	-	15.346 ^a
		MS	2.639 ^a	2.592 ^a	2.575 ^b	2.625 ^a	-	-	2.641 ^a
		TS	5.922 ^a	5.907 ^a	5.885 ^a	5.912 ^a	-	-	5.928 ^a
		Surf	35.786 ^a	35.785 ^a	-	-	35.785 ^a	35.786 ^a	35.786 ^a
	Enet	Point	-122.332 ^a	-122.332 ^a	-54.400 ^a	-	-	-	-122.332 ^a
		MS	2.069 ^b	2.332 ^b	2.561 ^b	-0.196	-	-	2.069 ^b
		TS	5.172 ^a	5.823 ^a	5.885 ^a	5.172 ^a	-	-	5.172 ^a
		Surf	35.778 ^a	35.785 ^a	-	-	35.778 ^a	35.786 ^a	35.778 ^a
	Lasso	Point	-122.332 ^a	-122.332 ^a	-54.400 ^a	-	-	-	-122.332 ^a
		MS	2.069 ^b	2.080 ^b	2.552 ^b	-0.429	-	-	2.069 ^b
		TS	5.172 ^a	5.819 ^a	5.885 ^a	5.172 ^a	-	-	5.172 ^a
		Surf	35.778 ^a	35.785 ^a	-	-	35.778 ^a	35.786 ^a	35.778 ^a
	RF	Point	-46.531 ^a	-18.324 ^a	-93.676 ^a	-	-	-	5.892 ^a
		MS	2.534 ^b	2.588 ^a	2.560 ^b	1.376	-	-	2.620 ^a
		TS	5.794 ^a	5.847 ^a	5.846 ^a	5.783 ^a	-	-	5.914 ^a
		Surf	35.784 ^a	35.784 ^a	-	-	35.770 ^a	35.786 ^a	35.785 ^a
	LSTM	Point	-46.548 ^a	-	-	-	-	-	23.133 ^a
		MS	2.633 ^a	-	-	1.246	-	-	2.646 ^a
		TS	5.920 ^a	-	-	5.503 ^a	-	-	5.914 ^a
		Surf	35.779 ^a	-	-	-	30.768 ^a	-	35.778 ^a
	Attention	Point	-2.899 ^a	-	-	-	-	-	-4.875 ^a
		MS	2.627 ^a	-	-	-58.304 ^a	-	-	2.634 ^a
		TS	5.847 ^a	-	-	5.924 ^a	-	-	5.908 ^a
		Surf	35.775 ^a	-	-	-	35.784 ^a	35.786 ^a	35.771 ^a
	SVR	Point	26.883 ^a	19.228 ^a	-57.719 ^a	-	-	-	25.431 ^a
		MS	2.649 ^a	2.608 ^a	2.486 ^b	-49.374 ^a	-	-	2.649 ^a
		TS	5.923 ^a	5.523 ^a	5.131 ^a	5.874 ^a	-	-	5.923 ^a
		Surf	-	-	-	-	-	-	35.716 ^a

TABLE A.6: The table reports the DM statistic for the out of sample dataset on the *put options* using *Dumas interpolation*. *a, b, c* denote statistical significance at the 1%, 5%, and 10% levels respectively.

Lags	Predictor	Feature	Encoder
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			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	0.058 ^a	-0.061 ^a	-0.068 ^a	-	-	-	0.044 ^a
		MS	0.118 ^a	-0.346 ^a	-0.157 ^a	-0.075 ^a	-	-	0.112 ^a
		TS	0.966^a	0.942 ^a	0.959 ^a	0.960 ^a	-	-	0.966 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000^a	1.000 ^a
	Enet	Point	-12.076	-12.076	-0.068 ^a	-	-	-	-12.117
		MS	-11.949	-11.677	-0.379 ^a	-52.679	-	-	-11.987
		TS	0.480 ^a	0.932 ^a	0.958 ^a	0.480 ^a	-	-	0.477 ^a
		Surf	0.998 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	0.998 ^a
	Lasso	Point	-12.076	-12.076	-0.068 ^a	-	-	-	-12.117
		MS	-11.949	-11.949	-0.380 ^a	-57.889	-	-	-11.987
		TS	0.480 ^a	0.922 ^a	0.958 ^a	0.480 ^a	-	-	0.477 ^a
		Surf	0.998 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	0.998 ^a
	RF	Point	0.017 ^a	-0.600 ^a	-0.640 ^a	-	-	-	-0.056 ^a
		MS	-0.040 ^a	-0.807 ^a	-0.404 ^a	-15.234	-	-	-0.292 ^a
		TS	0.959 ^a	0.932 ^a	0.946 ^a	0.943 ^a	-	-	0.957 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	0.999 ^a	1.000 ^a	1.000 ^a
	LSTM	Point	0.132 ^a	-	-	-	-	-	0.120 ^a
		MS	0.140 ^a	-	-	-90.859 ^a	-	-	0.178 ^a
		TS	0.956 ^a	-	-	0.938 ^a	-	-	0.944 ^a
		Surf	0.998 ^a	-	-	-	0.752 ^a	-	0.998 ^a
	Attention	Point	0.129 ^a	-	-	-	-	-	-0.035 ^a
		MS	0.027 ^a	-	-	-61.331 ^a	-	-	0.075 ^a
		TS	0.952 ^a	-	-	0.964 ^a	-	-	0.946 ^a
		Surf	0.997 ^a	-	-	-	1.000 ^a	1.000 ^a	0.991 ^a
	SVR	Point	0.134^a	-0.072 ^a	-0.111 ^a	-	-	-	0.105 ^a
		MS	0.188 ^a	-0.343 ^a	-2.491 ^a	-389.982 ^b	-	-	0.189 ^a
		TS	0.964 ^a	0.695 ^a	0.441 ^a	0.956 ^a	-	-	0.962 ^a
		Surf	0.984 ^a	0.998 ^a	0.998 ^a	-	0.974 ^a	0.999 ^a	0.984 ^a
10	Ridge	Point	0.005 ^a	-0.362 ^a	-0.442 ^a	-	-	-	0.044 ^a
		MS	0.130 ^a	-0.582 ^a	-0.519 ^a	-0.158 ^a	-	-	0.112 ^a
		TS	0.965 ^a	0.931 ^a	0.949 ^a	0.957 ^a	-	-	0.966 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	1.000 ^a	1.000 ^a	1.000 ^a
	Enet	Point	-12.071	-12.071	-0.442 ^a	-	-	-	-12.117
		MS	-11.952	-9.462	-0.752 ^a	-51.388 ^a	-	-	-11.987
		TS	0.480 ^a	0.923 ^a	0.949 ^a	0.480 ^a	-	-	0.477 ^a
		Surf	0.998 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	0.998 ^a
	Lasso	Point	-12.071	-12.071	-0.442 ^a	-	-	-	-12.117
		MS	-11.952	-11.952	-0.757 ^a	-85.404	-	-	-11.987
		TS	0.480 ^a	0.918 ^a	0.948 ^a	0.480 ^a	-	-	0.477 ^a
		Surf	0.998 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	0.998 ^a
	RF	Point	0.053 ^a	-1.016 ^a	-1.012 ^a	-	-	-	-0.056 ^a
		MS	-0.041 ^a	-0.692 ^a	-0.947 ^a	-14.741	-	-	-0.292 ^a
		TS	0.960 ^a	0.926 ^a	0.932 ^a	0.935 ^a	-	-	0.957 ^a
		Surf	1.000 ^a	1.000 ^a	1.000 ^a	-	0.998 ^a	1.000 ^a	1.000 ^a
	LSTM	Point	0.126 ^a	-	-	-	-	-	0.120 ^a
		MS	0.113 ^a	-	-	-109.069 ^a	-	-	0.178 ^a

20	Attention	TS	0.938 ^a	-	-	0.935 ^a	-	-	0.944 ^a
		Surf	0.998 ^a	-	-	-	0.760 ^a	-	0.998 ^a
		Point	0.110 ^a	-	-	-	-	-	-0.035 ^a
		MS	-0.112 ^a	-	-	-120.563 ^b	-	-	0.075 ^a
	SVR	TS	0.933 ^a	-	-	0.964 ^a	-	-	0.946 ^a
		Surf	0.995 ^a	-	-	-	1.000 ^a	1.000 ^a	0.991 ^a
		Point	0.129 ^a	-0.455 ^a	-0.493 ^a	-	-	-	0.105 ^a
		MS	0.196^a	-0.304 ^a	-2.735 ^a	-720.568 ^b	-	-	0.189 ^a
	Ridge	TS	0.963 ^a	0.682 ^a	0.441 ^a	0.952 ^a	-	-	0.962 ^a
		Surf	0.984 ^a	0.998 ^a	0.998 ^a	-	0.972 ^a	0.999 ^a	0.984 ^a
		Point	-0.012 ^a	-0.621 ^a	-1.135 ^a	-	-	-	0.044 ^a
		MS	0.135 ^a	-0.673 ^a	-1.280 ^a	-0.203 ^a	-	-	0.112 ^a
	Enet	TS	0.963 ^a	0.929 ^a	0.935 ^a	0.956 ^a	-	-	0.966 ^a
		Surf	1.000 ^a	1.000 ^a	-	-	1.000 ^a	1.000 ^a	1.000 ^a
		Point	-12.117	-12.117	-1.135 ^a	-	-	-	-12.117
		MS	-11.987	-6.193	-1.372 ^a	-52.491 ^a	-	-	-11.987
	Lasso	TS	0.477 ^a	0.898 ^a	0.934 ^a	0.477 ^a	-	-	0.477 ^a
		Surf	0.998 ^a	1.000 ^a	-	-	0.998 ^a	1.000 ^a	0.998 ^a
		Point	-12.117	-12.117	-1.135 ^a	-	-	-	-12.117
		MS	-11.987	-11.850	-1.419 ^a	-78.845	-	-	-11.987
	RF	TS	0.477 ^a	0.896 ^a	0.934 ^a	0.477 ^a	-	-	0.477 ^a
		Surf	0.998 ^a	1.000 ^a	-	-	0.998 ^a	1.000 ^a	0.998 ^a
		Point	0.017 ^a	-0.690 ^a	-1.804 ^a	-	-	-	-0.056 ^a
		MS	-0.036 ^a	-1.202 ^a	-1.906 ^a	-14.655	-	-	-0.292 ^a
	LSTM	TS	0.960 ^a	0.912 ^a	0.921 ^a	0.923 ^a	-	-	0.957 ^a
		Surf	1.000 ^a	0.999 ^a	-	-	0.998 ^a	1.000 ^a	1.000 ^a
		Point	0.117 ^a	-	-	-	-	-	0.120 ^a
		MS	-0.104 ^a	-	-	-155.628 ^a	-	-	0.178 ^a
	Attention	TS	0.954 ^a	-	-	0.951 ^a	-	-	0.944 ^a
		Surf	0.998 ^a	-	-	-	0.289 ^a	-	0.998 ^a
		Point	0.058 ^a	-	-	-	-	-	-0.035 ^a
		MS	-0.047 ^a	-	-	-269.818 ^a	-	-	0.075 ^a
	SVR	TS	0.942 ^a	-	-	0.963 ^a	-	-	0.946 ^a
		Surf	0.998 ^a	-	-	-	0.999 ^a	1.000 ^a	0.991 ^a
		Point	0.127 ^a	-0.330 ^a	-1.189 ^a	-	-	-	0.105 ^a
		MS	0.193 ^a	-0.601 ^a	-3.210 ^a	-373.961 ^b	-	-	0.189 ^a
		TS	0.963 ^a	0.681 ^a	0.436 ^a	0.950 ^a	-	-	0.962 ^a
		Surf	-	-	-	-	-	-	0.984 ^a

TABLE A.7: The table reports the $R^2_{\mathcal{M}_1, \mathcal{M}_2}$ statistic for the out of sample dataset on the *put* options using *Nadaraya Watson interpolation*. *a, b, c* denote statistical significance at the 1%, 5%, and 10% levels respectively.

Lags	Predictor	Feature	Encoder						
			SAM	PCA	CCA	NS	ADNS	SSVI	HAR
5	Ridge	Point	11.800 ^a	-10.929 ^a	-13.263 ^a	-	-	-	9.750 ^a

	MS	42.269 ^a	-65.875 ^a	-28.460 ^a	-20.019 ^a	-	-	37.418 ^a
	TS	45.059^a	44.338 ^a	44.748 ^a	44.826 ^a	-	-	44.792 ^a
	Surf	33.648^a	33.643 ^a	33.646 ^a	-	33.645 ^a	33.647 ^a	33.513 ^a
Enet	Point	-139.118 ^a	-139.118 ^a	-13.263 ^a	-	-	-	-139.047 ^a
	MS	-138.780 ^a	-153.269 ^a	-69.244 ^a	-27.158 ^a	-	-	-138.722 ^a
	TS	22.777 ^a	43.994 ^a	44.731 ^a	22.770 ^a	-	-	22.506 ^a
	Surf	33.580 ^a	33.644 ^a	33.646 ^a	-	33.580 ^a	33.646 ^a	33.445 ^a
Lasso	Point	-139.118 ^a	-139.118 ^a	-13.263 ^a	-	-	-	-139.047 ^a
	MS	-138.780 ^a	-138.780 ^a	-69.619 ^a	-31.408 ^a	-	-	-138.722 ^a
	TS	22.777 ^a	43.556 ^a	44.720 ^a	22.770 ^a	-	-	22.506 ^a
	Surf	33.580 ^a	33.644 ^a	33.646 ^a	-	33.580 ^a	33.646 ^a	33.445 ^a
RF	Point	4.200 ^a	-109.950 ^a	-92.113 ^a	-	-	-	-11.373 ^a
	MS	-7.165 ^a	-69.481 ^a	-57.897 ^a	-233.850 ^a	-	-	-20.231 ^a
	TS	44.850 ^a	43.865 ^a	44.273 ^a	44.038 ^a	-	-	44.521 ^a
	Surf	33.646 ^a	33.642 ^a	33.642 ^a	-	33.608 ^a	33.647 ^a	33.511 ^a
LSTM	Point	29.675 ^a	-	-	-	-	-	25.687 ^a
	MS	47.625 ^a	-	-	-69.362 ^a	-	-	54.563 ^a
	TS	44.607 ^a	-	-	43.792 ^a	-	-	43.960 ^a
	Surf	33.582 ^a	-	-	-	24.022 ^a	-	33.445 ^a
Attention	Point	31.518 ^a	-	-	-	-	-	-8.074 ^a
	MS	7.270 ^a	-	-	-56.994 ^a	-	-	25.841 ^a
	TS	44.367 ^a	-	-	44.997 ^a	-	-	43.789 ^a
	Surf	33.567 ^a	-	-	-	33.644 ^a	33.647 ^a	33.224 ^a
SVR	Point	32.833^a	-13.907 ^a	-21.334 ^a	-	-	-	27.840 ^a
	MS	65.726^a	-47.809 ^a	-114.786 ^a	-39.196 ^a	-	-	64.080 ^a
	TS	44.979 ^a	32.463 ^a	19.848 ^a	44.627 ^a	-	-	44.687 ^a
	Surf	33.063 ^a	33.581 ^a	33.580 ^a	-	32.705 ^a	33.638 ^a	32.928 ^a
10 Ridge	Point	0.974	-38.304 ^a	-53.060 ^a	-	-	-	9.750 ^a
	MS	44.660 ^a	-88.865 ^a	-55.744 ^a	-40.788 ^a	-	-	37.418 ^a
	TS	44.912 ^a	43.904 ^a	44.253 ^a	44.608 ^a	-	-	44.792 ^a
	Surf	33.623 ^a	33.618 ^a	33.621 ^a	-	33.620 ^a	33.623 ^a	33.513 ^a
Enet	Point	-139.090 ^a	-139.090 ^a	-53.060 ^a	-	-	-	-139.047 ^a
	MS	-138.761 ^a	-262.200 ^a	-89.655 ^a	-28.756 ^a	-	-	-138.722 ^a
	TS	22.714 ^a	43.556 ^a	44.226 ^a	22.708 ^a	-	-	22.506 ^a
	Surf	33.556 ^a	33.618 ^a	33.621 ^a	-	33.555 ^a	33.622 ^a	33.445 ^a
Lasso	Point	-139.090 ^a	-139.090 ^a	-53.060 ^a	-	-	-	-139.047 ^a
	MS	-138.761 ^a	-138.761 ^a	-88.834 ^a	-28.999 ^a	-	-	-138.722 ^a
	TS	22.714 ^a	43.327 ^a	44.205 ^a	22.708 ^a	-	-	22.506 ^a
	Surf	33.556 ^a	33.618 ^a	33.621 ^a	-	33.555 ^a	33.622 ^a	33.445 ^a
RF	Point	12.877 ^a	-112.051 ^a	-110.670 ^a	-	-	-	-11.373 ^a
	MS	-7.467 ^a	-65.757 ^a	-64.210 ^a	-237.336 ^a	-	-	-20.231 ^a
	TS	44.757 ^a	43.487 ^a	43.622 ^a	43.569 ^a	-	-	44.521 ^a
	Surf	33.622 ^a	33.618 ^a	33.618 ^a	-	33.581 ^a	33.623 ^a	33.511 ^a
LSTM	Point	26.676 ^a	-	-	-	-	-	25.687 ^a
	MS	32.646 ^a	-	-	-62.050 ^a	-	-	54.563 ^a
	TS	43.841 ^a	-	-	43.500 ^a	-	-	43.960 ^a
	Surf	33.556 ^a	-	-	-	24.373 ^a	-	33.445 ^a

20	Attention	Point	25.580 ^a	-	-	-	-	-	-8.074 ^a
		MS	-23.559 ^a	-	-	-59.891 ^a	-	-	25.841 ^a
		TS	43.494 ^a	-	-	44.889 ^a	-	-	43.789 ^a
		Surf	33.467 ^a	-	-	-	33.622 ^a	33.623 ^a	33.224 ^a
	SVR	Point	30.613 ^a	-53.899 ^a	-59.559 ^a	-	-	-	27.840 ^a
		MS	65.215 ^a	-44.157 ^a	-113.462 ^a	-38.085 ^a	-	-	64.080 ^a
		TS	44.855 ^a	31.823 ^a	19.774 ^a	44.358 ^a	-	-	44.687 ^a
		Surf	33.039 ^a	33.556 ^a	33.556 ^a	-	32.615 ^a	33.614 ^a	32.928 ^a
	Ridge	Point	-2.436 ^b	-47.118 ^a	-68.317 ^a	-	-	-	9.750 ^a
		MS	40.525 ^a	-75.143 ^a	-82.762 ^a	-46.672 ^a	-	-	37.418 ^a
		TS	44.700 ^a	43.626 ^a	43.599 ^a	44.376 ^a	-	-	44.792 ^a
		Surf	33.512 ^a	33.508 ^a	-	-	33.510 ^a	33.512 ^a	33.513 ^a
	Enet	Point	-139.047 ^a	-139.047 ^a	-68.317 ^a	-	-	-	-139.047 ^a
		MS	-138.722 ^a	-232.035 ^a	-89.353 ^a	-32.142 ^a	-	-	-138.722 ^a
		TS	22.506 ^a	42.452 ^a	43.574 ^a	22.500 ^a	-	-	22.506 ^a
		Surf	33.445 ^a	33.508 ^a	-	-	33.445 ^a	33.512 ^a	33.445 ^a
	Lasso	Point	-139.047 ^a	-139.047 ^a	-68.317 ^a	-	-	-	-139.047 ^a
		MS	-138.722 ^a	-146.146 ^a	-86.357 ^a	-38.315 ^a	-	-	-138.722 ^a
		TS	22.506 ^a	42.343 ^a	43.554 ^a	22.500 ^a	-	-	22.506 ^a
		Surf	33.445 ^a	33.508 ^a	-	-	33.445 ^a	33.512 ^a	33.445 ^a
	RF	Point	3.669 ^a	-74.835 ^a	-103.668 ^a	-	-	-	-11.373 ^a
		MS	-6.922 ^a	-66.153 ^a	-69.474 ^a	-240.388 ^a	-	-	-20.231 ^a
		TS	44.617 ^a	42.881 ^a	42.932 ^a	42.950 ^a	-	-	44.521 ^a
		Surf	33.511 ^a	33.499 ^a	-	-	33.470 ^a	33.512 ^a	33.511 ^a
	LSTM	Point	24.689 ^a	-	-	-	-	-	25.687 ^a
		MS	-13.668 ^a	-	-	-47.109 ^a	-	-	54.563 ^a
		TS	44.286 ^a	-	-	44.127 ^a	-	-	43.960 ^a
		Surf	33.448 ^a	-	-	-	8.102 ^a	-	33.445 ^a
	Attention	Point	13.513 ^a	-	-	-	-	-	-8.074 ^a
		MS	-10.893 ^a	-	-	-47.858 ^a	-	-	25.841 ^a
		TS	43.623 ^a	-	-	44.696 ^a	-	-	43.789 ^a
		Surf	33.441 ^a	-	-	-	33.499 ^a	33.512 ^a	33.224 ^a
	SVR	Point	30.070 ^a	-46.214 ^a	-72.478 ^a	-	-	-	27.840 ^a
		MS	62.774 ^a	-59.860 ^a	-107.780 ^a	-37.687 ^a	-	-	64.080 ^a
		TS	44.692 ^a	31.664 ^a	19.538 ^a	44.100 ^a	-	-	44.687 ^a
		Surf	-	-	-	-	-	-	32.928 ^a

TABLE A.8: The table reports the DM statistic for the out of sample dataset on the *put options* using *Nadaraya Watson interpolation*. *a, b, c* denote statistical significance at the 1%, 5%, and 10% levels respectively.

Appendix B

Peseran-Timmermann statistic

The following figures present the results of the PT statistic for the out-of-sample forecasting of the models described in Chapter 4.3. The results are presented for both call and put options contracts. The statistical significance results are not shown as all figures showed statistical significance at the 1% level. Thus, only the heat maps of the forecasting accuracy for the directional change in volatility are presented. Each heat map plots the average forecast accuracy for the directional change in implied volatility for each moneyness and term structure combination. Moneyness is on the x-axis and term structure is on the y-axis. The color intensity indicates the average forecast accuracy, with darker colors indicating higher accuracy.

B.1 Call Options Contracts

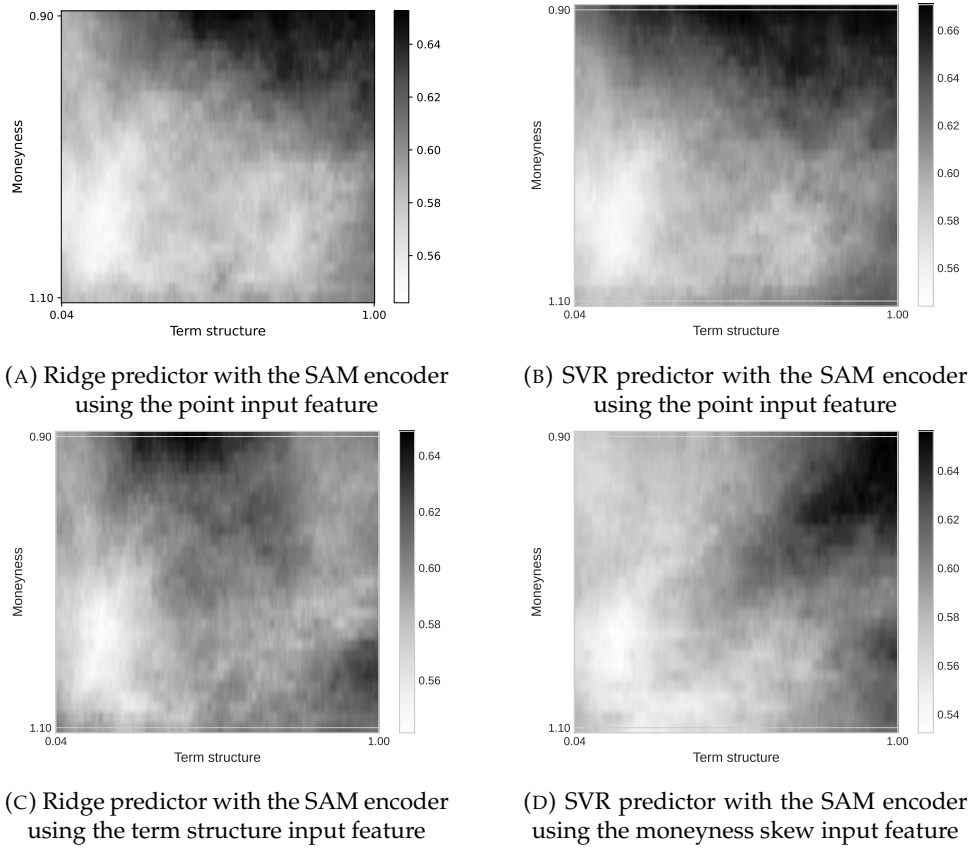


FIGURE B.1: Heat maps of directional results across moneyness and term structure for call options. Heat maps shown are for models trained on *5 day lagged features*. Results all show statistical significance at the 1% level.

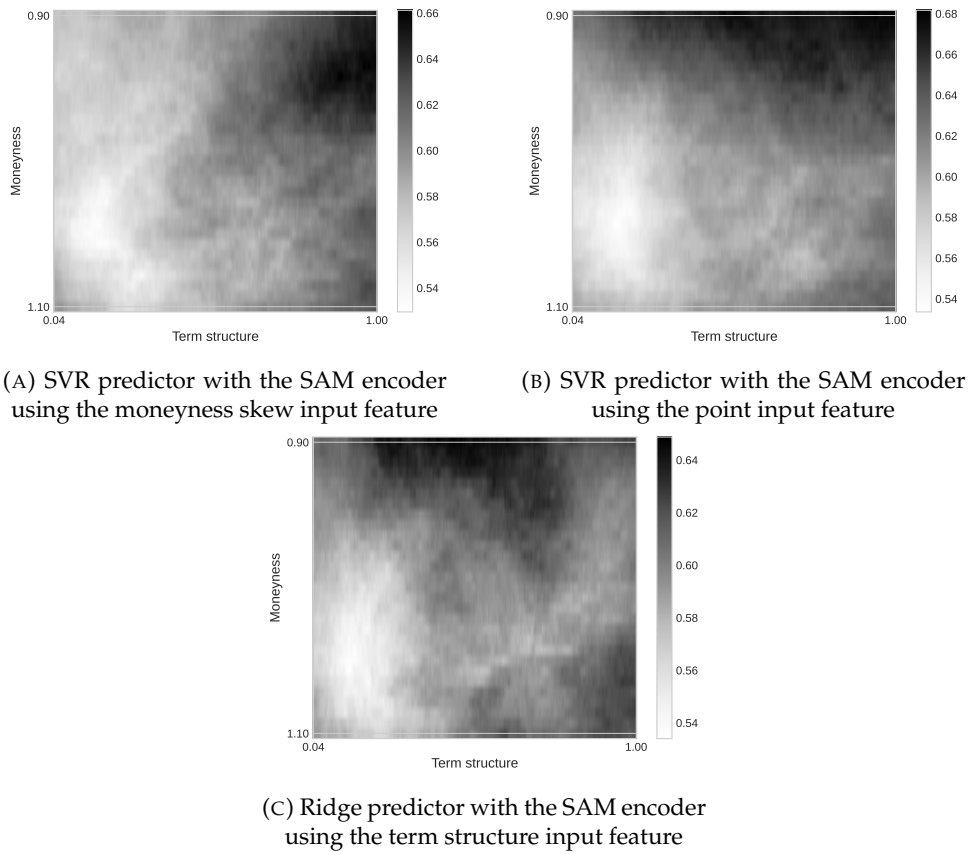


FIGURE B.2: Heat maps of directional results across moneyness and term structure for call options. Heat maps shown are for models trained on *10 day lagged features*. Results all show statistical significance at the 1% level.

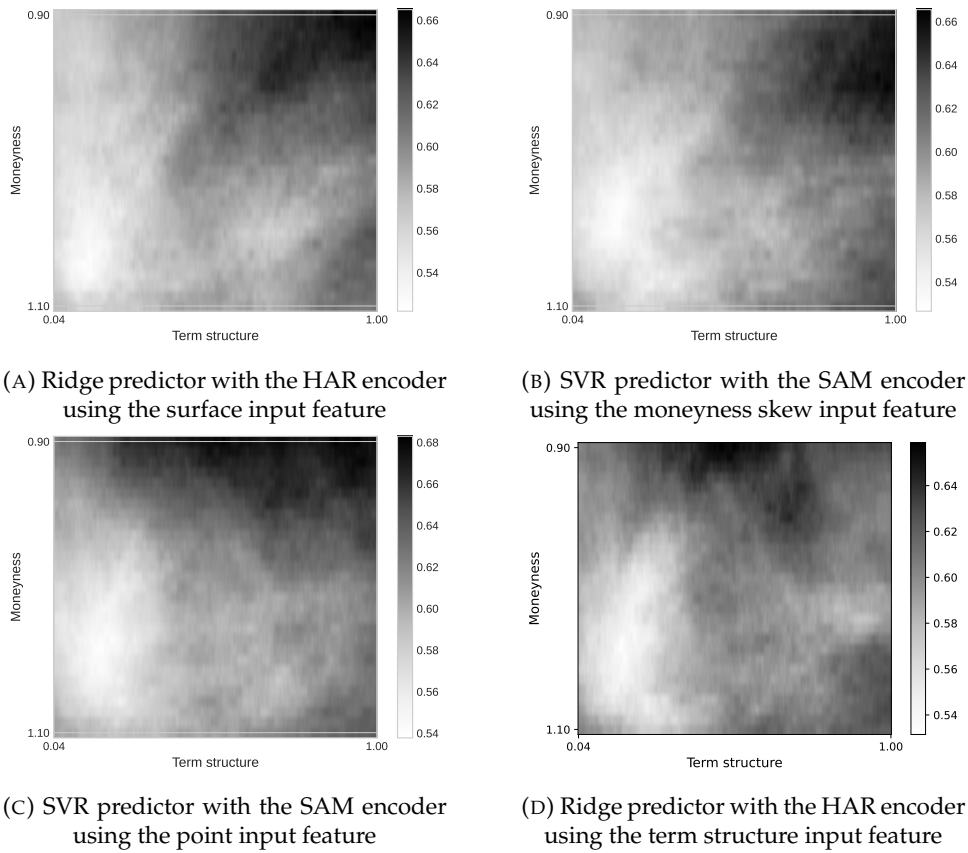


FIGURE B.3: Heat maps of directional results across moneyness and term structure for call options. Heat maps shown are for models trained on *20 day lagged features*. Results all show statistical significance at the 1% level.

B.2 Put options

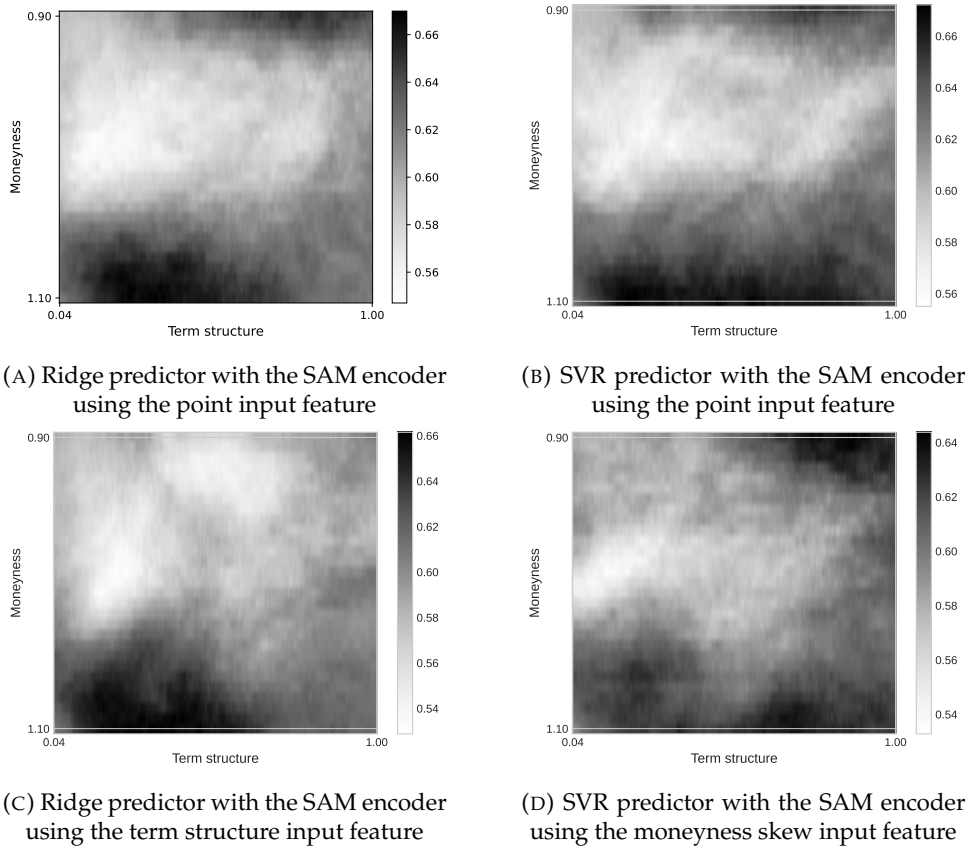


FIGURE B.4: Heat maps of directional results across moneyness and term structure for put options. Heat maps shown are for models trained on *5 day lagged features*. Results all show statistical significance at the 1% level.

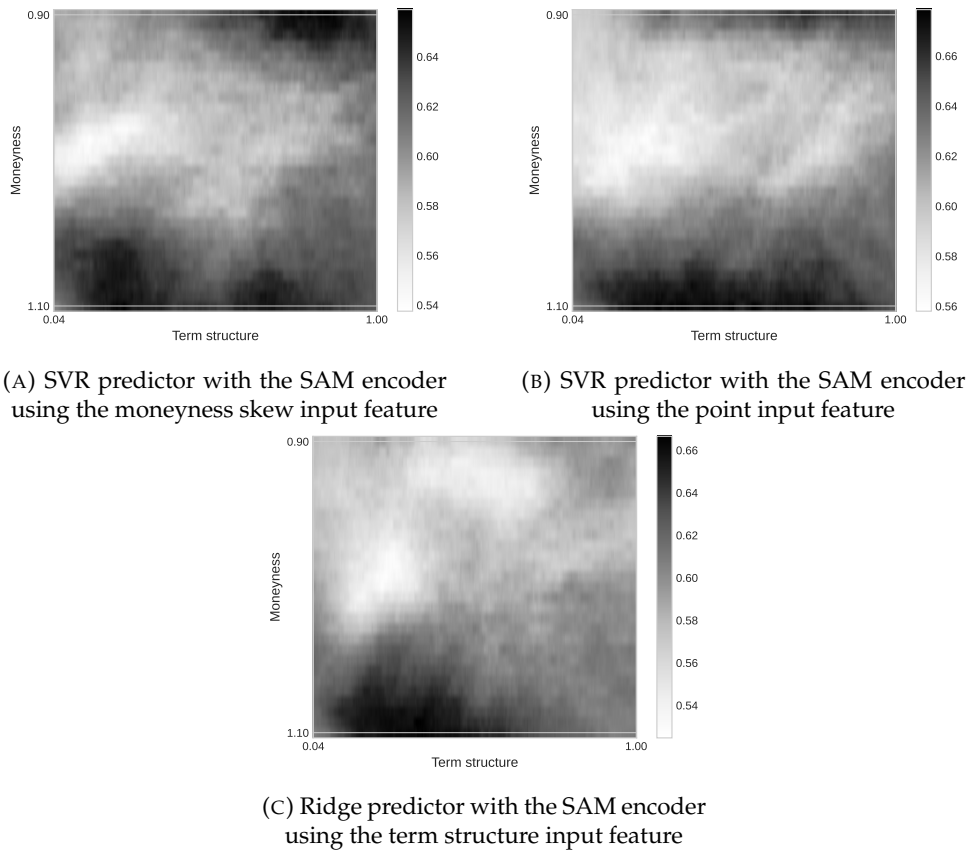


FIGURE B.5: Heat maps of directional results across moneyness and term structure for put options. Heat maps shown are for models trained on *10 day lagged features*. Results all show statistical significance at the 1% level.

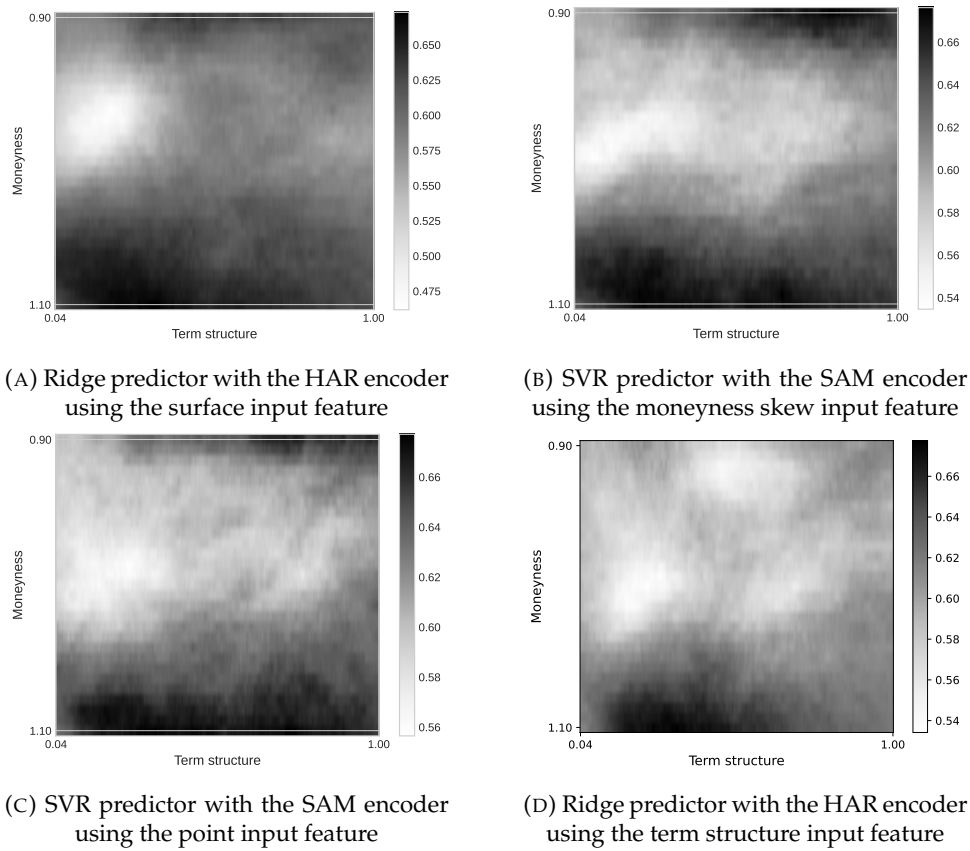
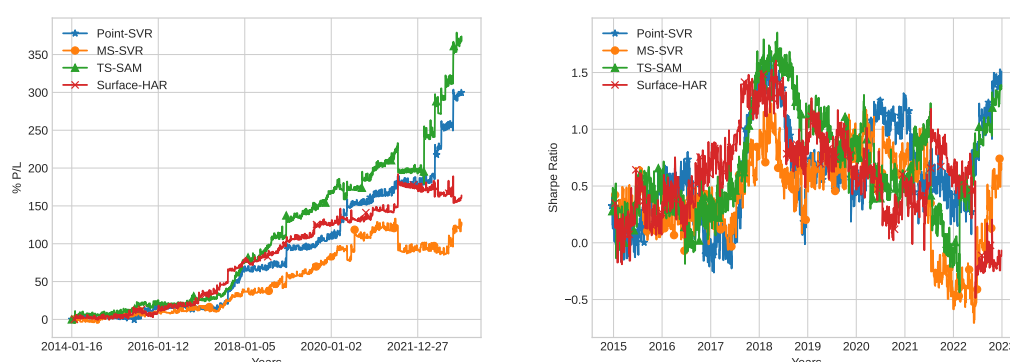


FIGURE B.6: Heat maps of directional results across moneyness and term structure for put options. Heat maps shown are for models trained on *20 day lagged features*. Results all show statistical significance at the 1% level.

Appendix C

Trading results for put options

The following figures and tables present the results of the trading strategy presented in Chapter 6 for put options contracts. The models used are the same as those used for call options, which are the best performing models from the empirical study in Chapter 5.



(A) Portfolio growth over the nine year out-of-sample test period for the put option IVS trading strategy using different forecast models.

(B) Difference in rolling sharpe ratio over the benchmark model

FIGURE C.1: Trading results — Growth of portfolio and rolling Sharpe ratio for volatility trading on the put option IVS for out-of-sample test period: 2014–2022.

	Point-SVR	MS-SVR	Surface-HAR	TS-SAM	Point-AR(1)	MSK-AR(1)	TSK-AR(1)	Surface-AR(1)
# Trades	610	624	596	606	656	638	483	532
Longs Pos	407	468	476	497	640	622	433	506
Shorts Pos	203	156	120	109	16	16	50	26
Long Wins (%)	65.11	63.25	60.71	69.42	65.31	58.36	63.28	57.51
Short Wins (%)	62.07	60.26	70.00	70.64	100.00	75.00	82.00	80.77
Wins (%)	64.10	62.50	62.58	69.64	66.16	58.78	65.22	58.65
Direction Wins (%)	53.09	53.69	50.56	50.44	42.57	46.72	44.10	41.84
Max. win (\$)	30,467.57	23,518.45	30,139.95	28,996.67	30,619.18	30,427.47	30,139.95	23,224.16
Max. loss (\$)	-2,496.99	-35,024.07	-26,195.13	-35,024.07	-35,024.07	-35,024.07	-23,972.68	-35,024.07
Avg. win (\$)	485.73	192.42	267.18	617.20	413.93	184.31	799.77	246.00
Median win (\$)	200.52	189.21	191.90	291.08	285.63	162.46	241.69	185.55
std (\$)	2399.27	2450.44	2316.93	3637.44	3713.28	4027.26	3793.14	4594.00
Alpha (%)	14.82	6.54	6.12	18.99	14.16	7.55	17.78	7.53
Beta	-0.474	-0.220	0.321	-0.882	-0.434	-0.473	-0.479	-0.308
CAGR (%)	16.53	9.39	11.39	18.87	16.05	9.26	19.47	9.98
Avg. SR	0.638	0.413	0.605	0.690	0.985	0.456	0.850	0.480
Gain on Alpha	0.66	-1.00	-1.41	1.21	-	-	-	-

TABLE C.1: Trading statistics

Trade algorithm statistics. The volatility trading algorithm on the put option IVS for nine years out-of-sample test period. A 1.5% of the option price transaction cost is considered for each option buy/sell. SR stands for Sharpe-Ratio. Gain on Alpha refers to the models gain on the bench mark AR(1) model. Bench mark models are shown below the midrule

Appendix D

Interpolation example surfaces

The following figures present examples of the results of the interpolation methods applied to the implied volatility surfaces for both call and put options contracts. The figures show the interpolated implied volatility surfaces using Dumas et al.'s method and the Nadaraya-Watson kernel regression method, along with the true implied volatility surface points.

D.1 Call Options

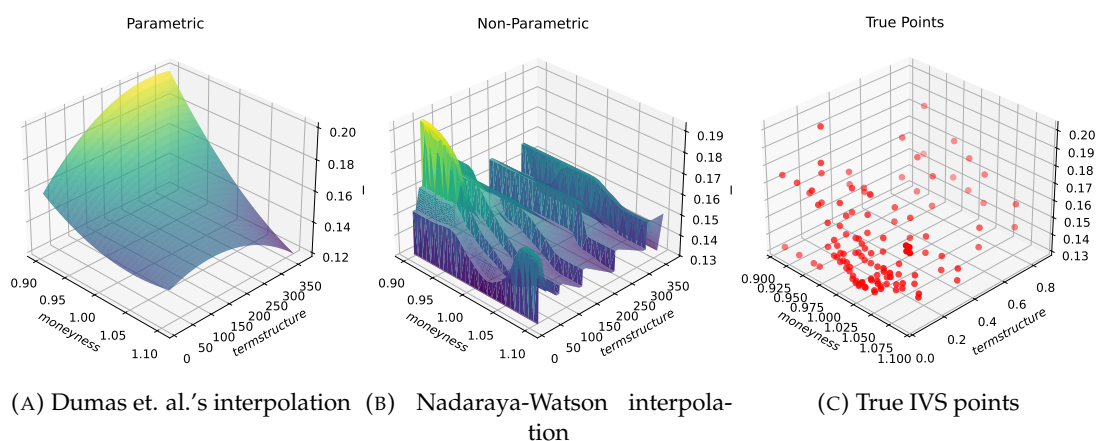


FIGURE D.1: Interpolated IVS for 2004-02-05

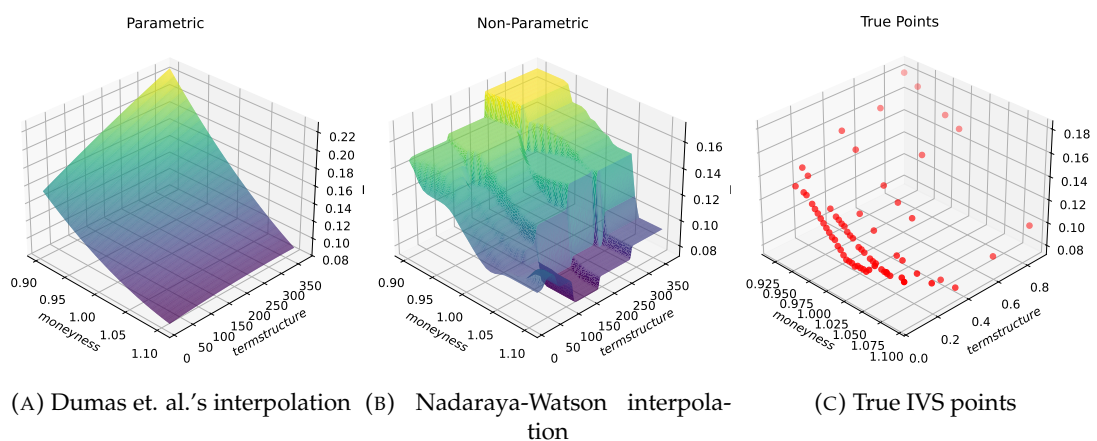


FIGURE D.2: Interpolated IVS for 2006-01-31

D.2 Put Options

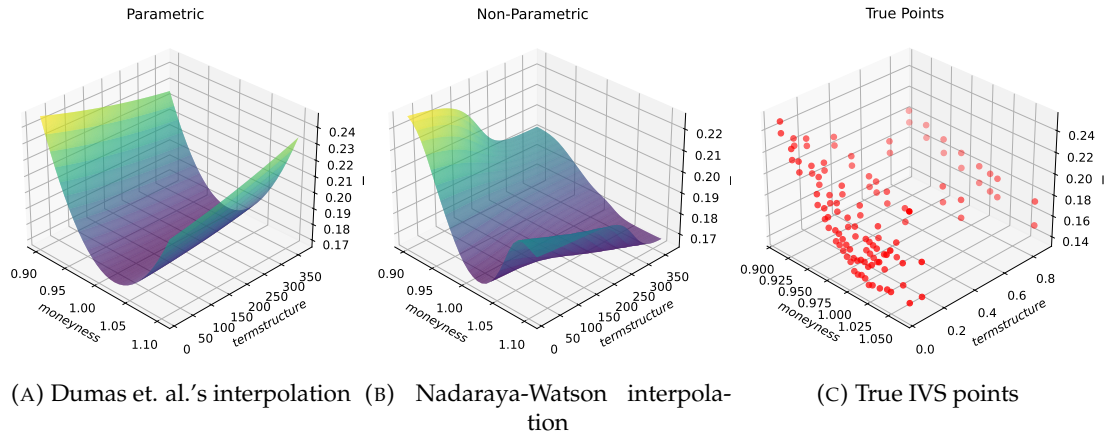


FIGURE D.3: Interpolated IVS for 2004-02-05

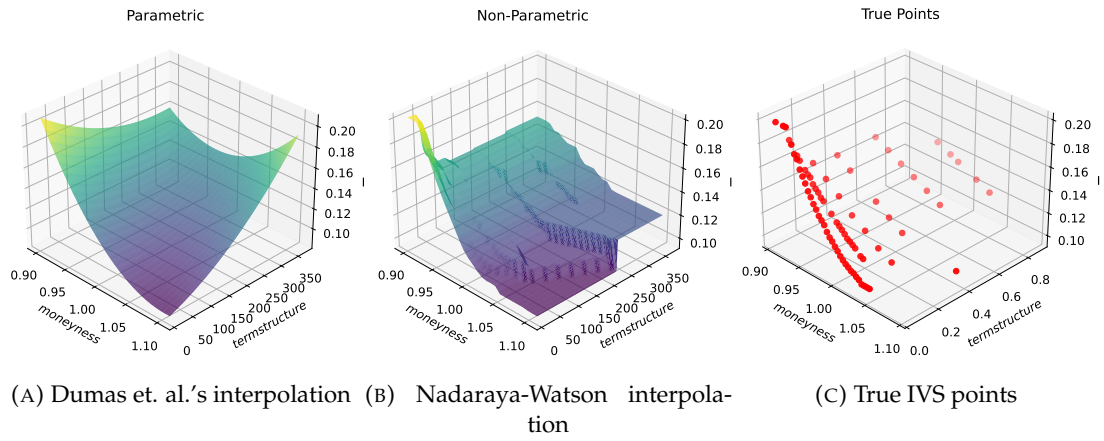


FIGURE D.4: Interpolated IVS for 2006-01-31

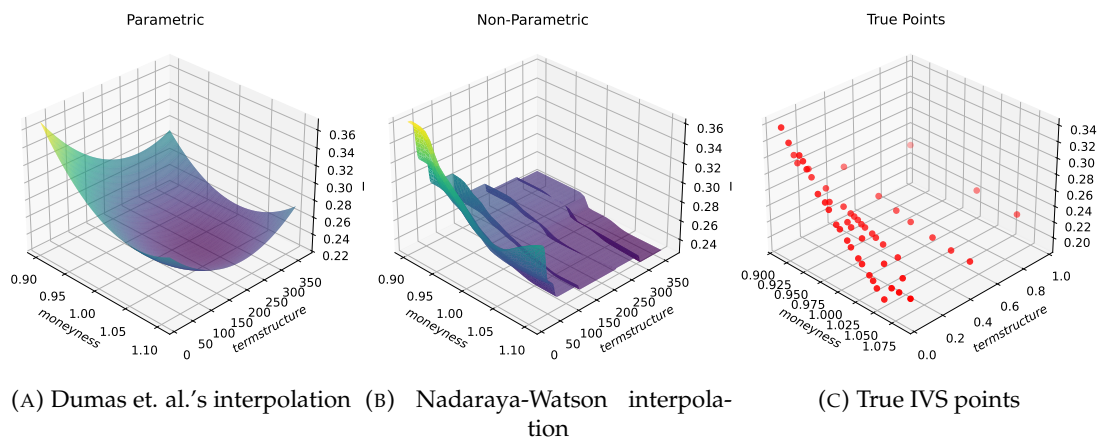


FIGURE D.5: Interpolated IVS for 2006-01-31

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